Quantum Reference Algebra for Information Flow

World structure and physical control

Complete Collected Volume

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October 2026

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Introduction

Why should the algebra through which a system acts be related to the structure through which its world propagates? Q-RAIF studies this interface directly. A positive control cost, an information metric and a Lorentzian form are distinct objects; a bridge has to follow the actual generators and their representation.

A derives the common quadratic shell and Clifford relations under its local propagation assumptions. B shows when a centred field-pulse cost is independent of the unknown carrier state, and when generator alignment relates that cost to the propagation form. C contrasts control discarded into a continuous-spectrum port with control retained in the closed propagation. The former restricts the exact multiplicative domain; the latter permits an exact represented algebra and a conditional Lorentzian extension.

Finite field actuators, material sources, clocks and receiver renewal provide physical witnesses and resource tests for these interfaces. They do not prove arbitrary world–control alignment or indefinitely cyclic operation. Mass–Energy, Null Cluster and No View remain separate companion papers: positive spectral readout, a matched null-access task and conservation-constrained source selection complete the six-paper responsibility without changing this volume's three-paper membership.

Reading map

QR-A Local propagation and the single quadratic shell constrain Clifford and Lorentzian readout. A common cone still needs the stated speed-alignment dynamics.

QR-B A centred physical coupling gives a state-independent cost criterion and a conditional bridge to the propagation form. Alignment is an additional physical question.

QR-C Discarded control ports and closed propagation give different exact representation results. Finite source, clock and renewal witnesses test their physical implementation.

Lorentzian Read-out from Local Unitary Propagation: Clifford Relations, Causal Cones and Spectral Mass

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October 2026

Abstract

A local unitary propagation law can supply the relative sign of a Lorentzian read-out without turning a positive control metric into an indefinite one. For a first-order Hermitian symbol with independent spatial coefficients, all bands share one quadratic shell exactly when those coefficients satisfy the corresponding Euclidean Clifford relations. The resulting form is $\omega^2 - Q(q)$; it describes the full cone when both frequency branches occur. The evolution time is supplied. The internal dimension bounds the number of anticommuting directions, while calibrated coefficient errors give a forward certificate for approximate shells.

A shared shell for one species does not imply one cone for all matter. Fixed-block one-particle mixing cannot merge distinct shells. In a mediator model, sufficiently accumulated interactions align speeds, even when different edges are active on disjoint scale intervals. At a supplied Lorentz-invariant conformal fixed point, nonconserved spin-2 perturbations are irrelevant; additional conserved currents are the marginal exceptions. Neither argument creates the assumed infrared regime.

On a lattice, support, group and rest-and-curvature speeds differ. An actual record protocol certifies the angle of one coined walk and thereby its ideal dictionary parameters, not its noisy long-time transport. The interfaces to physical control and spectral mass require, respectively, calibrated generator alignment and a positive-energy representation. These conditions make the structural connection testable while leaving the formation of the carrier and accessible momentum window as dynamical questions.

Keywords: Lorentzian signature, Clifford algebra, hyperbolic polynomials, quantum cellular automata, quantum walks, Weyl and Dirac equations, causal cone, spectral mass, Q-RAIF

1 Introduction

Programmes that reconstruct geometry from quantum data usually take the Lorentzian signature as given. Holographic entanglement formulas and thermodynamic derivations of the Einstein equation relate entropies to areas and energy fluxes in a spacetime whose causal

structure is already Lorentzian [2, 3]. The Holographic Alaya-Field Framework (HAFF) studies which distinctions a physical reduction can resolve and what geometric data can be reconstructed from them [42, 43]; a spacetime metric needs a further dictionary there as well. This paper asks where the indefinite direction of such a read-out can come from in a model whose dynamics is given explicitly.

Two facts fix the starting point. A positive definite form stays positive under every linear change of variables (Proposition 1), so no control metric, information metric or work form becomes Lorentzian by reparameterization. The companion paper on persistence makes the same distinction physically: its descent metric and field-derived work form are positive. Under an additional all-state work condition and calibrated generator alignment it relates work to the positive spatial shell, not directly to the Lorentzian form [45]. Second, imposing Clifford relations $\{e_\mu, e_\nu\} = 2q_{\mu\nu}\mathbf{1}$ for a chosen Lorentzian q puts the signature into the premise; the universal property of $Cl(V, q)$ then supplies an encoding, rather than a signature-selection mechanism [4, 5].

The model class is local unitary propagation: a particle with an internal Hilbert space $\mathbb{C}^s$ hops on a lattice by a translation-invariant, finite-range unitary step, as in quantum walks and the one-particle sector of quantum cellular automata [9, 10, 11, 12, 13]. Its low-energy generator is a first-order Hermitian matrix symbol. In this setting four objects can be defined separately: a positive norm on generators, which plays the role of a control metric; the causal cones of the dynamics; the algebra of the symbol's coefficients; and the spectral mass. For a first-order Hermitian symbol, unitarity alone makes the characteristic polynomial hyperbolic in the frequency (Theorem 2). For linearly independent spatial coefficients, all bands lie on one quadratic shell if and only if the coefficients satisfy the shell relations of Theorem 4; the characteristic form is then ω^2 minus the Hilbert–Schmidt norm of the spatial symbol and has signature $(1, d)$, and the Lorentzian Clifford generators are $\gamma^0 = \beta$, $\gamma^i = \beta\alpha_i$ (Theorem 4, Corollary 5). The internal dimension bounds the spatial one, $d + [\mu \neq 0] \leq 2\nu_2(s) + 1$. For a qubit carrier three is the largest dimension with an isotropic massless cone, and the only one in which every internal rotation axis is coupled to a direction of motion (Theorem 8, Proposition 10).

On the lattice, group velocities lie in the convex hull of the neighbourhood, so the speed of any massless read-out cone is at most the inradius of that polytope per step (Proposition 12). Exact formulas for the split-step walk and computations for the Weyl automaton then show that the read-out map is not fixed by the dynamics: a band point, a momentum window and the observable that defines the speed all have to be chosen (Section 5).

The mathematical ingredients are standard: Dirac's square root of the Klein–Gordon operator [6], Gårding's hyperbolic polynomials [7], the representation theory of complex Clifford algebras [8], and the derivations of Weyl and Dirac equations from quantum walks and automata [9, 11, 13]. Hyperbolicity as the defining property of a physical dispersion relation is developed in Ref. [14], and an information-theoretic argument linking three spatial dimensions to the qubit is given in Ref. [15]. This paper separates the four objects in one model, identifies which inputs of a Lorentzian read-out follow from propagation, and calculates how the remaining read-out choices enter. The several-species analysis adds a quantitative fixed-block mixing obstruction and applies cut-balanced convergence to interacting speeds that need not couple simultaneously.

The argument proceeds from the symbol and its shell to dimension, lattice windows and

several-species dynamics. A residual bound connects the exact shell criterion to calibrated approximate symbols; a finite angle certificate then connects one such dictionary to actual records. The concluding sections distinguish these contributions from the physical selection of a carrier, an interacting fixed point and a complete source-and-return apparatus.

2 Local unitary propagation and four objects

2.1 The model

Let $\Lambda \subset \mathbb{R}^d$ be a lattice of physical positions and let the one-particle Hilbert space be $\ell^2(\Lambda) \otimes \mathbb{C}^s$. A translation-invariant step with finite neighbourhood $N \subset \Lambda$ acts as

$$(W\psi)(x) = \sum_{h \in N} A_h \, \psi(x - h), \qquad A_h \in M_s(\mathbb{C}), \tag{1}$$

and is unitary exactly when its symbol

$$W(k) = \sum_{h \in N} A_h \, e^{-ik \cdot h} \tag{2}$$

is unitary for every k in the Brillouin zone. One step takes time τ. The eigenvalues of $W(k)$ are $e^{-i\omega_b(k)\tau}$ with quasi-energies $\omega_b(k)$ defined modulo $2\pi/\tau$, and the group velocity of band b is $\nabla_k \omega_b$. With this convention the pure shift $A_h = \mathbf{1}$ for a single h moves every state by $+h$ per step.

Near a point k_* where a gap closes or nearly closes, the step is described by an effective generator. If a parameter θ controls the gap, we say that the walk has *first-order symbol*

$$H(q) = \sum_{i=1}^{d} q_i \alpha_i + \mu\beta, \qquad \alpha_i, \beta \in \mathrm{Herm}(s), \tag{3}$$

at k_* if $(i/\tau) \operatorname{Log} W_\theta(k_* + q) = H(q) + O(|q|^2 + |\mu|^2)$ with μ proportional to θ. This is the scaling limit studied for quantum walks and automata [11, 12, 13]: H generates the continuum dynamics $i\partial_t \psi = H(-i\nabla)\psi$ when lattice spacing and step time go to zero at fixed ratio. For example, the split-step walk of Section 5.2 has $W_\theta(k) = \mathrm{diag}(e^{-ika}, e^{ika}) R(\theta)$ with $R(\theta) = \cos\theta - i \sin\theta \, \sigma_y$, and at $k_* = 0$ its first-order symbol is $H(q) = (a/\tau) q \sigma_z + (\theta/\tau)\sigma_y$.

2.2 Four objects

A control metric. The generators of the internal space form the real vector space $\mathrm{Herm}(s)$, which carries the positive Hilbert–Schmidt norm

$$G_{\mathrm{HS}}(A) = \frac{1}{s} \operatorname{tr}(A^2). \tag{4}$$

Restricted to the span of the coefficients of H, it is the natural positive metric on the controls that the symbol couples to translations. Other positive forms of the same type appear in the series: the descent metric G of a gradient controller and the work form K derived from the stress of a field-coupled actuator [45], and monotone information metrics on states [29].

Causal cones. The step (1) has a strict causal region. After n steps a state supported in S is supported in $S + nN$, the n-fold Minkowski sum, so all influence lies in the cone over the polytope $C_N = \tau^{-1} \operatorname{conv} N$. A first-order symbol has a second cone, its characteristic cone, defined in Section 3. The two need not agree.

A representation. The coefficients α_i, β generate a real algebra inside $M_s(\mathbb{C})$. Whether they satisfy Clifford relations, and with which quadratic form, is a property of the dynamics, not a choice of encoding.

Spectral mass. For a selected logarithm and energy zero, the positive band's value at k_* is the rest energy $E_0 = \hbar|\omega(k_*)|$, and its curvature defines an inertial mass m_* through $\omega(k_* + q) \simeq \omega(k_*) + \hbar q^2/(2m_*)$. A measured phase difference alone does not fix an unknown common energy offset. A Lorentzian read-out identifies $E_0 = m_* c^2$, which defines a speed c; whether this c agrees with either causal cone is a question about the model.

Proposition 1 (Positive forms stay positive)**.** *Let G be a positive definite form on a real vector space V and $T : U \to V$ linear. Then T^*G, given by $(T^*G)(u, u') = G(Tu, Tu')$, is positive semidefinite, and positive definite if T is injective. In particular no positive definite form is congruent to a form with a negative direction.*

Proof. $(T^*G)(u, u) = G(Tu, Tu) \geq 0$, with equality only if $Tu = 0$. The last statement is Sylvester's law of inertia. □

The Lorentzian form of a read-out is therefore not a pullback or a reparameterization of G_{HS}, G or K. In the model it arises as a different kind of object, a form on frequency–momentum space built from the spectrum of the evolution.

3 Where the indefinite direction comes from

For a first-order symbol (3) write $P(\omega, q) = \det(\omega \mathbf{1} - H(q))$ for its characteristic polynomial and $P_0(\omega, q) = \det(\omega \mathbf{1} - \sum_i q_i \alpha_i)$ for the principal part. A homogeneous polynomial p on $\mathbb{R}^n$ is *hyperbolic with respect to e* if $p(e) \neq 0$ and $t \mapsto p(\xi + te)$ has only real roots for every ξ [7].

Theorem 2 (Hyperbolicity in the evolution frequency)**.** *For Hermitian α_i, β, every root in ω of $P(\omega, q)$ is real for every real q, and P_0 is hyperbolic with respect to the frequency direction $e = (1, 0, \ldots, 0)$. The propagation speed of the principal part in a unit direction u is the largest eigenvalue of $\sum_i u_i \alpha_i$.*

Proof. The roots of $P(\cdot, q)$ are the eigenvalues of the Hermitian matrix $H(q)$. For P_0, $P_0(e) = 1$ and the roots of $t \mapsto P_0(\xi_0 + t, \xi')$ are $-\xi_0$ plus the eigenvalues of $\sum_i \xi'_i \alpha_i$, which are real. The characteristic speeds of $i\partial_t \psi = \sum_i \alpha_i (-i\partial_i)\psi$ in direction u are the eigenvalues of $\sum_i u_i \alpha_i$. □

The evolution parameter is supplied by the model. Unitarity makes its conjugate frequency a hyperbolicity direction; it does not choose how many fundamental time parameters

a theory should have. The next lemma determines the signature when the characteristic form is quadratic.

Lemma 3 (Hyperbolic quadratic forms are Lorentzian)**.** *A real quadratic form p on $\mathbb{R}^n$ with $p(e) > 0$ is hyperbolic with respect to e if and only if it has exactly one positive eigenvalue. Then $p = \ell^2 - r$ with ℓ linear and r positive semidefinite, and if p is non-degenerate its signature is $(1, n-1)$.*

Proof. Decompose $\xi = se + \zeta$ with ζ in the p-orthogonal complement of e. Then $p(\xi + te) = p(e)(s+t)^2 + p(\zeta)$, which has real roots in t for every ξ if and only if $p(\zeta) \le 0$ on the complement. That is the statement that p has one positive direction and is negative semidefinite on a complementary hyperplane. □

Theorem 4 (Single shell, Clifford relations and the Lorentzian form)**.** *Let $\alpha_1, \dots, \alpha_d, \beta \in \mathrm{Herm}(s)$ with $\alpha_1, \dots, \alpha_d$ linearly independent, and fix $\mu \in \mathbb{R}$. The following are equivalent.*

(a) There is a quadratic form Q on $\mathbb{R}^d$ such that every eigenvalue ω of $H(q)$ satisfies $\omega^2 = Q(q) + \mu^2$ for all q.

(b) $H(q)^2 = (Q(q) + \mu^2)\mathbf{1}$ for all q.

(c) $\{\alpha_i, \alpha_j\} = 2Q_{ij}\mathbf{1}$, $\mu\{\alpha_i, \beta\} = 0$ and $\mu^2(\beta^2 - \mathbf{1}) = 0$.

If they hold, then:

1. *Q is positive definite and $Q(q) = G_{\mathrm{HS}}\big(\sum_i q_i \alpha_i\big)$;*
2. *the read-out form $\eta(\omega, q) = \omega^2 - Q(q)$ on frequency–momentum space has signature $(1, d)$, and the characteristic set of P_0 lies on its null cone; the sets coincide if $d \ge 2$, if $\mu \neq 0$, or if $d = 1$ and α_1 has both positive and negative eigenvalues;*
3. *if $d + [\mu \neq 0] \ge 2$, then s is even and $P(\omega, q) = (\omega^2 - Q(q) - \mu^2)^{s/2}$;*
4. *every group velocity $v = \nabla_q \omega$ satisfies $Q^{-1}(v, v) = Q(q)/(Q(q) + \mu^2) \le 1$, with equality only for $\mu = 0$.*

Proof. (a)⇒(b): $H(q)$ is Hermitian, hence diagonalizable, and all its eigenvalues square to $Q(q) + \mu^2$. (b)⇒(c): expanding,

$$H(q)^2 = \sum_{i,j} q_i q_j \tfrac{1}{2}\{\alpha_i, \alpha_j\} + \mu \sum_i q_i \{\alpha_i, \beta\} + \mu^2 \beta^2.$$

Comparing the parts of degree two, one and zero in q with $(Q(q) + \mu^2)\mathbf{1}$ gives the three relations. (c)⇒(a): the same expansion gives (b), and (b) gives (a).

For item 1, $Q(q)\mathbf{1} = (\sum_i q_i \alpha_i)^2 \ge 0$; if $Q(q) = 0$ the Hermitian matrix $\sum_i q_i \alpha_i$ squares to zero and so vanishes, and linear independence gives $q = 0$. Taking the normalized trace gives $Q(q) = G_{\mathrm{HS}}(\sum_i q_i \alpha_i)$. For item 2, η is ω^2 minus a positive definite form, and every root of $P_0(\cdot, q)$ belongs to $\{\pm\sqrt{Q(q)}\}$. When $d \ge 2$, the principal symbol at nonzero q anticommutes with an invertible symbol in a Q-orthogonal direction, pairing its two signs.

If $\mu \neq 0$, β provides the pairing. In the remaining one-dimensional case both signs occur exactly when α_1 has both signs. For item 3, choose coordinates with $Q = c^2\mathbf{1}$. The matrices α_i/c together with β (if $\mu \neq 0$) pairwise anticommute and square to $\mathbf{1}$. If at least two are present, each x anticommutes with an invertible g, giving $\operatorname{tr} x = \operatorname{tr}(g^{-1}xg) = -\operatorname{tr} x$. Thus $H(q)$ is traceless, and its eigenvalues $\pm\sqrt{Q+\mu^2}$ have equal multiplicity. For item 4, at a differentiable band point with $\omega \neq 0$, $\nabla_q \omega = Qq/\omega$ and $Q^{-1}(v,v) = Q(q)/\omega^2$. At a massless node the same statement applies to directional band limits. □

Corollary 5 (Lorentzian Clifford generators)**.** *Under Theorem 4 with $\mu \neq 0$, the matrices*

$$\gamma^0 = \beta, \qquad \gamma^i = \beta\alpha_i \tag{5}$$

satisfy $\{\gamma^0,\gamma^0\} = 2$, $\{\gamma^0,\gamma^i\} = 0$ and $\{\gamma^i,\gamma^j\} = -2Q_{ij}$, so they generate a representation of $Cl(1,d)$ on $\mathbb{C}^s$; γ^0 is Hermitian and the γ^i are anti-Hermitian. The eigenvalue equation $H(q)\psi = \omega\psi$ is equivalent to $(\omega\gamma^0 - \sum_i q_i\gamma^i - \mu)\psi = 0$, and $(\omega\gamma^0 - \sum_i q_i\gamma^i)^2 = \eta(\omega,q)\mathbf{1}$. For $\mu = 0$ and $d \geq 1$, the matrices

$$\Gamma^0 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \qquad \Gamma^i = \begin{pmatrix} 0 & \alpha_i \\ -\alpha_i & 0 \end{pmatrix} \tag{6}$$

generate a representation of $Cl(1,d)$ on $\mathbb{C}^s \oplus \mathbb{C}^s$.

Proof. $(\beta\alpha_i)^2 = -\beta^2\alpha_i^2 = -Q_{ii}$ and $\{\beta,\beta\alpha_i\} = \alpha_i - \alpha_i = 0$ use $\{\alpha_i,\beta\} = 0$ and $\beta^2 = 1$; similarly $\{\beta\alpha_i,\beta\alpha_j\} = -\{\alpha_i,\alpha_j\}$. Multiplying $H(q) - \omega$ by β gives the second statement. The block matrices are checked by direct multiplication. □

What has been derived. In the model the indefinite direction of the read-out form is the frequency ω, the spectral variable of the one-parameter group of time steps. The positive part is the Hilbert–Schmidt norm of the spatial symbol, a control-type metric. The Lorentzian form is the difference $\omega^2 - G_{\mathrm{HS}}(\sum_i q_i\alpha_i)$, which by Proposition 1 could not be obtained from any positive form alone. Theorem 2 makes the supplied evolution frequency a hyperbolicity direction for every first-order Hermitian symbol; Lemma 3 says that a quadratic cone with one time direction has signature $(1,d)$; and Theorem 4 says that the cone is quadratic, with all internal states on it, exactly when the propagation coefficients satisfy Euclidean Clifford relations. The Lorentzian Clifford algebra of Corollary 5 is then constructed, not supplied: the time-like generator is the mass term β.

The signature is no longer an input. The Clifford relations are equivalent to an observable property, a single shell, which means no birefringence and no internal state that propagates differently from the others. That property, isotropy of Q (which can be absorbed into the choice of spatial coordinates for one species) and the existence of a band point with a linear symbol remain inputs.

Remark 6 (Two boundary cases). For $d = 1$ and $\mu = 0$, $\alpha_1^2 = Q_{11}\mathbf{1}$ allows unequal multiplicities of $\pm Q_{11}^{1/2}$, and for $s = 1$ the symbol $H(q) = cq$ has a single branch. Its characteristic set is one chiral branch, with one future null direction. This is the chiral field used as an actuator in Ref. [45]: it supplies one null direction, and a full $(1,1)$ cone needs both chiralities. In the other direction, a quadratic cone can occur without a single shell. The Maxwell

field in Riemann–Silberstein form has the symbol $c\,\mathbf{S}\cdot q$ with spin-one matrices; its characteristic polynomial is $\omega(\omega^2 - c^2|q|^2)$, and the flat factor is removed by the Gauss constraint. Theorem 4 concerns symbols for which every internal state lies on the cone.

3.1 A forward certificate for approximate shells

The exact operator identity has a finite-error consequence. It requires bounds on implemented coefficients or their calibration errors, not merely access to effects whose ideal algebra contains those coefficients.

Proposition 7 (Squared-band certificate)**.** *Choose a positive quadratic form Q and Hermitian coefficients α_i, β. Let*

$$R_{ij} = \tfrac{1}{2}\{\alpha_i, \alpha_j\} - Q_{ij} I, \qquad R_{i\beta} = \{\alpha_i, \beta\}, \qquad R_\beta = \beta^2 - I.$$

Suppose their operator norms are bounded by calibrated numbers $\epsilon_{ij}, \epsilon_{i\beta}, \epsilon_\beta$. If the implemented symbol is $\widetilde{H}(q) = H(q) + E(q)$ with Hermitian E and $\|E(q)\| \le e(q)$, put

$$\Delta(q) = \sum_{i,j} |q_i q_j| \epsilon_{ij} + |\mu| \sum_i |q_i| \epsilon_{i\beta} + \mu^2 \epsilon_\beta,$$
$$L(q) = \sum_i |q_i| \|\alpha_i\| + |\mu| \|\beta\|, \qquad B(q) = \Delta(q) + 2L(q)e(q) + e(q)^2.$$

Every eigenvalue $\widetilde{\omega}$ obeys

$$\left|\widetilde{\omega}^2 - Q(q) - \mu^2\right| \le B(q).$$

With $S = Q(q) + \mu^2 \ge 0$, this gives $\left||\widetilde{\omega}| - \sqrt{S}\right| \le \sqrt{B(q)}$. If $S > B(q)$, a sharper radius is $B(q)/(\sqrt{S} + \sqrt{S - B(q)})$. The bounds hold uniformly on any specified window where their input bounds hold.

Proof. Expanding $H^2 - SI$ gives $\sum q_i q_j R_{ij} + \mu \sum q_i R_{i\beta} + \mu^2 R_\beta$, whose norm is at most Δ. The additional square error is $HE + EH + E^2$, of norm at most $2Le + e^2$. Evaluate on an eigenvector of the Hermitian $\widetilde{H}$. Finally, $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$ for nonnegative x, y; away from zero use the difference of squares. □

This is a forward dispersion certificate. It does not say that nearly anticommuting coefficients can be repaired to an exact Clifford representation, that the two frequency multiplicities are balanced, or that a finite effect menu implements operator-norm tomography. A calibration valid on a restricted state set is not a norm bound on the entire carrier. The distinction matters: a Hilbert–Schmidt trace identity can hold exactly while different internal states retain different propagation speeds.

3.2 Outputs to physical control and mass read-out

Two physical interfaces use the shell theorem, but require different additional inputs. For control, the spatial map $J(q) = \sum_i q_i \alpha_i$ is an isometry into internal generators with G_{HS} because its square is scalar. The trace identity alone is weaker: the local unitary walk

$W(k) = \mathrm{diag}(e^{-ik}, e^{-2ik})$ has $H(q) = q\,\mathrm{diag}(1,2)$ and $G_{\mathrm{HS}}(H(q)) = 5q^2/2$, yet two distinct speeds. The operator square, not a mean over internal states, supplies one shell.

Q-RAIF B [45] gives that distinction an operational test. For its centered independent field input, a coupling $A(a) = J(Ca)/E_*$ has switching work

$$W_\rho(a) = \kappa_g \,\mathrm{tr}[\rho A(a)^2].$$

If the physical calibration C is invertible, independence of this charge from every unknown internal state is equivalent to a scalar spatial shell. Its value is $\kappa_g Q(Ca)/E_*^2$. The field profile, calibrated map and energy scale are supplied; the encounter requires vanishing or commuting free internal evolution. This is a bridge between two calibrated uses of the same generators. It neither equates the cost with $\omega^2 - Q$ nor identifies the chiral control field with a general multi-dimensional propagation carrier. Q-RAIF C [46] uses Corollary 5's symbol to represent the Lorentzian quadratic space on the closed carrier; discarding a drive gives a different, reduced-channel representation problem.

For mass–energy read-out, the massive continuum shell must also carry the positive-energy Poincare representation specified in Q-RAIF ME [47]. That paper constructs it on the positive band of the Dirac symbol, with the boost Jacobian and spin transport on momentum-space wavefunctions. The symmetric pair of eigenvalues in Theorem 4 alone is not such a representation or a preparation of its positive sector. A finite lattice window is not invariant under arbitrary boosts. The actual coin experiment in Section 5.5 provides a narrower, priced parameter certificate, which remains useful without upgrading it to exact relativistic covariance.

4 Spatial dimension and the internal space

Theorem 8 (Dimension bound)**.** *Suppose Theorem 4 holds on $\mathbb{C}^s$, and let $n = d + [\mu \neq 0]$. Write $s = 2^\nu m$ with m odd, $\nu = \nu_2(s)$. Then $n \leq 2\nu + 1$. Conversely, for every s and every $n \leq 2\nu_2(s) + 1$ there are coefficients satisfying Theorem 4.*

Proof. Choose spatial coordinates in which $Q = c^2\mathbf{1}$; the coefficients transform linearly, so the relations are preserved. Then $e_i = \alpha_i/c$ and, if $\mu \neq 0$, $e_{d+1} = \beta$ are n pairwise anticommuting Hermitian involutions, which define a representation of the complex Clifford algebra $\mathbb{C}l_n$ on $\mathbb{C}^s$. This algebra is isomorphic to $M_{2^{n/2}}(\mathbb{C})$ for even n and to $M_{2^{(n-1)/2}}(\mathbb{C}) \oplus M_{2^{(n-1)/2}}(\mathbb{C})$ for odd n [8]. Every finite-dimensional representation is a direct sum of irreducible ones of dimension $2^{\lfloor n/2 \rfloor}$, so $2^{\lfloor n/2 \rfloor}$ divides s and $\lfloor n/2 \rfloor \leq \nu$. For the converse take an irreducible representation of $\mathbb{C}l_{2\nu+1}$, which has dimension 2^ν, tensor it with $\mathbf{1}_m$, and use any n of its generators. □

Corollary 9 (Qubit carriers)**.** *For $s = 2$: an isotropic single-shell symbol exists for massless propagation in $d \leq 3$ and for massive propagation in $d \leq 2$. Massive single-shell propagation in $d = 3$ requires $4 \mid s$; the minimal case $s = 4$ is the Dirac symbol $\alpha_i = \sigma_x \otimes \sigma_i$, $\beta = \sigma_z \otimes \mathbf{1}$, a pair of Weyl qubits coupled by the mass term.*

For $s = 2$ and $d = 3$ the symbol is $H(q) = c\,\sigma \cdot q$ up to a unitary and an orthogonal change of spatial coordinates, and

$$\det\big(\omega\mathbf{1} - c\,\sigma \cdot q\big) = \omega^2 - c^2|q|^2. \tag{7}$$

This is the Lorentzian form of signature $(1,3)$, obtained from the Pauli relations $\{\sigma_i, \sigma_j\} = 2\delta_{ij}$, which are Euclidean. No operator on $\mathbb{C}^2$ anticommutes with all three Pauli matrices, so $Cl(1,3)$ has no representation on $\mathbb{C}^2$; the Lorentzian form exists anyway, as the characteristic form of the evolution.

Two forms on the same carrier. The identity $\det(x_0 I + x \cdot \sigma) = x_0^2 - |x|^2$ is preserved by determinant-one filtering, not by the positive Hilbert–Schmidt norm. For the massive Dirac carrier, the time-like generator in Corollary 5 is the mass coefficient, not a modular generator.

Proposition 10 (Full coupling of a qubit)**.** *Let $s = 2$, $\mu = 0$, and let the symbol satisfy Theorem 4 with traceless coefficients. The map $q \mapsto \sum_i q_i \alpha_i$ is an injective linear map from $\mathbb{R}^d$ into the three-dimensional space of traceless Hermitian 2×2 matrices. It is onto, so that every internal rotation generator is coupled to some direction of motion, if and only if $d = 3$.*

Proof. Injectivity is the linear independence assumed in Theorem 4; the target has dimension three. □

Proposition 10 is a precise version of a familiar observation: momentum space of a Weyl particle is identified with the Lie algebra of internal rotations of its qubit. It is a characterization under a stated premise, not a derivation of three dimensions. The premise, that no internal axis is left uncoupled, is a physical input. The following conjecture states what would have to be shown to turn it into a selection principle.

Conjecture 11 (Minimal carriers and three dimensions)**.** *In local unitary dynamics whose elementary internal carrier is a qubit and whose low-energy propagation is isotropic and single-shell, propagation that uses the full internal space selects $d = 3$, and massive propagation requires two carriers coupled by a chirality-exchanging term.*

Theorem 8 and Proposition 10 reduce this selection proposal to a physical reason for coupling every internal axis to momentum. Irreducibility alone does not supply it: $H = q_x\sigma_x + q_y\sigma_y$ generates the full matrix algebra, while the omitted σ_z is not a conserved charge. The three-dimensional Weyl automaton [13] and the minimal-directional-system argument of Ref. [15] motivate the proposal, but do not select three dimensions within the present class. A selection principle must distinguish the two-dimensional counterexample by stability, resources or formation, and must explain why a qubit is the relevant elementary carrier.

5 The lattice level: causal polytope, three speeds and nodes

5.1 Group velocities lie in the causal polytope

Proposition 12 (Group velocities and the neighbourhood)**.** *For the step (1), let b be a band and k_0 a point where the eigenvalues of $W(k_0)$ are distinct. Then $\nabla_k \omega_b(k_0) \in \tau^{-1} \operatorname{conv} N$.*

Proof. Take $\tau = 1$. Let X be the position operator and ψ finitely supported. Then $\langle X\rangle_n = \langle W^n\psi, XW^n\psi\rangle$ satisfies $\langle X\rangle_n - \langle X\rangle_0 = \sum_{j<n}\langle W^j\psi, VW^j\psi\rangle$ with $V = W^*[X,W]$. From Eq. (1), $([X,W]\psi)(x) = \sum_h hA_h\psi(x-h)$, so V is bounded and translation invariant with symbol $V(k) = W(k)^* i\nabla_k W(k)$. Write $W(k) = \sum_b e^{-i\omega_b(k)}P_b(k)$ wherever the eigenvalues are distinct, which is almost everywhere once it holds at one point, since the discriminant of the characteristic polynomial is real-analytic in k. For each such k the Cesàro means $n^{-1}\sum_{j<n} W(k)^{-j}V(k)W(k)^j$ converge to $\sum_b P_bVP_b$, and differentiating $W = \sum_b e^{-i\omega_b}P_b$ with $P_b(\nabla P_b)P_b = 0$ gives $P_bVP_b = \nabla\omega_b\, P_b$. By bounded convergence,

$$\frac{\langle X\rangle_n - \langle X\rangle_0}{n} \longrightarrow \bar v(\psi) = \sum_b \int \nabla\omega_b(k)\,\langle\hat\psi(k), P_b(k)\hat\psi(k)\rangle\,\frac{dk}{|B|},$$

where B is the Brillouin zone. On the other hand $W^n\psi$ is supported in $\operatorname{supp}\psi + nN$, so for every unit vector u, $u\cdot\langle X\rangle_n \le \max_{x\in\operatorname{supp}\psi} u\cdot x + n\max_{h\in N} u\cdot h$. Hence $u\cdot\bar v(\psi) \le h_N(u) := \max_{h\in N} u\cdot h$. The map $\hat\psi \mapsto \bar v(\psi)$ is continuous in L^2 because $|\nabla\omega_b| \le \|V(k)\|$ is bounded, and finitely supported states are dense, so the inequality holds for all ψ. Taking $\hat\psi = P_b(k)\varphi$ with φ concentrating at k_0 gives $u\cdot\nabla\omega_b(k_0) \le h_N(u)$ for every u, which is membership in $\operatorname{conv} N$. □

The ballistic scaling used in the proof is the expectation-value form of the weak limit theorem for quantum walks [16].

Corollary 13 (Speed of a massless read-out cone)**.** *If a band has a first-order massless symbol at k_* whose cone is round with speed c, then $c \le \tau^{-1} r_{\text{in}}(\operatorname{conv} N)$, where r_{in} is the inradius.*

Proof. Near k_* the group velocity tends to $c\,\hat q$ along every direction $\hat q$, so $c\hat q \in \tau^{-1}\operatorname{conv} N$ for all $\hat q$. □

5.2 The split-step walk

Consider the coined walk that applies a coin rotation and then a conditional shift,

$$W_\theta(k) = \begin{pmatrix} e^{-ika}\cos\theta & -e^{-ika}\sin\theta \\ e^{ika}\sin\theta & e^{ika}\cos\theta \end{pmatrix}, \qquad 0 < \theta < \frac{\pi}{2}, \tag{8}$$

with neighbourhood $N = \{a, -a\}$ and step time τ.

Proposition 14 (Three speeds of one walk)**.** *For the walk* (8)*:*

1. $\cos(\omega\tau) = \cos\theta\cos(ka)$*;*
2. *the support speed is a/τ, and the largest group velocity is $(a/\tau)\cos\theta$, attained at $ka = \pm\pi/2$;*
3. *near $k = 0$, $\omega(k)^2\tau^2 = \theta^2 + \theta\cot\theta\,(ka)^2 + O((ka)^4)$; the rest energy is $E_0 = \hbar\theta/\tau$, the inertial mass $m_* = \hbar\tau\tan\theta/a^2$, and the Lorentzian read-out speed defined by $E_0 = m_* c_{\text{eff}}^2$ is $c_{\text{eff}} = (a/\tau)(\theta\cot\theta)^{1/2}$;*

4. $\cos\theta < (\theta\cot\theta)^{1/2} < 1$.

Proof. Item 1: $\det W_\theta = 1$ and $\operatorname{tr} W_\theta = 2\cos\theta\cos(ka)$. Item 2: the support statement is Eq. (1) with $N = \{\pm a\}$. Differentiating item 1, $v = (a/\tau)\cos\theta\sin(ka)/\sin(\omega\tau)$, and $v^2 \le (a/\tau)^2\cos^2\theta$ is equivalent to $\cos^2\theta\cos^2(ka) \le \cos^2(ka)$, with equality only at $\cos(ka) = 0$. Item 3: differentiating item 1 twice at $k = 0$, where $\omega\tau = \theta$ and $\omega' = 0$, gives $\tau\sin\theta\,\omega'' = a^2\cos\theta$; then $\frac{1}{2}(\omega^2)'' = \omega\omega''$, $m_* = \hbar/\omega''$, and $E_0/m_* = \omega(0)\omega''(0)$. Item 4 is $\sin\theta\cos\theta < \theta < \tan\theta$. □

The Lorentzian speed read from the rest energy and the inertia of slow particles lies strictly between the largest transport speed and the support speed. No wave packet reaches it, because $v \le (a/\tau)\cos\theta < c_{\text{eff}}$; and the strict causal cone is wider. In the scaling limit $a, \tau \to 0$ with $a/\tau = c$ and $\theta/\tau = \mu$ fixed, all three speeds tend to c and the dispersion tends to $\omega^2 = c^2q^2 + \mu^2$ [11, 12]. At $\theta = 0$ the walk is a pair of exact shifts, with $\omega = \pm k$ and one speed. The deviations at finite mass are $1 - c_{\text{eff}}^2\tau^2/a^2 = \theta^2/3 + O(\theta^4)$ and $1 - \cos^2\theta = \theta^2 + O(\theta^4)$. The consequences for mass read-outs are developed in the companion paper on mass and energy [47].

5.3 The Weyl automaton on the body-centred cubic lattice

The minimal three-dimensional automaton of D'Ariano and Perinotti [13] has $s = 2$, neighbourhood $N = \{\ell(\pm1, \pm1, \pm1)/\sqrt{3}\}$ of eight vectors of length ℓ, and symbol

$$W(k) = e^{-i\kappa_x\sigma_x}\, e^{-i\kappa_y\sigma_y}\, e^{-i\kappa_z\sigma_z}, \qquad \kappa = \frac{\ell k}{\sqrt{3}}, \tag{9}$$

with $\cos(\omega\tau) = c_xc_yc_z - s_xs_ys_z$, where $c_i = \cos\kappa_i$, $s_i = \sin\kappa_i$. Expanding, $c_xc_yc_z = 1 - |\kappa|^2/2 + O(|\kappa|^4)$ and $s_xs_ys_z = \kappa_x\kappa_y\kappa_z + O(|\kappa|^5)$, so

$$\omega\tau = |\kappa| + \frac{\kappa_x\kappa_y\kappa_z}{|\kappa|} + O(|\kappa|^3). \tag{10}$$

At leading order $W(k) \simeq \mathbf{1} - i\sigma\cdot\kappa$, the symbol is $c\,\sigma\cdot k$ with $c = \ell/(\sqrt{3}\tau)$, and Eq. (7) gives a round Lorentzian cone. Isotropy fails at relative order $|\kappa|$ through the parity-odd term in Eq. (10). Table 1 collects analytic properties and numerical unitary wave-packet diagnostics. It is not a table of records obtained from a physical measuring instrument.

5.4 What the read-out map adds

The lattice results show three ways in which the dynamics leaves the Lorentzian read-out underdetermined.

First, at finite mass the speed of the read-out depends on the observable that defines it. In the walk (8), rest energy and inertia give c_{eff}, the fastest wave packets give $(a/\tau)\cos\theta$, and the support front gives a/τ (Proposition 14).

Second, the Lorentzian cone governs low-momentum states only. In the Weyl automaton a state prepared on one site contains all momenta and leaves almost half of its probability

Property	**Value**
Causal polytope $\tau^{-1}\operatorname{conv} N$	cube $\lvert x_i\rvert \le \ell/(\sqrt{3}\tau)$ per unit time (analytic)
Inradius of the cube	$\ell/(\sqrt{3}\tau) = c$: Corollary 13 is saturated (analytic)
Group velocity components	at most c, attained on the cube faces
Largest group speed	$(3/2)^{1/2}c$; the cube corners, at speed $\sqrt{3}c$, are not reached by group velocities
Band-touching points in the Brillouin zone	four: at quasi-energy 0, $k = 0$ (chirality $+1$) and $\kappa = -\frac{\pi}{2}(1,1,1)$ (chirality -1); at quasi-energy π/τ, one of each chirality
One-site initial state	probability outside the cube: zero to machine precision; outside the round cone of radius ct: 0.50 after 16 steps, 0.47 after 24
Gaussian packets near $k = 0$, 24 steps	probability outside radius $ct + R_0$ (R_0 the 99% radius at $t = 0$): 0.084, 0.023, 0.008 for momentum widths $\sigma_\kappa = 0.4$, 0.25, 0.15

Table 1: Weyl automaton (9). Chirality is the sign of the determinant of the linear map $q \mapsto$ coefficient of $-i\sigma$ in $\pm W(k_0 + q)$. The packets carry a fixed spinor and therefore populate both bands.

outside the round cone, while it never leaves the cube; the leakage falls as the momentum width shrinks (Table 1).

Third, the Brillouin zone contains several band-touching points. In the Weyl automaton there are two at quasi-energy zero with opposite chiralities, as the doubling theorem for lattice Hamiltonians would lead one to expect [17], although that theorem does not apply directly to automata. A low-energy observer reads a Weyl cone of one chirality or the other depending on which point the prepared states are near. The dynamics does not choose; the preparation and the instruments do. In one dimension a single chiral cone is possible, since the shift by one site is a legitimate automaton; whether a strictly local isotropic automaton in three dimensions can carry a single node is a question we leave open.

So in this model the read-out map cannot be removed. What can be said precisely is what it consists of: a band point, a momentum window around it, and an observable defining the speed, each with computable consequences. This matches the treatment of access in HAFF, where what a physical reduction resolves depends on the channel and not only on the algebra it generates [42]. Here the Lorentzian geometry read by an observer is the characteristic geometry of the momentum window that its couplings reach.

5.5 Finite records certify one walk's dictionary

The choice of a read-out need not leave all its parameters inaccessible. The companion mass–energy paper realizes the same coin rotation by a fresh Gaussian oscillator, computes its exact open-channel linewidth, and reads its rest phase through controlled mixed-memory instruments [47]. On a supplied finite periodic ring, a normalized zero mode is preserved by a spatially uniform field coupling. Independent X and Z record trials at one and ten steps resolve the phase alias and certify θ. The ring, uniform support, axes, gain, clock, fresh ports

and record archive are priced inputs.

Let that actual protocol give $\theta \in [\theta_-, \theta_+]$ at its stated confidence, with supplied a, τ. Monotonicity of $\tan\theta$, $\cos\theta$ and $C(\theta) = \theta\cot\theta$ gives simultaneous intervals

$$E_0 \in \frac{\hbar}{\tau}[\theta_-, \theta_+], \qquad m_* \in \frac{\hbar\tau}{a^2}[\tan\theta_-, \tan\theta_+],$$
$$c_{\text{eff}}^2 \in \frac{a^2}{\tau^2}[C(\theta_+), C(\theta_-)], \qquad v_{\max} \in \frac{a}{\tau}[\cos\theta_+, \cos\theta_-].$$

At $\theta_+ = \pi/2$ the inertial upper bound is infinite. With the priced angle radius $b = .004296823$ at 95% confidence derived in Ref. [47], true angle .3 and $a = \tau = 1$, every confidence interval on that event lies inside $[.3 - 2b, .3 + 2b]$. The implied intervals are

$$c_{\text{eff}} \in [.9838969, .9856636], \qquad v_{\max} \in [.9527616, .9578408], \qquad c_{\text{support}} = 1.$$

The support speed is supplied, not measured by the phase protocol. The other two are model implications of its angle certificate. The full preparation, pulse, read-memory and reference budget is given with that protocol.

These are parameter inferences for the prescribed ideal walk, informed by an actual open-drive experiment. They are not its noisy long-time transport velocities: nonzero-momentum transport requires the real Bloch channel and inter-momentum coherences. Nor does calibration prove that the dictionary is Lorentzian throughout the band. The companion finite-angle window theorem gives precisely where a Lorentzian mass reading becomes negative and must be rejected. Thus an observer's interface can be calibrated, can distinguish otherwise conflated speeds, and can fail at a testable boundary.

6 Several species: when do the cones become one?

Theorem 4 gives every single-shell species its own form Q. For isotropic species in one frame, a common light cone requires their speeds to agree. In a free model nothing makes them agree: each speed is a parameter of the step. This section asks whether coupling the species can make the speeds equal, so that one cone is an output of the dynamics. The answer depends on the kind of coupling. Off-diagonal mixing cannot produce a single quadratic principal shell while its distinct diagonal shells are held fixed. Interactions can change those diagonal propagation laws: in the mediator model their speeds converge when each coupling has divergent accumulated strength toward low energy.

6.1 Mixing does not merge cones

Write the principal symbol of two species in a direction u as $A_1(u)$ on $\mathbb{C}^{s_1}$ and $A_2(u)$ on $\mathbb{C}^{s_2}$, both single-shell, $A_a(u)^2 = Q_a(u)\mathbf{1}$. A one-particle coupling adds a block off-diagonal term, which may itself depend linearly on q.

Theorem 15 (Mixing and the single shell)**.** *For every $s_1 \times s_2$ matrix V, the symbol $A = \begin{pmatrix} A_1 & V \\ V^* & A_2 \end{pmatrix}$ satisfies*

$$\min_{\lambda\in\mathbb{R}} \big\|A^2 - \lambda\mathbf{1}\big\|_{\text{HS}}^2 \;\geq\; \tfrac{1}{2}\min(s_1, s_2)\,\big(Q_1(u) - Q_2(u)\big)^2. \tag{11}$$

Hence the coupled system has a single shell only if $Q_1 = Q_2$. Moreover the largest eigenvalue of A is at least the largest eigenvalue of either block and the smallest is at most the smallest of either block.

Proof. The diagonal blocks of $A^2 - \lambda$ are $Q_1 - \lambda + VV^*$ and $Q_2 - \lambda + V^*V$, and the Hilbert–Schmidt norm of a block matrix is at least that of its diagonal blocks. If V has rank r, then VV^* and V^*V share the nonzero eigenvalues $\sigma_1^2, \dots, \sigma_r^2$ and have $s_1 - r$ and $s_2 - r$ zero eigenvalues. Using $(x+a)^2 + (x+b)^2 \geq \frac{1}{2}(a-b)^2$ for each shared eigenvalue, and for $k = \min(s_1, s_2) - r$ pairs of zeros, the squared norm is at least $\frac{1}{2}(r+k)(Q_1 - Q_2)^2$. The last statement is Cauchy interlacing for the principal submatrices A_1 and A_2. □

The bound (11) does not depend on the strength of the coupling. A mass-type coupling, independent of q, leaves the principal symbol untouched, but can change low-momentum band slopes. For example,

$$H(q) = \begin{pmatrix} v_1 q & g \\ g & v_2 q \end{pmatrix}, \qquad \omega_\pm(q) = \tfrac{1}{2}(v_1 + v_2)q \pm \sqrt{g^2 + \tfrac{1}{4}(v_1 - v_2)^2 q^2}.$$

For $g \neq 0$ both slopes at $q = 0$ equal $(v_1 + v_2)/2$, although the two distinct principal speeds remain v_1, v_2. Equal slopes at one band point are not a common quadratic characteristic cone. An on-site lattice coin can likewise change the effective logarithm of a step; the fixed-block theorem applies only after its principal blocks have been specified. For a chiral principal velocity matrix with fixed diagonal entries, the Schur–Horn theorem makes the eigenvalues majorize those entries, so off-diagonal mixing cannot reduce their variance. The interacting route below instead changes the propagation coefficients themselves.

6.2 Loops average speeds

Lemma 16 (Loops of one speed)**.** *Let a self-energy be a loop whose internal lines all have Lorentzian propagators of speed c and whose vertices are covariant under the Lorentz group of speed c. Then the coefficient of* $\ln \Lambda$ *in it is Lorentz covariant with speed c, for any fixed regular cutoff shape enclosing the origin and scaled by Λ.*

Proof. The substitution $k \mapsto (k_0, k/c)$ makes the integrand covariant under the standard Lorentz group. For a homogeneous integrand f of degree $-D$, the coefficient of $\ln \Lambda$ is the flux of the divergence-free field $xf(x)$ through a surface enclosing the origin, and it is the same for every such surface. □

The smallest interacting model is a real scalar of speed u coupled to N Dirac fermions of speed v by $g\phi\bar{\psi}\psi$ in $3+1$ dimensions. We integrate out a shell of spatial momenta with the frequency unrestricted; the logarithmic coefficients are scheme independent at this order. By Lemma 16 the fermion loop adds $\frac{Ng^2}{4\pi^2 v^3}(p_0^2 + v^2 p^2)\, dl$ to the inverse scalar propagator $p_0^2 + u^2 p^2$, where $l = \ln(\Lambda/E)$, so the squared scalar speed is replaced by a convex combination of u^2 and v^2. The fermion self-energy contains one fermion and one scalar line. Its frequency integrals are

$$J_0 = \int \frac{t^2\, dt}{(t^2+v^2)(t^2+u^2)^2} = \frac{\pi}{2u(v+u)^2}, \qquad J_s = \int \frac{dt}{(t^2+v^2)(t^2+u^2)^2} = \frac{\pi(v+2u)}{2vu^3(v+u)^2}, \tag{12}$$

and the ratio of the spatial to the temporal coefficient gives the speed toward which v moves.

Proposition 17 (Speed flow in the Yukawa model)**.** *At one loop,*

$$\frac{dv}{dl} = a\Big(\frac{v+2u}{3} - v\Big), \quad a = \frac{g^2}{4\pi^2 u(u+v)^2}, \qquad \frac{du^2}{dl} = \frac{Ng^2}{4\pi^2 v^3}\,(v^2-u^2), \tag{13}$$

and the coupling obeys $dg^2/dl = -(g^4/2\pi^2)\big[1/(vu(v+u)) + N/(2v^3) + 1/(u(v+u)^2)\big]$*. At* $u = v$ *the linearized flow of* $u - v$ *has rate* $(6N+1)g^2/(24\pi^2 v^3)$*, and* $u + 6Nv$ *is conserved to linear order.*

For $N = 1$ these equations coincide with those of Anber and Donoghue [18], who derived them by dimensional regularization. The momentum-shell calculation gives the same logarithmic coefficients. Each speed moves toward a target that lies between the speeds of the lines in its loop: the fermion toward $(v+2u)/3$, the scalar toward v. The scalar moves $6N$ times faster, so the common speed lies close to the initial fermion speed. Since the Yukawa coupling is marginally irrelevant, $g^2 \simeq 8\pi^2 c_*^3/[(2N+3)l]$, where c_* is the common limiting speed and

$$\frac{|u-v|}{v} \;\propto\; \Big[\ln\frac{\Lambda}{E}\Big]^{-(6N+1)/(3(2N+3))}, \tag{14}$$

with exponent $7/15$ for one Dirac fermion. Integrating the full flow to $l = 10^9$ gives log–log slopes -0.4667 for $N = 1$ and -0.7576 for $N = 4$, as predicted.

Theorem 18 (Speeds converge through a common mediator)**.** *Let fermion species of speeds* $v_1, \dots, v_n$ *couple to one scalar of speed* u *with prescribed nonnegative, locally integrable functions* $g_1^2(l), \dots, g_n^2(l)$*, each fermion obeying the speed equations in* (13) *and* u^2 *receiving the sum of the fermion loops. The theorem is conditional on these coupling histories; a multi-species beta function must separately establish their accumulated strengths. Assume positive initial speeds. Let* D *be the difference between the largest and the smallest of* $u, v_1, \dots, v_n$*, and* M_0 *the initial largest speed. Then the largest speed never increases, the smallest never decreases, and*

$$D(l) \le D(0)\exp\Big(-\frac{1}{24\pi^2 M_0^3}\int_0^l \min_a g_a^2\, dl'\Big). \tag{15}$$

If $\int g_a^2\, dl < \infty$*, the speed* v_a *changes in total by at most* $D(0)\int g_a^2\, dl/(24\pi^2 m_0^3)$*, with* m_0 *the initial smallest speed.*

Proof. Each v_a moves toward u at rate $\frac{2}{3}a_a$, and u^2 moves toward a weighted mean of the v_a^2, so u decreases when it is the largest speed and increases when it is the smallest. Consider the upper Dini derivative D^+ of the maximum minus the minimum. If both extremes are fermions $v_a \ge v_b$, then $D^+ \le \frac{2}{3}[a_a(u-v_a) - a_b(u-v_b)] \le -\frac{2}{3}(\min_c a_c)D$; if the maximum is u, then $D^+ \le -\frac{2}{3}a_b(u-v_b)$; the case of u minimal is symmetric. All speeds stay in $[m_0, M_0]$, where $\frac{2}{3}a_a \ge g_a^2/(24\pi^2 M_0^3)$, and Grönwall's inequality gives (15). The last statement follows from $|v_a'| \le \frac{2}{3}a_a D(0)$ and $a_a \le g_a^2/(16\pi^2 m_0^3)$. □

The estimate gives a rate when the weakest simultaneous coupling has divergent integral. Convergence itself needs less.

Corollary 19 (Intermittent interactions)**.** *In the one-loop star model of Theorem 18, suppose the nonnegative functions g_a^2 are locally integrable and initial speeds are positive. Every speed converges. Species connected to the mediator by edges with $\int_0^\infty g_a^2\,dl = \infty$ have the same limiting speed as the mediator. In particular, all speeds converge to one value if each such integral diverges, even when the couplings are never active simultaneously.*

Proof. Write $v_a' = A_a(u - v_a)$ and $u' = \sum_a B_a(v_a - u)$, with

$$A_a = \frac{g_a^2}{6\pi^2 u(u+v_a)^2}, \qquad B_a = \frac{g_a^2(u+v_a)}{8\pi^2 u v_a^3}.$$

For an active edge, $B_a/A_a = \frac{3}{4}(1 + u/v_a)^3$. All speeds stay in $[m_0, M_0]$, so the two directional rates are uniformly comparable, and their integrals diverge exactly when $\int g_a^2$ does. This is a type-symmetric continuous-time consensus system. The cut-balanced convergence theorem [1], applied to the coefficients along the solution, gives convergence and equality of limits within each component of the infinite-integral interaction graph. The theorem allows state-dependent coefficients. It does not provide the same rate as (15) when $\min_a g_a^2$ vanishes. □

The accumulated interaction across renormalization scales decides the outcome. The variable $l = \ln(\Lambda/E)$ is scale, not laboratory time; intermittent edges here are active on different scale intervals. An identically uncoupled species keeps its cone; a finite integrated coupling need not erase an initial difference. Marginally irrelevant couplings remove differences as a power of $\ln(\Lambda/E)$; near an interacting fixed point they fall as a power of E/Λ, with the one-loop exponent $7g_*^2/(24\pi^2 c_*^3)$ for $N = 1$. In $4 - \varepsilon$ dimensions the Yukawa fixed point gives $(6N + 1)\varepsilon/[3(2N + 3)]$ at this order. The weak-coupling model must remain applicable along this flow. A mediator that breaks rotations can instead drive speeds apart, as in nematic criticality of d-wave superconductors [20].

The common speed lies between the smallest and the largest initial speed. Each initial speed of a massless round cone is at most the inradius of conv N per step (Corollary 13), so the emergent cone lies inside the causal polytope.

6.3 Fixed points and the number of cones

Near an interacting fixed point the question has an answer that does not rely on perturbation theory.

Theorem 20 (Speed splittings at a conformal fixed point)**.** *Let the infrared fixed point be a unitary conformal field theory in $D = d + 1 \geq 3$ dimensions, Lorentz invariant for one cone, and consider local perturbations invariant under translations, spatial rotations and CPT. Up to total derivatives the Lorentz-violating perturbations are $\int d^D x\, O_{0\cdots0}$ for symmetric traceless primaries of even spin $\ell \geq 2$. Their couplings satisfy $D - \Delta_O \leq 2 - \ell \leq 0$; equality in the first inequality means that the current is conserved, while a marginal Lorentz-violating coupling requires a conserved spin-2 current. If the stress tensor is the only conserved spin-2 current, all independent spin-2 speed perturbations are irrelevant. When the excited non-conserved spin-2 modes have a smallest gap $\gamma_2 = \min(\Delta_O - D) > 0$, their slowest linearized decay is $(E/\Lambda)^{\gamma_2}$.*

Proof. Descendants integrate to zero. A primary transforms in a finite-dimensional irreducible representation of the Lorentz group, whose complexification is that of $SO(D)$, and because $(SO(D), SO(D-1))$ is a Gelfand pair only the symmetric traceless representations contain a rotation-invariant component, once each. CPT acts on a Hermitian rank-ℓ tensor by $(-1)^\ell$, which removes odd ℓ, and scalars are Lorentz invariant. The unitarity bound $\Delta \geq \ell + D - 2$ for $\ell \geq 1$ holds with equality exactly for conserved currents [21, 22]. The perturbation $\int T_{00}$ rescales time and changes no relative speed. □

The stability argument is established in Refs. [23, 24]; it assumes the Lorentz-invariant fixed point and does not prove that every microscopic dynamics reaches it. Additional conserved spin-2 currents are the marginal exceptions. In three dimensions, for exactly two such currents, a suitable basis generates separate translations and the two families of current correlators factorize [25, Section 8]. This is dynamical separation of the current sectors, not necessarily a tensor product of the full theories: a shared singlet constraint can remain. Nor does counting currents count sectors in general; two free scalar fields already have three conserved spin-2 currents. The result needed here is the local criterion for suppressing relative-speed perturbations, while identifying all independently propagating sectors requires further structure.

In two dimensions several propagation sectors can retain different speeds, as in spin–charge separation. A Sugawara decomposition gives an additional sector only when $T - T_J$ is nonzero. Under its positivity, spectrum and finite-signal-speed assumptions, the scale-to-Lorentz theorem permits either one Lorentz factor or several [26]; it does not forbid a common cone from being selected dynamically. Establishing one cone therefore requires information about the actual sectors and their interactions, rather than the dimensionality alone.

6.4 What the lattice adds

For number-conserving automata whose interaction preserves the supplied neighbourhood, each distinguishable particle stays in $x_j + nN$ after n steps. The centre-of-mass version of Proposition 12 therefore bounds every isolated bound-state band by $\operatorname{conv} N$ per step. Changing the neighbourhood changes that bound. Equalization of low-energy cones does not itself change the support cone.

For a loop dominated by one linear band point, the logarithmic coefficient carries that point's read-out speed by Lemma 16. Finite ultraviolet matching terms need not have the same ratio of temporal and spatial coefficients. A Lorentz-violating regulator can leave corrections of order $g^2/16\pi^2$ [28], subsequently reduced only at the infrared rates above.

Composites need not inherit a constituent's read-out cone. Take distinguishable walkers with $\theta_A = .3$, $\theta_B = .5$, $a = \tau = 1$, first apply both free coined steps, and then multiply the coincident-position sector by $e^{+.4i}$. With total momentum K conjugate to $(x_A + x_B)/2$, the isolated positive band at $K = 0$ has $\omega_0 = .7614121$ and $m_* = 1/\omega_0'' = .9023$, compared with the free threshold .8 and constituent mass sum .8556. Thus $c_{\text{eff}}^2 = \omega_0 \omega_0'' = .8439$, below the constituent values $\theta_A \cot \theta_A = .9698184$ and $\theta_B \cot \theta_B = .9152439$. Periodic relative-coordinate calculations with 91 and 131 sites agree at the quoted precision. They also track a localized band beyond $|K| = .68$; no such endpoint or global maximum speed is

inferred from this local mass calculation. Contact interactions can produce bound molecules in quantum walks [19]. In the example above, the separately computed quasienergy and curvature do not obey the relativistic binding-energy–inertia relation.

A common cone can also be kinematic. The minimal isotropic Weyl automaton is unique on the body-centred cubic lattice [13], so all massless species built from it share $c = \ell/(\sqrt{3}\tau)$; a Dirac automaton built from two of them has first-order speed $c(1-m^2)^{1/2}$ in lattice units, and photons built from pairs of Weyl automata have the dispersion $2|n_{k/2}|$ and the same low-energy cone [27]. For an electron on a lattice of Planck spacing, $m \approx 4 \times 10^{-23}$ and the mass correction is of order 10^{-45}. Loop matching can generate relative-speed corrections of order $g^2/16\pi^2$ unless a symmetry forbids them. Contact binding also changes the composite cone in the example above, but its size is model dependent and carries no general loop suppression factor.

6.5 How much running is needed

With marginal couplings the logarithmic rate (14) is far too slow to produce the observed equality of speeds. Reducing a difference by a factor 10^{-13} requires $\ln(\Lambda/E) \approx 10^{28}$ for one Dirac fermion with $g^2 = 4\pi$ at the cutoff, and about 10^{12} even for a hundred fermions, while $\ln(M_{\mathrm{Pl}}/m_e) = 51.5$. Anber and Donoghue reached the same qualitative conclusion [18].

Conjecture 21 (One cone needs a strongly coupled regime)**.** *In local unitary dynamics with several interacting gapless species, the relative speed difference of two sectors at scale E is $\delta c(E) \simeq \delta c(\Lambda) \exp(-\int_E^\Lambda \gamma \, d\ln\mu)$, where γ is the anomalous dimension of their relative stress tensor and $\delta c(\Lambda)$ is of order $g^2/16\pi^2$ from lattice matching even when the step is common. Equalities of order 10^{-15} then require $\int \gamma \, d\ln\mu \gtrsim 30$ above accessible energies, that is, a strongly coupled regime with γ of order one over many decades, or a symmetry that removes $\delta c(\Lambda)$. The largest residual differences sit between the most weakly coupled sectors.*

Theorems 18 and 20 tie equalization to interaction strength; lattice matching regenerates relative-speed differences, while marginal running removes them only logarithmically. The quantitative estimates above and the power-law emergence in strongly coupled holographic models [24] support the conjecture. Anber and Donoghue's observation that the weakest interactions leave the largest differences also makes gravitational waves against light a sensitive comparison [18].

A viable ultraviolet realization must place the strongly coupled range consistently with the observed weakly coupled particle description and experimental bounds. A composite realization is one possibility, not a necessary consequence of the conjecture. A symmetry route, such as non-relativistic supersymmetry protecting dimension-four terms, can avoid strong coupling but must survive symmetry breaking. Gravity is shared by all species yet is irrelevant at low energy; this flow cannot align its cone there. The relation of the gravitational-wave and matter cones therefore requires an ultraviolet mechanism or a protecting symmetry.

6.6 Consequences

The mediator model makes a common speed a conditional output of divergent accumulated interactions. The conformal argument establishes stability against relative-speed perturbations near a supplied Lorentz-invariant fixed point; it does not establish formation of that fixed point. This separates three inputs to the higher-dimensional Consciousness construction: each species' shell and signature, the alignment of their cones, and the actual infrared regime. Its boosted-ball family requires a common cone. In the two-dimensional HAFF gravity model [44], identifying the matter and Jackiw–Teitelboim cones remains an input; the two-dimensional sector theorem neither supplies nor forbids that identification.

7 Relations to other frameworks and papers

7.1 Algebraic frameworks

Sequential operations are represented here by linear maps under ordinary composition, which is associative by definition. Jordan and octonionic models need additional bracketing and composition rules before tensor products, subsystems or error correction can be transferred [30, 31, 32]; this is a statement of scope, not a no-go theorem. Von Neumann and C^*-algebras are compatible with everything above, since the step (1) and its symbol live in a concrete C^*-algebra; such algebras do not single out a quadratic form until a dynamics is given. Clifford algebras are distinguished in this paper by a dynamical property, the single shell of Theorem 4, rather than by being chosen as an encoding.

7.2 Thermodynamic gravity, modular flow and spectral triples

Jacobson's derivation of the Einstein equation assumes local Lorentzian causal structure, local equilibrium and an entropy–area relation [3], and entanglement derivations of linearized gravity work on a Lorentzian background [33, 34]. The present model supplies local Lorentzian cones for the low-momentum sector of a local unitary dynamics. It supplies neither equilibrium nor the entropy–area relation, and Section 5.4 shows that its cones depend on the momentum window.

Local algebras in relativistic quantum field theory are of Type III [35, 36]. For the vacuum and wedge regions the Bisognano–Wichmann theorem identifies modular flow with the boost subgroup [37, 48]; the identification presupposes the Lorentzian structure it describes. Whether the modular data of states of a lattice automaton reproduce the boosts of its low-energy cone in the scaling limit is open and could be computed in the walk of Section 5.2.

In Connes' noncommutative geometry a Dirac operator with Euclidean Clifford symbol encodes Riemannian geometry [38], and Lorentzian versions are an active problem [39, 40, 41]. Theorem 4 says that using such a Dirac operator as a Hamiltonian, $i\partial_t\psi = D\psi$, produces the Lorentzian characteristic cone of signature $(1, d)$ on time times space. We have not worked out the relation to the Krein-space approaches cited.

7.3 The Q-RAIF papers and HAFF

Section 3.2 specifies the separate outputs to control and spectral mass. The actuator field of Q-RAIF B is the single-chirality case of Remark 6; its outgoing stress is used in HAFF-D [44]. Identifying that matter cone with the gravitational cone is a further physical input, not a consequence of an isometry of internal generators.

8 Discussion

Ingredient of a Lorentzian read-out	**Status in local unitary propagation**
Evolution time	supplied; unitarity makes its frequency a hyperbolicity direction (Theorem 2)
Signature $(1, d)$	derived for any quadratic cone (Lemma 3)
Clifford relations	equivalent to a single shell (Theorem 4); Lorentzian generators constructed (Corollary 5)
Single shell, isotropy, linear band point	inputs; observable and testable in a given model
Spatial dimension d	input; bounded by the internal space (Theorem 8)
Speed c	a property of the dynamics; at finite mass it depends on the defining observable (Proposition 14)
Read-out map	input: band point, momentum window and speed observable (Section 5.4)
Common speed of several species	conditional output of the mediator model; stable against irrelevant relative-speed perturbations near a supplied Lorentz-invariant fixed point, whose formation is not derived (Theorems 18, 20)
Lattice, neighbourhood, step	inputs

Table 2: Inputs of a Lorentzian read-out in local unitary propagation.

Table 2 separates the derived signature from the remaining inputs: lattice data, absence of birefringence and the observer's read-out map. Deriving the first two requires a formation model; no property of the dynamics alone specifies the third.

The problems that follow have the most leverage. Section 6 answers the question of a common cone for several species under stated conditions; what remains there is the anomalous dimension γ_2 in a strongly coupled lattice model, which Conjecture 21 needs. Another is chirality: whether a strictly local, isotropic three-dimensional automaton can have a single node, which would remove one component of the read-out map. The mass-setting interface now has a nonempty field realization and finite certificate, but renewed drives, full noisy spatial transport and covariant support remain physical tasks. A fourth is the modular route: computing the modular flow of a lattice state and comparing it with the boosts of the low-energy cone. Position-dependent coins, which produce effective curved metrics, would connect the model to the gravitational questions of HAFF.

9 Conclusion

Propagation supplies the relative sign in $\omega^2 - G_{\mathrm{HS}}$; a positive control metric alone cannot. The single-shell criterion, dimension bound and calibrated residuals connect this form to a specified carrier. Finite angle records imply separated ideal dictionary-speed intervals, while noisy transport remains a different observable. For several species, the mediator model aligns cones under accumulated-coupling conditions; the conformal argument tests their stability, not the creation of a fixed point. These are the respective interfaces to mass–energy read-out, the common-cone geometry of Consciousness II and the still-supplied matter cone of HAFF-D. The calibrated work bridge constrains how propagation generators can be reused for control, without turning positivity into an indefinite signature. Formation of the carrier, clock and accessible momentum window remains a dynamical question.

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Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost

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(Dated: October 2026)

What can make the structure of physical control agree with the structure of propagation? Persistence alone does not select a Clifford algebra. A more specific condition does: for a linear Hermitian control family coupled to one centered, independent field input, switching work is independent of the unknown controlled state exactly when the generators square to a scalar quadratic form. Polarization then gives Clifford relations. If the same generators enter a calibrated propagation symbol, the measured work form is a positive pullback of its spatial shell, not the indefinite spacetime form. This conditional connection separates a physical restriction from a choice of notation.

We realize the coupling with a degenerate carrier and a free chiral field. The same interaction determines the control channel, its error and its added stress. An accurate bit flip requires drive occupation of order $\epsilon^{-1/2}$; squeezed inputs nearly attain the bound, while classical inputs need order ϵ^{-1}. A feedback task with scheduled fresh independent drives joins this actuator to calibration records and has an exact stationary loss and a stability proof. A separate sharp energy bound prices translation-only response readout. For inquiry while quantum content is held, a coherent echo and finite Gibbs bank provide explicit protection, reading and renewal operations. Their clocks, drive preparation and spent outputs remain in the resource account. These finite implementations constrain the control–world interface without deriving the supplied field, propagation time or an indefinitely reusable apparatus.

I. INTRODUCTION

Interaction with a large environment drives a quantum system towards decoherence and thermalization [1, 6]. A subsystem that keeps a non-equilibrium steady state does so through exchanges with its surroundings, which may be regulated by feedback [7–10]. The Q-RAIF question is not only what such regulation costs, but why its control structure should bear any relation to the structure in which the subsystem propagates.

Two distinctions are needed. Positivity of a gradient descent metric does not select a Clifford algebra, and no change of coordinates turns a positive work form into a Lorentzian form. Standard dissipative and coherent controls can stabilize states without a Clifford presentation [15–18]. A connection has to come from an actual interaction and from an operational condition on its use.

Here one field coupling supplies both the operation and its energetic response. For a fixed linear family of Hermitian control generators, a centered independent field input gives a work proportional to the expectation of the generator's square. Requiring that charge to be the same for every unknown carrier state is equivalent to scalar-square, hence Clifford, relations (Proposition 6). This is a condition on a specified calibration experiment, not a consequence of persistence or a claim that all useful controls must satisfy it. A one-state average can conceal its failure. When these generators are also the calibrated generators of propagation, Corollary 7 relates the work form to the spatial propagation shell. Evolution time and the physical alignment of the two uses remain inputs.

The finite actuator also makes the cost of maintaining an interface explicit. A degenerate bit coupled to a free chiral field has a flip channel, outgoing stress and switching work determined by the same unitary. Theorem 8 bounds the drive occupation needed for a given error; squeezed inputs nearly attain it, whereas classical drives do not. Theorem 13 composes this imperfect actuator with bath noise and calibration, and Proposition 14 gives one feasibility condition for records and drive energy. Translation-only response readout has its own sharp mean-energy bound, including retained ancillas and joint decoding (Theorem 16).

An interface can be queried while it carries unknown quantum content. A coherent controlled-parity echo protects that content; a mixed port produces a binary record, and a finite Gibbs bank renews the active oscillator while retaining its spent outputs. Collection loss, control and readout have separate charges. The detailed resonant, packet and finite-band collection constructions are grouped in Appendix A: they support the required access rather than define the main control–propagation question.

The outgoing stress is the matter source used in HAFF-D [29]; the feedback task realizes the supplied flip of the Imprint Chain [36]. These are interfaces between stated models. The chiral field, spatial propagation carrier and external clock are not generated by the persistence calculation. Sections II–III state its algebraic obligations; Sections IV–V establish the physical coupling and the conditional bridge; the energy, persistence and inquiry results then describe what that interface can do at finite resources.

II. PERSISTENCE AS A FEEDBACK PROBLEM

Consider $\mathcal{H}_{\text{tot}} = \mathcal{H}_R \otimes \mathcal{H}_E$ with reduced dynamics

$$\dot{\rho}_R = -i[H_{\text{eff}}, \rho_R] + \mathcal{D}[\rho_R]. \tag{1}$$

The controls must be specified in the composed system dynamics, rather than inferred from a formal stationarity condition on a protocol cost.

Standard quantum control writes $H_{\text{ctrl}} = \sum_k u_k(t) G_k$ with generators of a Lie algebra $\mathfrak{g}$, $[G_i, G_j] = i f_{ijk} G_k$ [14]. The Lie algebra fixes the accessible directions. The size of a correction is quantified by a positive cost, norm or information metric supplied separately. This separation is standard and is not a defect of Lie-algebraic control. Section IV shows where such a cost can come from physically: from the stress of the field that implements the control.

III. WHAT THE OPERATOR ALGEBRA MUST SUPPLY

A. Associativity as a boundary of the model

Remark 1 (Associativity as a control-model boundary)**.** *The master-equation model represents sequential controls by linear maps under ordinary composition, which is associative by definition [1]. This is a declared scope condition, not a theorem excluding every non-associative dynamics. Jordan or octonionic models would need additional bracketing, subsystem and composition rules before a control calculation could be transferred [12, 13]. Q-RAIF Paper A [30] uses the same operator-composition boundary for propagation.*

B. A positive metric for Lyapunov descent

For an auxiliary gradient controller on parameters λ consider the descent rule

$$\dot{\lambda} = -\Gamma\, G^{-1} \nabla_\lambda V, \qquad \Gamma > 0. \tag{2}$$

Proposition 2 (Positive descent metric)**.** *The controller contribution to $\dot{V}$ under Eq. (2) is $-\Gamma(\nabla_\lambda V)^T G^{-1} (\nabla_\lambda V)$. It is nonpositive for every gradient exactly when G^{-1}, equivalently the invertible symmetric matrix G, is positive definite. If G has a negative direction, a gradient along it makes the contribution positive.*

Proof. The first statement is the chain rule. A symmetric quadratic form is nonnegative on all vectors precisely when it is positive semidefinite, and an invertible positive semidefinite matrix is positive definite. □

The descent mechanism therefore selects positive definiteness, not Lorentzian signature. Channel types such as unitary, dephasing and dissipative can be labelled by a discrete index, a direct-sum decomposition or supplied weights; their qualitative difference does not need a negative norm.

Parameter descent is not yet the physical state equation. With drift H_0, dissipator $\mathcal{D}$ and controls $H_{\text{ctrl}}(\lambda) = \sum_i u_i(\lambda) G_i$, the closed loop is

$$\dot{\rho} = -i[H_0 + H_{\text{ctrl}}(\lambda), \rho] + \mathcal{D}[\rho], \qquad \dot{\lambda} = -\Gamma G^{-1} \nabla_\lambda V, \tag{3}$$

and for differentiable $V(\rho, \lambda)$

$$\dot{V} = D_\rho V[-i[H_0 + H_{\text{ctrl}}, \rho] + \mathcal{D}[\rho]] - \Gamma(\nabla_\lambda V)^T G^{-1} (\nabla_\lambda V). \tag{4}$$

A stability proof must control the first term as well. Setting a control amplitude equal to a gradient component gives no general sign.

Dissipative and coherent control can stabilize target states with positive costs and Lie or operator-algebraic controls [15–18]. Von Neumann and C^*-algebras are compatible with this model; neither supplies a preferred quadratic form without a dynamics. These examples also show why persistence alone imposes no indefinite Clifford structure.

C. Clifford presentation and a physical criterion

If one chooses a quadratic space (V, q) and requires control generators to satisfy $v^2 = q(v)\mathbf{1}$, the universal property identifies $Cl(V, q)$ as the minimal algebra for that presentation [11]. This is a definitional fact. It does not show that persistence, channel discrimination or Lyapunov descent imposes those relations. Pauli controls of a qubit satisfy $\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbf{1}$, the relations of $Cl(3, 0)$, so the Clifford package is a convenient description of qubit controls. A particular feedback model may favour some representation because of its controls and costs, but that preference has to be shown in its closed-loop dynamics. Proposition 6 below gives a different, explicit criterion: the centered-input switching charge must be independent of the unknown carrier state for every command. That requirement is stronger than positivity or persistence.

a. State stability. Equation (4) is the continuous-time obligation: a parameter descent controls only its own contribution, while drift, dissipator and controlled state evolution remain. For full-rank states and a fixed full-rank target, $\dot{D}(\rho\|\rho_*) = \mathrm{tr}[\dot{\rho}(\log\rho - \log\rho_*)]$. Theorem 13 establishes stability instead for a discrete composed channel with a physically realized, imperfect correction.

IV. A CONTROL FLIP REALIZED BY A FIELD COUPLING

A. The coupling and its channel

The field-coupling and drive bounds in this and the next section use $\hbar = 1$: $\bar{\omega}N$ and $\bar{\omega}\lambda^2$ are energies in these units. In physical units they are $\hbar\bar{\omega}N$ and $\hbar\bar{\omega}\lambda^2$; the field Hamiltonian and stress carry the same overall $\hbar$. Quadratures, occupation and pulse areas are dimensionless. The later delay, entropy and retained-oscillator accounts display $\hbar$ explicitly.

The controlled system is a two-level system with zero free Hamiltonian, Pauli operators X and Z. A bosonic mode with $[b, b^\dagger] = 1$ and quadratures $Q = b + b^\dagger$, $P = i(b^\dagger - b)$, $[Q, P] = 2i$, interacts through

$$U = \exp[-i\lambda XQ], \qquad \lambda > 0. \tag{5}$$

In the eigenbasis $X|x\rangle = x|x\rangle$, $U = \sum_x |x\rangle\langle x| \otimes e^{-i\lambda xQ}$, and $e^{-i\lambda xQ}$ is the displacement $D(-i\lambda x)$. Write $\chi_\sigma(k) = \mathrm{tr}(\sigma e^{-ikQ})$ for the characteristic function of Q in a mode state σ.

Proposition 3 (Channel of the coupling)**.** *Let the mode start in a state σ independent of the bit. The bit channel multiplies the matrix element $\rho_{xx'}$ in the X basis by $\chi_\sigma(\lambda(x - x'))$. For bit states diagonal in Z, its output Z statistics agree with a flip having failure probability*

$$\epsilon = \tfrac{1}{2}\big(1 + \mathrm{Re}\,\chi_\sigma(2\lambda)\big). \tag{6}$$

For a coherent input $|\alpha\rangle$, $\alpha > 0$, the factor is

$$e^{-\lambda^2(1-xx')-2i\alpha\lambda(x-x')}. \tag{7}$$

If $2\lambda\alpha = \pi/2$ and $N = \alpha^2$, the channel on every bit state is

$$\mathcal{F}_N(\rho) = s_N X\rho X + (1 - s_N)\rho, \qquad s_N = \frac{1 + e^{-2\lambda^2}}{2}, \qquad \lambda^2 = \frac{\pi^2}{16N}, \tag{8}$$

so that $\epsilon_N = 1 - s_N = \big(1 - e^{-\pi^2/(8N)}\big)/2$.

Proof. Tracing the mode gives the factor $\mathrm{tr}(\sigma\, e^{i\lambda x'Q}e^{-i\lambda xQ}) = \chi_\sigma(\lambda(x - x'))$. A Z-diagonal state $(I + zZ)/2$ has $\rho_{+-} = z/2$ in the X basis, so the output mean of Z is $z\,\mathrm{Re}\,\chi_\sigma(2\lambda)$, which is $(1-2s)z$ for a flip with success probability s. For the coherent input, $D(\beta)D(\alpha) = e^{(\beta\alpha^* - \beta^*\alpha)/2}D(\alpha + \beta)$ gives $e^{-i\lambda xQ}|\alpha\rangle = e^{-i\alpha\lambda x}|\alpha - i\lambda x\rangle$. The overlap is the coherent-state inner product $\langle\alpha - i\lambda x'|\alpha - i\lambda x\rangle = e^{-\lambda^2(1-xx')-i\alpha\lambda(x-x')}$ times the Weyl phase $e^{-i\alpha\lambda(x-x')}$ from composing the displacements; this is Eq. (7). At $2\lambda\alpha = \pi/2$ the off-diagonal factor is $-e^{-2\lambda^2}$, and $1 - 2s_N = -e^{-2\lambda^2}$ yields Eq. (8) on all inputs. □

The channel $\mathcal{F}_N$ differs from the ideal Ad_X by $\epsilon_N(\mathrm{id} - \mathrm{Ad}_X)$, so its half diamond-norm distance from the ideal flip is ϵ_N, and its average gate infidelity is $2\epsilon_N/3$. These are different error conventions. The coherent family above needs infinite occupation for an exact flip; Section VI shows that this holds for every input state.

B. A local switch on a free field

Equation (5) becomes a local operation once the mode is defined by a switch. Take a right-moving free massless scalar in $1+1$ dimensions with null coordinate u,

$$\phi(u) = \int_0^\infty \frac{d\omega}{\sqrt{4\pi\omega}} \big(a_\omega e^{-i\omega u} + a_\omega^\dagger e^{i\omega u}\big), \qquad [a_\omega, a_{\omega'}^\dagger] = \delta(\omega - \omega'), \tag{9}$$

current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$. The current obeys $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$. For a real smooth compactly supported switch g define

$$Q_g = \int g(u)\, j(u)\, du = \lambda(b + b^\dagger), \qquad \lambda^2 = \int_0^\infty \frac{\omega}{4\pi} |\hat{g}(\omega)|^2 d\omega, \qquad \hat{g}(\omega) = \int g(u) e^{-i\omega u} du, \tag{10}$$

where $b = \int_0^\infty f^*(\omega) a_\omega d\omega$ with $f(\omega) = i\sqrt{\omega/(4\pi)}\, \hat{g}^*(\omega)/\lambda$ is a normalized mode. Its mean frequency is $\bar{\omega} = \int \omega |f|^2 d\omega$, and the current contains it as $j(u) = \psi(u) b + \psi^*(u) b^\dagger + j_\perp(u)$ with $\psi(u) = -i \int_0^\infty \sqrt{\omega/(4\pi)}\, f(\omega) e^{-i\omega u} d\omega$.

A probe at fixed position with interaction Hamiltonian $g(t) X j(t)$ evolves by $U = e^{i\vartheta} \exp(-iXQ_g)$ for a real phase ϑ. The reason is that $[g(t)Xj(t), g(t')Xj(t')]$ is a multiple of the identity, because $X^2 = 1$ and the current commutator is a c-number, so the Magnus series stops at second order and the second-order term is a phase. This uses three conditions: a single probe, a fixed coupling operator X, and a probe free Hamiltonian that is zero or commutes with X. Nonzero field frequencies are harmless. A probe Hamiltonian that does not commute with X makes $X(t)$ time dependent, and several probes keep relative phases; neither case is covered. A point-like limit of the switch needs regularization, so we use smooth profiles. This exact Weyl evolution of a degenerate detector is a known method [21, 22]; what we add is its use as a priced feedback actuator.

C. Stress, work and a derived quadratic cost

Proposition 4 (Stress of one encounter)**.** *For the unitary $U = \exp(-iXQ_g)$,*

$$U^\dagger j(u) U = j(u) + \tfrac{1}{2} X g'(u), \qquad U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u)\, Xj(u) + \tfrac{1}{4} g'(u)^2. \tag{11}$$

The expected added stress is $g'(u)\langle Xj(u)\rangle + g'(u)^2/4$. The cross term vanishes if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis. A Z-diagonal bit marginal alone is not sufficient.

Proof. $[Q_g, j(u)] = \int g(v)[j(v), j(u)] dv = -\frac{i}{2} g'(u)$ is a c-number, so $e^{iXQ_g} j(u) e^{-iXQ_g} = j(u) + \frac{1}{2} X g'(u)$ exactly. Squaring, using $X^2 = 1$, and noting that a c-number shift leaves normal ordering unchanged gives the stress identity. Under independence $\langle Xj\rangle = \mathrm{tr}(\rho X)\langle j\rangle = 0$, even when the input beam has $\langle j\rangle \neq 0$. If the joint state is block diagonal in the bit's Z basis, $X \otimes A$ has zero expectation for every field operator A. For the last claim, $(|{\uparrow}\rangle|0\rangle + |{\downarrow}\rangle|1_b\rangle)/\sqrt{2}$ has bit marginal $I/2$ but $\langle Xb\rangle = 1/2$, hence $\langle Xj(u)\rangle = \mathrm{Re}\,\psi(u) \neq 0$; with a relative phase i the integrated cross term equals $-\lambda\bar{\omega}$ and changes the work below. □

Under the stated joint condition the added stress is

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2, \tag{12}$$

the second form following from $2\lambda\,\mathrm{Im}\,\psi = g'/2$. If the interaction is off before and after the encounter and the bit Hamiltonian is zero, the energy delivered by the external switching agent equals the change of field energy,

$$W_{\mathrm{switch}} = \frac{1}{4} \int g'(u)^2 du = \frac{1}{4\pi} \int_0^\infty \omega^2 |\hat{g}(\omega)|^2 d\omega = \lambda^2 \bar{\omega}, \tag{13}$$

by Parseval's identity and $|f|^2 = \omega |\hat{g}|^2/(4\pi\lambda^2)$. This holds for every input state of the field that satisfies the joint condition.

Proposition 5 (Derived quadratic cost)**.** *For $g = \sum_i a_i g_i$ with real smooth compactly supported g_i, $W_{\mathrm{switch}} = \sum_{ij} a_i K_{ij} a_j$ with $K_{ij} = \frac{1}{4}\int g_i' g_j' du$. If the g_i are linearly independent, K is positive definite.*

Proof. $a^T K a = \frac{1}{4}\int (\sum_i a_i g_i')^2 du$ vanishes only if $\sum_i a_i g_i$ is constant; compact support makes the constant zero, and linear independence gives $a = 0$. □

D. Coherent drive

Let the mode b carry a coherent state $|\alpha\rangle$, $\alpha = \sqrt{N}$, with all orthogonal modes in vacuum, independent of a Z-diagonal bit. The output field marginal is then

$$\rho_F^{\text{out}} = \tfrac{1}{2}|\alpha - i\lambda\rangle\langle\alpha - i\lambda| + \tfrac{1}{2}|\alpha + i\lambda\rangle\langle\alpha + i\lambda|, \tag{14}$$

independently of the bit's bias, although bit and field are correlated afterwards. The incoming field energy is $E_{\text{in}} = N\bar{\omega}$ and the outgoing energy is $(N+\lambda^2)\bar{\omega}$. A wave packet is not an eigenmode of the field Hamiltonian, so these are mean values; the field Hamiltonian is not $\bar{\omega}b^\dagger b$. The incoming stress is $4N[\operatorname{Re}\psi(u)]^2$ and the encounter adds F_g of Eq. (12). The gravitational use of these profiles is developed in Ref. [29].

V. CONTROL STRUCTURE AND PROPAGATION

A. State-independent charging selects scalar squares

The fixed Pauli coupling can be generalized without changing the field. Let $A(a) = \sum_i a_i A_i$ be a real linear family of dimensionless Hermitian operators on a finite carrier. During one encounter a is fixed, and the carrier free Hamiltonian vanishes or commutes with $A(a)$. The switch g is fixed, nonzero, smooth and compactly supported. The input field is independent of the carrier and centered, $\langle j(u)\rangle = 0$, with finite energy; the vacuum is one example. Put $\kappa_g = \frac{1}{4}\int g'(u)^2 du > 0$, in the $\hbar = 1$ convention of Section IV.

Proposition 6 (Content-independent switching work)**.** *For the interaction $g(t)A(a)j(t)$ the field shift and switching work are*

$$U^\dagger j(u) U = j(u) + \tfrac{1}{2}A(a)g'(u), \tag{15}$$

$$W_\rho(a) = \kappa_g \operatorname{tr}\big[\rho A(a)^2\big]. \tag{16}$$

The following conditions are equivalent:

1. *$W_\rho(a)$ is independent of the carrier density matrix ρ for every command a;*
2. *$A(a)^2 = Q_{\text{ctrl}}(a)I$ for every a;*
3. *for a real symmetric matrix Q_{ctrl}, $\{A_i, A_j\} = 2(Q_{\text{ctrl}})_{ij}I$.*

The form is positive semidefinite, and positive definite if $a \mapsto A(a)$ is injective. In that case the generators give a representation of its Euclidean Clifford algebra.

Proof. The field commutator is a c-number. The second Magnus term is $i\vartheta A(a)^2$ and all higher terms vanish, since $A(a)^2$ commutes with $A(a)$ and with the field. Thus $U = e^{i\vartheta A(a)^2}e^{-iA(a)Q_g}$. The first factor need not be a global phase, but it commutes with field observables. The Weyl commutator calculation of Proposition 4 gives the stated shift and

$$U^\dagger T_{uu} U - T_{uu} = g'(u)A(a)j(u) + \tfrac{1}{4}g'(u)^2 A(a)^2.$$

Independence and centering remove the cross term for every ρ. The carrier free energy is unchanged and the interaction is off at the endpoints, so integration gives Eq. (16). A Hermitian operator has the same expectation in every density matrix exactly when it is a scalar multiple of I. This proves the first equivalence. Expanding $A(a)^2$ and comparing polynomial coefficients gives the second. Finally $Q_{\text{ctrl}}(a)I = A(a)^2 \geq 0$, and equality implies $A(a) = 0$. □

The all-state condition matters. For $A(a) = a_1 Z + a_2 I$ a maximally mixed carrier has work $\kappa_g(a_1^2 + a_2^2)$, although $A(a)^2 = (a_1^2 + a_2^2)I + 2a_1 a_2 Z$ is not scalar. The two Z eigenstates instead cost $\kappa_g(a_1 \pm a_2)^2$. Thus a positive average charge does not impose Clifford relations. Correlated inputs or a noncentered field introduce the cross term in the proof and need their own account. In particular this centered-input calibration is not a claim of state-independent work for every coherent drive used to rotate the carrier; nor does it guarantee a useful unitary gate.

B. A calibrated bridge to the spatial shell

Write a propagation symbol on the same internal carrier as $J(p) = \sum_i p_i \alpha_i$, with linearly independent Hermitian α_i and $J(p)^2 = Q_{\mathrm{prop}}(p) I$. The scalar-square argument makes Q_{prop} positive definite. Here J has energy units and the spatial carrier is supplied. Let C be a calibrated linear map from commands to momenta and $E_* > 0$ a fixed energy scale. Coupling the actual control through $A(a) = J(Ca)/E_*$ is a physical alignment condition, not a relabelling of two unrelated quadratic forms.

Corollary 7 (Aligned work and propagation)**.** *Under the encounter assumptions of Proposition 6 and this alignment,*

$$W(a) = a^T K_{\mathrm{op}} a, \qquad K_{\mathrm{op}} = \frac{\kappa_g}{E_*^2}\, C^T Q_{\mathrm{prop}} C. \tag{17}$$

Conversely, if C is invertible and this aligned family satisfies the all-state charge condition, it imposes a scalar quadratic shell on J. If in addition a supplied mass generator satisfies $\beta^2 = I$ and $\{\beta, \alpha_i\} = 0$, then

$$[\,\omega\beta - \beta J(p)\,]^2 = [\,\omega^2 - Q_{\mathrm{prop}}(p)\,] I. \tag{18}$$

Proof. Substitution in Eq. (16) proves the work identity. An invertible C ranges over all momenta, so the converse is the scalar-square criterion. For the last identity, $\beta J = -J\beta$ cancels the cross terms and $(\beta J)^2 = -J^2$. □

In Eq. (18), ω denotes energy conjugate to the independently supplied time, with $\hbar = 1$. The relative sign comes from propagation and the mass anticommutator, not from turning a positive work metric into an indefinite one. This is the closed-carrier symbol map of Q-RAIF C [31] with an explicit calibrated cost attached to it.

A nonempty case uses $\alpha_1 = X$, $\alpha_2 = Z$ and $\beta = Y$. In units $E_* = 1$, $Q_{\mathrm{prop}}(p) = p_1^2 + p_2^2$. For $C = \mathrm{diag}(1, 2)$, $K_{\mathrm{op}}/\kappa_g = \mathrm{diag}(1, 4)$ while the propagation shell is unchanged. The unknown calibration cannot be inferred from positivity of the cost. During the encounter the carrier must still obey the commuting-free-Hamiltonian condition. A general massive propagation Hamiltonian does not commute with $A(a)$; pausing or compensating it requires physical control and a separate cost. The example therefore supplies two compatible, calibrated uses of the carrier, not an autonomous spacetime-generating device.

C. Work, descent and spacetime

The descent metric G in Proposition 2 prices changes of controller parameters. Proposition 5 derives a work form K on switch profiles; Eq. (17) instead prices operator amplitudes for one fixed switch. These are different ways to vary a coupling. If controller parameters are the amplitudes being priced and the corresponding work form is positive definite, choosing it as their descent metric makes $-K^{-1}\nabla V$ the steepest direction at fixed quadratic charge, by Cauchy–Schwarz in the K inner product. That is a design choice; the feedback of Section VII has a stability proof without any gradient metric.

The new restriction is not equality of all three forms. It is the testable relation (17), under known coupling, calibration and input conditions. Its failure rejects that aligned model. Without those conditions, persistence supplies no route from a positive cost to a spatial shell or to Lorentzian signature.

VI. THE ENERGY COST OF AN ACCURATE FLIP

How much drive is needed for a flip of given accuracy? The coherent family gives $N = \pi^2/(16\ell)$ with $\ell = -\frac{1}{2}\log(1 - 2\epsilon)$, about $\pi^2/(16\epsilon)$. The following theorem bounds every input state.

Theorem 8 (Energy–accuracy bound)**.** *Let σ be any state of the mode b with finite mean occupation $N = \mathrm{tr}(\sigma b^\dagger b)$, let $\lambda > 0$, and let the coupling* (5) *flip Z-diagonal bit states with failure probability $\epsilon \in (0, \frac{1}{2})$ as in Eq.* (6)*. With $\theta_\epsilon = \arccos(2\epsilon - 1) \in (\pi/2, \pi)$,*

$$N \geq \frac{\theta_\epsilon(1 - 2\epsilon)}{4\sqrt{\epsilon(1 - \epsilon)}} - \frac{1}{2}, \tag{19}$$

$$N + \lambda^2 \geq \frac{\theta_\epsilon}{4\sqrt{\epsilon(1 - \epsilon)}} - \frac{1}{2}. \tag{20}$$

Both right-hand sides equal $\pi/(4\sqrt{\epsilon}) - 1 + O(\sqrt{\epsilon})$. The second inequality is a bound on an occupation-plus-activation budget. It equals the outgoing single-mode energy in units of $\bar{\omega}$ only with the independent zero-cross-term field inputs of Section IV C; correlations can change the actual switching work. An exact flip, $\epsilon = 0$, is impossible at finite occupation.

Proof. Write $C = \cos(2\lambda Q)$ and $U_2 = e^{-2i\lambda Q}$, so that $\langle C\rangle = \operatorname{Re}\chi_\sigma(2\lambda) = 2\epsilon - 1$ and $b^\dagger b = (Q^2 + P^2 - 2)/4$. Finite N makes both second moments finite.

Position quadrature. The function $G(c) = \arccos^2 c$ is convex on $[-1, 1]$: with $c = \cos y$, $y \in (0, \pi)$, one finds $G''(c) = 2(1 - y\cot y)/\sin^2 y \geq 0$, since $y\cot y \leq 1$. For every real q, $(2\lambda q)^2 \geq G(\cos 2\lambda q)$, with equality when $|2\lambda q| \leq \pi$ and strict inequality otherwise because $G \leq \pi^2$. Jensen's inequality for the spectral distribution of Q gives

$$\langle Q^2\rangle \geq \frac{G(\langle C\rangle)}{4\lambda^2} = \frac{\theta_\epsilon^2}{4\lambda^2}. \tag{21}$$

Momentum quadrature. Since $U_2^\dagger P U_2 = P - 4\lambda$, we have $[U_2, P] = 4\lambda U_2$ on the domain of P. Writing $\delta U_2 = U_2 - \langle U_2\rangle$, $\delta P = P - \langle P\rangle$ and applying Cauchy–Schwarz to the Hilbert–Schmidt products $\operatorname{tr}(\sigma\,\delta U_2\,\delta P)$ and $\operatorname{tr}(\sigma\,\delta P\,\delta U_2)$ gives $|\langle [U_2, P]\rangle| \leq 2\,\Delta U_2\,\Delta P$ with $(\Delta U_2)^2 = 1 - |\langle U_2\rangle|^2$. Hence $\Delta P \geq 2\lambda|\chi|/\sqrt{1 - |\chi|^2}$ with $\chi = \chi_\sigma(2\lambda)$. Because $|\chi| \geq |\operatorname{Re}\chi| = 1 - 2\epsilon$ and $x/\sqrt{1 - x^2}$ increases,

$$\langle P^2\rangle \geq (\Delta P)^2 \geq \frac{\lambda^2(1 - 2\epsilon)^2}{\epsilon(1 - \epsilon)}. \tag{22}$$

In particular $|\chi| < 1$ at finite N, which excludes $\epsilon = 0$.

Combination. Adding the two bounds,

$$N \geq \frac{1}{4}\left[\frac{\theta_\epsilon^2}{4\lambda^2} + \frac{\lambda^2(1 - 2\epsilon)^2}{\epsilon(1 - \epsilon)}\right] - \frac{1}{2}, \tag{23}$$

and minimizing over λ by the arithmetic–geometric mean inequality gives Eq. (19). Adding λ^2 changes the coefficient of λ^2 to $\frac{1}{4}[(1 - 2\epsilon)^2/(\epsilon(1 - \epsilon)) + 4] = 1/[4\epsilon(1 - \epsilon)]$, and the same step gives Eq. (20). The expansion uses $\theta_\epsilon = \pi - 2\sqrt{\epsilon} + O(\epsilon^{3/2})$. □

The bound is nearly attained by Gaussian drives.

Proposition 9 (Coherent and Gaussian families)**.** *(a) The coherent family with $2\lambda\alpha = \pi/2$ needs $N = \pi^2/(16\ell)$, $\ell = -\frac{1}{2}\log(1 - 2\epsilon)$. (b) For a displaced pure Gaussian state whose Q has mean 2α and variance $V > 0$, $\chi(2\lambda) = e^{-4i\lambda\alpha}e^{-2\lambda^2 V}$ and $N = \alpha^2 + (V + V^{-1} - 2)/4$. At $2\lambda\alpha = \pi/2$ and failure ϵ,*

$$\min N = \frac{\sqrt{1 + \pi^2/(4\ell)} - 1}{2}, \qquad \min(N + \lambda^2) = \frac{1}{2}\sqrt{\Big(1 + \frac{\pi^2}{4\ell}\Big)(1 + 4\ell)} - \frac{1}{2}, \tag{24}$$

attained at $V = (1 + \pi^2/(4\ell))^{-1/2}$ and $V = [(1 + 4\ell)/(1 + \pi^2/(4\ell))]^{1/2}$ respectively. Both equal $\pi/(4\sqrt{\epsilon}) - \frac{1}{2} + O(\sqrt{\epsilon})$, so their gap to Theorem 8 tends to half a quantum.

Proof. The Gaussian characteristic function gives χ. The failure condition reads $\lambda^2 V = \ell$, so $\alpha^2 = \pi^2/(16\lambda^2) = aV$ with $a = \pi^2/(16\ell)$ and $N = (1 + 4a)V/4 + 1/(4V) - \frac{1}{2}$, minimized at $V = (1 + 4a)^{-1/2}$. Adding $\lambda^2 = \ell/V$ replaces $1/(4V)$ by $(1 + 4\ell)/(4V)$. □

Squeezing is what makes the better scaling possible.

Proposition 10 (Classical drives)**.** *If $\sigma = \int P(\beta)|\beta\rangle\langle\beta|\,d^2\beta$ with a probability density $P \geq 0$, then a flip with failure ϵ needs*

$$N \geq \frac{(\pi - \arccos(1 - 2\epsilon))^2}{16\ell} = \frac{\pi^2}{16\epsilon}\big(1 - 4\sqrt{\epsilon}/\pi + O(\epsilon)\big). \tag{25}$$

Proof. Each coherent component has $\chi_\beta(2\lambda) = e^{-2\lambda^2}e^{-4i\lambda\operatorname{Re}\beta}$, so $\int P\cos(4\lambda\operatorname{Re}\beta) = -(1 - 2\epsilon)e^{2\lambda^2}$. This must be at least -1, hence $\lambda^2 \leq \ell$. The convexity argument of Theorem 8 applied to $y = 4\lambda\operatorname{Re}\beta$ gives $16\lambda^2\int P(\operatorname{Re}\beta)^2 \geq (\pi - \arccos[(1 - 2\epsilon)e^{2\lambda^2}])^2 \geq (\pi - \arccos(1 - 2\epsilon))^2$. Finally $N = \int P|\beta|^2 \geq \int P(\operatorname{Re}\beta)^2$ and $\lambda^2 \leq \ell$. □

ϵ	Classical bound (25)	Coherent, $2\lambda\alpha = \pi/2$	Bound (19)	Grid variational	Gaussian half-turn (24)
0.1	3.496	5.529	1.165	1.847	1.904
0.05	8.589	11.709	2.278	2.921	2.958
0.01	53.53	61.07	6.742	7.314	7.330
0.001	591.7	616.2	23.80	24.32	24.33

TABLE I. Mean drive occupation needed for a flip with failure probability ϵ. Including one activation's switching work at $\epsilon = 0.01$: bound (20) 6.890, grid variational minimum 7.470, Gaussian half-turn 7.487, coherent 61.08. The grid minima change by less than 2×10^{-3} when the spacing is halved; this is not a certified global error bound.

At fixed λ, a constrained minimum can be certified by a supporting ground-state problem. If ψ_μ is the ground state of $b^\dagger b + \mu \cos(2\lambda Q)$ and $\langle C \rangle_{\psi_\mu} = 2\epsilon - 1$, then every σ with the same $\langle C \rangle$ satisfies $\mathrm{tr}(\sigma b^\dagger b) \geq \langle b^\dagger b \rangle_{\psi_\mu}$, because $\mathrm{tr}\, \sigma(b^\dagger b + \mu C)$ is at least the ground energy. Table I lists numerical variational minima on a position grid, further minimized over λ, alongside the analytic bounds and the explicitly attainable Gaussian half-turn family. Halving the grid checks resolution stability, not the continuous global minimum. The computed states are close to displaced squeezed states, while the coherent half-turn family uses substantially more occupation.

Remark 11 (Scope of the occupation)**.** *(i) For a multimode field state, Cauchy–Schwarz applied to $b = \int f^* a_\omega d\omega$ gives $\langle b^\dagger b \rangle \leq \omega_h^{-1} \langle H_F \rangle$ with $\omega_h^{-1} = \int |f|^2 \omega^{-1} d\omega$, so the field energy is at least ω_h times the bound* (19)*; under the zero-cross-term condition of Section IV C, $\omega_h \leq \bar{\omega}$ then makes field energy plus switching work at least ω_h times the bound* (20)*. (ii) The occupation is the drive level present at the encounter, not energy absorbed by the bit. The bit is degenerate; the beam leaves with energy $(N + \lambda^2)\bar{\omega}$ in the coherent case. Whether drive energy is lost depends on whether the outgoing beam is recovered. (iii) Squeezed inputs can have locally negative normal-ordered stress, and their preparation and phase reference are additional resources.*

Recovering the beam is not free of consequences. Two bits placed one after the other on the same chiral packet, with the same switch profile in u, experience $U = \exp[-i\lambda(X_1 + X_2)Q]$.

Proposition 12 (Shared drive correlates errors)**.** *For a shared drive each bit's marginal channel is the single-encounter channel. For two bits starting at $Z = +1$, $\langle Z_1 Z_2 \rangle = \frac{1}{2}[1 + \mathrm{Re}\, \chi(4\lambda)]$. For the coherent family of Proposition 3 this equals $(1 + e^{-8\lambda^2})/2$, which exceeds the product $\langle Z_1 \rangle \langle Z_2 \rangle = e^{-4\lambda^2}$ obtained with independent drives.*

Proof. In the X basis the joint element with indices $(x_1 x_2), (x_1' x_2')$ is multiplied by $\chi(\lambda(x_1 + x_2 - x_1' - x_2'))$; tracing one bit recovers Proposition 3. $Z_1 Z_2$ connects (x_1, x_2) with $(-x_1, -x_2)$, which gives arguments $\pm 4\lambda$ and 0. For the coherent family $\chi(4\lambda) = e^{-8\lambda^2} e^{-2\pi i}$, and $(1 + y^2)/2 > y$ for $0 < y < 1$. □

At $N = 16$ the single failure probability is 0.0371, and the probability that both flips fail is 0.00393, against 0.00138 for independent drives. Drive energy can be shared only at the price of correlated failures. The persistence analysis below uses a fresh drive at each opportunity.

VII. PERSISTENCE WITH A FINITE-ENERGY ACTUATOR

A. The task

We use the feedback task of the Imprint Chain [36] with the corrective flip replaced by the coupling of Section IV. The persistent subsystem is a bit restricted to Z-diagonal states, with value $W = \pm 1$ and target $W = +1$. A bath step flips W with probability $d \in (0, \frac{1}{2})$; put $c = 1 - 2d$. A reading prepares a fresh probe, copies W by a CNOT, rotates the probe by U_a with $a \in \{0, 1\}$, and measures it with effects $E_{y|s} = \frac{1}{2}(I + y\nu\sigma_s)$, $\sigma_0 = Z$, $\sigma_1 = X$, where the instrument label S is hidden and fixed. Then

$$\Pr(Y = y \mid W = w, S = s, a) = \begin{cases} (1 + yw\nu)/2, & a = s, \\ 1/2, & a \neq s. \end{cases} \tag{26}$$

A calibration stage selects a; its mismatch probability $q = \Pr(a \neq S)$ is fixed along a trajectory. In each task step the system reads Y, accepts feedback with an independent probability $h \in (0, 1]$, switches on the coupling if accepted and $Y = -1$, and then the bath acts. The loss is the occupation probability $L_t = \Pr(W_t = -1)$.

Two declarations fix the resource accounting. Each opportunity has its own drive, initially independent of the bit and of earlier records, and the outgoing field never returns to the controller's reading access. The drive is scheduled: the same incoming beam arrives at every opportunity and the command only decides whether g is switched on. On-demand emission of a whole packet would need a router and a timing scheme that we do not model.

B. Closed-loop stability

Theorem 13 (Persistence with a finite-energy actuator)**.** *Let each coupling act on the bit as the flip channel $sX\rho X + (1-s)\rho$, which is the case exactly when $\chi_\sigma(2\lambda)$ is real, for instance for the coherent and Gaussian families at $2\lambda\alpha = \pi/2$. Put $\Lambda = c(1-hs)$ and $B = chs\nu$, and let the task start from W uniform.*

(a) Conditional on $a = S$, the bit evolves by a fixed channel $\mathcal{T}$ on Z-diagonal states, $m \mapsto \Lambda m + B$ for the mean m of W. Conditional on $a \neq S$, $m \mapsto \Lambda m$.

(b) $\mathcal{T}$ has the unique fixed point $\rho_* = (I + m_* Z)/2$, $m_* = B/(1-\Lambda)$. The trace distance to ρ_* contracts by exactly Λ per step, and $D(\rho_t\|\rho_*)$ is nonincreasing and tends to zero.

(c) The stationary loss is

$$L_\infty(q) = \frac{1}{2} - \frac{(1-q)B}{2(1-\Lambda)}. \tag{27}$$

(d) The stationary rate of actual couplings is

$$p(q) = \frac{h}{2}\left[1 - \frac{(1-q)\nu B}{1-\Lambda}\right] = h\nu L_\infty(q) + \frac{h(1-\nu)}{2}, \tag{28}$$

of which a fraction s succeed. Over a window of T steps the expected number of couplings is $\frac{h}{2}\big[T - (1-q)\nu\frac{B}{1-\Lambda}\big(T - \frac{1-\Lambda^T}{1-\Lambda}\big)\big]$.

(e) For two calibrations with $q' < q$, $L_\infty(q) - L_\infty(q') = (q-q')B/[2(1-\Lambda)]$ and $p(q) - p(q') = h\nu[L_\infty(q) - L_\infty(q')]$.

Proof. With $a = S$ a reading has $\mathbb{E}[WY] = \nu$, and the coupling is attempted with probability $h(1-Y)/2$ and flips with probability s. Thus the mean after feedback is $m - 2s\,\mathbb{E}[Wh(1-Y)/2] = (1-hs)m + hs\nu$, and the bath multiplies it by c. With $a \neq S$, Y is independent of W and the mean after feedback is $(1-hs)m$. Fresh probes and fresh drives make each step the same channel on the bit marginal. Trace distance between Z-diagonal states is $|m-m'|/2$, which the affine map multiplies by $\Lambda < 1$. Data processing gives $D(\mathcal{T}\rho\|\rho_*) \leq D(\rho\|\rho_*)$, and $m_t \to m_*$ gives convergence. Starting from $m_0 = 0$, the branch means are $m_*(1-\Lambda^t)$ and 0, weighted by $1-q$ and q; this gives (c). The attempt probability is $h(1-\nu m)/2$ for $a = S$ and $h/2$ otherwise, which gives (d) after averaging and using $(1-q)m_* = 1 - 2L_\infty$. Part (e) follows from (c) and (d). □

Gate failure enters the recursion through hs, while the attempt rate keeps h: a failed gate uses the whole encounter. Stability here does not rely on a gradient rule or a metric G. It follows from the contraction of the composed channel, and the relative entropy to the stationary state is a Lyapunov function of the full closed loop, which is what Section III C 0 a required of a proof. Whether persistence is useful is a separate matter: the stationary polarization is positive only on the calibrated branch and grows with s.

Table II uses $\nu = 0.8$, $c = 0.9$, $h = 0.4$ and coherent drives. The calibrated column uses 32 passive two-read blocks, $q_{32} = 0.0404888$ (Section VII C). With an exact flip the calibrated loss would be 0.199631 [36].

N	ϵ_N	Loss, $q = \frac{1}{2}$	Loss, q_{32}	Rate p, $q = \frac{1}{2}$	Rate p, q_{32}	Added energy per opportunity
1	0.354394	0.360165	0.231654	0.155253	0.114129	0.070401
4	0.132699	0.348517	0.209300	0.151525	0.106976	0.016497
16	0.037104	0.344779	0.202127	0.150329	0.104680	0.004036
64	0.009546	0.343806	0.200259	0.150018	0.104083	0.001003

TABLE II. Stationary loss and coupling rate with a coherent drive of occupation N. The last column is $p\lambda^2$ at q_{32} in units of $\bar{\omega}$; the incoming energy $N\bar{\omega}$ is paid at every opportunity in addition. Reading, probe and calibration costs are not included.

C. Calibration with the same imperfect gate

The Imprint Chain separates calibration blocks by an exact randomization $\mathcal{R}(\rho) = \frac{1}{2}(\rho + X\rho X)$. With an imperfect gate, triggering it with probability $\frac{1}{2}$ leaves a fraction $1 - s$ of the bit's mean. Triggering with probability $1/(2s)$, which is at most one because $s \geq \frac{1}{2}$, and discarding the coin gives

$$\Big(1 - \frac{1}{2s}\Big)\rho + \frac{1}{2s}\big(sX\rho X + (1-s)\rho\big) = \tfrac{1}{2}(\rho + X\rho X) \tag{29}$$

exactly. This presumes that s is known. If the trigger probability is rounded to k binary digits, each randomization has flip probability within $s2^{-k-1}$ of $\frac{1}{2}$. Since $\mathrm{id} - \mathrm{Ad}_X$ has diamond norm 2, the record distributions of a protocol with $n+1$ randomizations differ from the exact ones by at most $(n+1)s2^{-k-1}$ in total variation, or 1.52×10^{-5} for $N = 16$, $n = 32$, $k = 20$.

Passive two-read blocks randomize, read, let the bath act and read again, without a feedback flip. With the compensated randomization they keep the Imprint Chain likelihood: the product of the two readings is positive with probability $p_{\mathrm{cal}} = (1 + c\nu^2)/2$ if $a = S$ and $\frac{1}{2}$ otherwise. After n blocks the optimal mismatch probability is

$$q_n = \frac{1}{2}\sum_{k=0}^{n}\binom{n}{k}\min\big\{p_{\mathrm{cal}}^k(1-p_{\mathrm{cal}})^{n-k}, 2^{-n}\big\}. \tag{30}$$

Active blocks, with a feedback flip after the first reading, change once the flip is imperfect. For $a = S$,

$$\Pr(y_1, y_2) = \tfrac{1}{4}\big[1 + c\nu^2 y_2\big(s + (1-s)y_1\big)\big], \tag{31}$$

and the pair is uniform for $a \neq S$. This law is obtained from the passive pair by an S-independent post-processing that keeps the first reading and relabels with probability s, so gate error creates no calibration information about the sensor. The success probability s itself is readable from the closed loop: in the persistence task the lag-one correlation of readings on gated steps differs from that on ungated steps by a term proportional to s, so the agent's own records determine s, the bath contraction and the squared gain without references, and never the sign [33]. Calibration with n blocks uses $2n$ readings and $n+1$ randomizations with an expected $(n+1)/(2s)$ couplings.

D. Records and drive in one acceptance condition

Proposition 14 (Joint record–drive condition)**.** *If $chs\nu > 0$, the stationary loss satisfies $L_\infty(q) \leq L_*$ exactly when*

$$q \leq 1 - \frac{(1-2L_*)(1-c+chs)}{chs\nu}. \tag{32}$$

If the right-hand side is negative, no amount of calibration achieves L_* with that actuator.

Proof. Rearrange Eq. (27) using $1 - \Lambda = 1 - c + chs$. □

Take $L_* = 0.2$. A coherent drive with $N = 1$ fails even with a known interface, whose loss is 0.220330. For $N = 4$, 16 and 64 the minimal numbers of passive blocks are 57, 36 and 33; since q_n is nonincreasing, checking that $n-1$ blocks fail and n succeed establishes the minimum. With 32 blocks fixed, Eq. (32) requires $s \geq 0.9943781$, that is $\epsilon \leq 0.0056219$. Coherent half-turn drives then need $N \geq 109.10$, classical drives need $N \geq 98.93$ by Proposition 10, and by Theorem 8 every single-mode drive acting as a flip channel needs $N \geq 9.390$ (and $N + \lambda^2 \geq 9.503$). The Gaussian half-turn family reaches 9.957 and the computed grid variational minimum is 9.945. These minima refer to the declared block protocol and feedback rule, not to every possible experiment.

Corollary 15 (Scaling of the persistence cost)**.** *Among drives acting as flip channels, holding the stationary loss within δ of its exact-flip value at a known interface requires a drive occupation per activation that grows as $\delta^{-1/2}$ for optimal drives and as δ^{-1} for classical drives.*

Proof. At $q = 0$, L_∞ is a smooth decreasing function of s with nonzero derivative at $s = 1$, so δ is proportional to ϵ to first order. Theorem 8 and Proposition 9 give occupation of order $\epsilon^{-1/2}$ for optimal drives, and Proposition 10 together with the coherent family gives order ϵ^{-1} for classical drives. □

E. Resource accounting

The resources of the task are kept as a vector. Each task opportunity uses one reading, one probe and CNOT, a gate coin, a bath interval and one incoming drive of occupation N, prepared whether or not it is used. Each actual coupling adds switching work $\lambda^2\bar{\omega}$ at rate p. For $N = 16$ and 32 blocks, calibration uses 64 readings and 33 randomization opportunities carrying incoming energy $528\bar{\omega}$, with an expected 17.14 couplings and switching work $0.6606\bar{\omega}$. Over a task window of 60 steps the expected number of couplings is 6.494. The occupations 61.07 and 7.33 at $\epsilon = 0.01$ are input levels, and 7.49 minimizes input plus one activation's work; none of these is the total energy of an experiment, which would also include reading, preparation, records and the switching apparatus.

F. Finite-energy response readout

An outgoing stress can change a boundary trajectory, which can in turn change signal delays. Consider the particular readout in which these delays are the only branch-dependent signal. Use one calibrated proper clock, nonnegative probe Hamiltonians H_j on separate ports, and a joint probe–ancilla state independent of the unknown branch. The ancilla may be correlated with the probes but is not acted on by the signal. In branch $k = 0, 1$ the probes undergo $U_k = \exp[-i\sum_j H_j\tau_{kj}/\hbar]$. Preparation and final joint measurement can be arbitrary. The charged probe input is $E = \sum_j \operatorname{Tr}(\rho H_j)$, above the declared zero levels.

Theorem 16 (Sharp signed-delay energy bound)**.** *Let* $\Delta\tau = \max_j |\tau_{1j} - \tau_{0j}| > 0$ *and define*

$$\zeta = \max_{x>0} \frac{1-\cos x}{x} = 0.724611\ldots, \qquad z = \frac{\zeta E \Delta\tau}{\hbar}. \tag{33}$$

The trace distance of the two output states, including the retained ancilla, is at most $\sqrt{z(2-z)}$ *for* $0 \le z \le 1$*, and at most one otherwise. For equal priors, error at most* $\epsilon < 1/2$ *therefore requires*

$$E \ge \frac{\hbar}{\zeta\Delta\tau}\left[1 - 2\sqrt{\epsilon(1-\epsilon)}\right]. \tag{34}$$

This lower bound is sharp across the permitted probe families, including oppositely signed delay differences. It need not be sharp for a specified Hamiltonian or a readout with all delay differences of the same sign. If $\Delta\tau = 0$*, no such protocol distinguishes the branches.*

Proof. The scalar maximum occurs at the unique $x_* \in (0, \pi)$ satisfying $\tan(x_*/2) = x_*$; for $x \ge \pi$, $(1-\cos x)/x \le 2/\pi$, whereas the value at $2\pi/3$ gives $\zeta \ge 9/(4\pi) > 2/\pi$. Thus $1 - \cos x \le \zeta|x|$ for every real x. Purify the probe–ancilla input. The overlap of the two resulting pure states is

$$A = \operatorname{Tr}\rho\exp(-iK), \qquad K = \sum_j H_j(\tau_{1j} - \tau_{0j})/\hbar.$$

The Hamiltonians commute strongly, and their positivity gives $|K| \le (\Delta\tau/\hbar)\sum_j H_j$ by joint spectral calculus. Consequently $\operatorname{Re} A \ge 1 - z$. For $z \le 1$, the pure-state trace distance is $\sqrt{1-|A|^2} \le \sqrt{z(2-z)}$; discarding the purifying reference cannot increase it. Equal-prior error is at least $(1-D)/2$ for trace distance D. Solving this inequality gives Eq. (34).

For saturation, take two probes with equal excited energy $e = x_*\hbar/\Delta\tau$, and delay differences $+\Delta\tau, -\Delta\tau$. On the three states $|00\rangle, |e0\rangle, |0e\rangle$ prepare

$$|\psi\rangle = \sqrt{1-p}\,|00\rangle + \sqrt{p/2}\,(|e0\rangle + |0e\rangle), \qquad p = \frac{1 - 2\sqrt{\epsilon(1-\epsilon)}}{1-\cos x_*}.$$

Here $0 \le p < 1$, $E = pe$, and the branch overlap is exactly $1 - p(1-\cos x_*) = 2\sqrt{\epsilon(1-\epsilon)}$. The optimal binary measurement attains error ϵ and Eq. (34). □

Mean-energy quantum speed limits constrain distinguishable evolution, including arbitrary-fidelity targets [34, 35]. Here the relevant resource is the sum of the physical probe energies, not the signed generator's ground-state shift; this accounts for opposite delay differences without assigning a negative probe cost. A readable branch marker, changed output frequency or amplitude, branch-dependent preparation, or a signal interaction other than translation

falls outside the theorem. For example, an unpriced orthogonal marker would distinguish the branches at zero probe energy. Receiver, clock and preparation costs remain additional resources.

In HAFF-D, three echoes determine the coefficients of a supplied JT solution. Its source-compensated inverse permits known stresses in both null directions and propagates timing and source-calibration errors to the next coupled projection [29]. This connects a selection-dependent source, its response and a subsequent operation, conditional on the declared instrument. For its compact-pulse example the largest proper-delay difference is 0.05001338; a translation-only instrument with error 0.1 requires input energy at least $11.0374\hbar$ in that clock's units. The geometric inverse does not construct such a probe or its reflecting boundary. Its supplied bulk source-compensation test is not an energy–resolution demonstration. A joint apparatus must meet the readout bound, the actuator bound and its actual stress and entropy accounts without counting the same outgoing energy twice.

G. Finite-rate entropy export

Drive occupation and exported heat obey different bounds. If the maintained subsystem and its finite reused controller remove a net dimensionless entropy current $\dot{s} > 0$, and all of it leaves through a chiral conformal exhaust of capacity c_{ex} above a thermal input at temperature T, HAFF-D gives [29]

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2.$$

The current must include retained side information and controller consumption; the number of attempted flips alone does not determine it. The bound assumes an extensive long-window entropy-current account, zero chemical potential or the matched conserved-charge account, and one common clock. It is not an exact bound for every short packet. It remains positive at $T = 0$ for a thermal input, where ordinary Landauer heat is zero. For independent parallel exhausts at one temperature, c_{ex} is their total capacity.

A nonthermal drive changes this comparison. If its incoming energy and dimensionless entropy currents are e_{in} and σ_{in}, the same Gibbs argument gives

$$\dot{Q} + e_{\mathrm{in}} - E_{\mathrm{th}}(\sigma_{\mathrm{in}}) \geq k_B T_\sigma \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2, \qquad E_{\mathrm{th}}(\sigma) = \frac{3\hbar\sigma^2}{\pi c_{\mathrm{ex}}}, \quad T_\sigma = \frac{6\hbar\sigma_{\mathrm{in}}}{\pi c_{\mathrm{ex}} k_B}.$$

The incoming energy above its entropy-matched thermal value can pay for export. In particular, a coherent beam with zero entropy is not a zero-energy reservoir. Its drive occupation and the additional heat must be accounted for separately.

This heat is neither the incoming occupation $N\bar{\omega}$ nor an extra charge to add to an outgoing energy already counted. It is a constraint on that energy when it carries the removed entropy. The scheduled drive calculation above does not establish its exhaust-capacity assumptions. A complete apparatus must price reading, recovery, sensing and entropy transport in the same cycle.

VIII. CALIBRATING AN INTERFACE WHILE IT CARRIES QUANTUM CONTENT

Preparing test states before storage is not the only way to learn an interface. The outgoing field can carry information about the coupling without carrying the logical content that the apparatus is trying to keep. We give the instrument, its loss under incomplete collection, and a coherent implementation.

On the two-qubit code $\mathcal{C} = \mathrm{span}\{|01\rangle, |10\rangle\}$, let Z_L have eigenvalues $+1, -1$. The coupling $Z_1 + \alpha a Z_2$ restricts to mZ_L, $m = 1 - \alpha a$, with unknown $\alpha > 0$ and chosen command a. Write

$$Q_f = \sqrt{k}(b + b^\dagger), \qquad U_m = e^{-imZ_L Q_f}, \qquad k > 0, \quad \Pi = (-1)^{b^\dagger b}.$$

The independently prepared incoming mode is vacuum. A beam splitter collects fraction ξ of this mode, leaving fraction $1 - \xi$ inaccessible.

Proposition 17 (A content-preserving calibration instrument)**.** *Measure the collected mode's parity and apply Z_L for an odd outcome. The corrected instrument, including an arbitrary retained reference, is*

$$\mathcal{I}_\pm(\rho) = p_\pm(a)\mathcal{D}_{\xi,m}(\rho), \qquad p_\pm(a) = \frac{1 \pm e^{-2k\xi m^2}}{2}, \tag{35}$$

where $\mathcal{D}_{\xi,m}$ fixes the populations and multiplies logical coherence by $e^{-2k(1-\xi)m^2}$. At full collection each outcome is a scalar multiple of the identity channel. Fresh-port queries chosen from earlier records can therefore calibrate the interface without replacing its held state.

Proof. The accessible displacement has the expansion $I_L \cos(m\sqrt{\xi}Q_A) - iZ_L \sin(m\sqrt{\xi}Q_A)$. On the vacuum the two field vectors have opposite parity, with squared norms p_+, p_-. The conditional pulse makes both logical factors I_L. The independent inaccessible displacement commutes with this correction and leaves $\mathcal{D}_{\xi,m}$. Each branch is thus Eq. (35), not merely an equality of its unconditional channel. Composition proves the adaptive statement; the same computation tensors with any reference. □

Environment-assisted correction of unitary Kraus branches is a known general mechanism [3, Proposition 2]. Here its records are also the data for learning the same interaction's gain and collection fraction. Repeated ancilla-based parity measurements have been implemented for a cavity mode [5]; collecting the full mode of a travelling chiral field is an additional requirement, not supplied by that experiment.

Partial collection prices the calibration itself. For N queries at each of $a = \pm h$, with fixed α, ξ, the held coherence is multiplied by

$$\exp[-4Nk(1-\xi)(1+\alpha^2 h^2)]. \tag{36}$$

This loss occurs before any subsequent operating benefit. With $k = .2$, $\alpha = 1.1$, $h = .5$ and $N = 4096$, keeping the calibration's dephasing error below .01 needs $1 - \xi \le 4.7335 \times 10^{-6}$. At $\xi = .99$ its error is essentially $1/2$. More records are not automatically better for held content.

The collection fraction in the preceding instrument is realized by the resonant and finite-band protocols of Appendix A. Their finite-packet, waveform and renewal estimates retain the assumptions needed for actual access. The following echo concerns the resulting available mode.

Proposition 18 (Coherent parity echo)**.** *Suppose the relevant mode remains available for joint control. The bounded involution*

$$R = |0_L\rangle\langle 0_L| \otimes I + |1_L\rangle\langle 1_L| \otimes \Pi = I_L \otimes P_{\text{even}} + Z_L \otimes P_{\text{odd}}$$

satisfies, on the quadrature domain,

$$R(Z_L \otimes Q_f)R = I_L \otimes Q_f, \qquad RU_m R = I_L \otimes e^{-imQ_f}. \tag{37}$$

This protects the body's marginal for every mode input and even for an initially correlated body–mode state. For an independent vacuum input, the first R acts trivially; the final R alone disentangles the body and mode.

Proof. $\Pi Q_f \Pi = -Q_f$. In the $+1$ logical block the conjugated generator is Q_f, and in the -1 block it is $-\Pi Q_f \Pi = Q_f$. Functional calculus gives the unitary identity. A unitary on the mode alone cannot change the body's reduced state. On vacuum $\Pi|0\rangle = |0\rangle$. □

One qubit meter implements R coherently. Prepare it in $|0\rangle_C$, apply

$$M = \exp[-i(\pi/2)P_{\text{odd}} \otimes Y_C], \qquad C_Z = I_L \otimes |0\rangle\langle 0|_C + Z_L \otimes |1\rangle\langle 1|_C,$$

and then apply $M^\dagger C_Z M$, with the tensor identities understood. The meter computes parity, controls the body phase and uncomputes parity; it returns exactly to $|0\rangle_C$. Bounded Hamiltonians $g(t)P_{\text{odd}} \otimes Y_C$ and $\gamma(t)(I_L - Z_L)/2 \otimes |1\rangle\langle 1|_C$ suffice, with pulse areas $\pi/2$ and π. Amplitude bounds impose corresponding nonzero durations. No classical syndrome bit has to be erased in this coherent correction. Producing a retained classical calibration record is a separate operation. Pulse area is not a heat bill: the control field, capture, clock and support still need their physical energy account. Half-diamond errors of supplied primitive channels accumulate at most by their sum, by telescoping compositions; such a bounded-channel error alone does not bound an unbounded stress observable.

A retained oscillator offers a direct alternative to the spectral meter. In its interaction picture, with $n = b^\dagger b$ and $P_1 = |1_L\rangle\langle 1_L|$, the dispersive Hamiltonian $H_R(t) = \hbar\chi(t)P_1 \otimes n$ implements R when $\int \chi(t)dt = \pi$. Such number-controlled parity gates are established cavity operations [5, pp. 2–3]. For a degenerate logical body, $\chi \ge 0$ and oscillator energy $\hbar\omega n$, the total Hamiltonian is bounded below. If χ starts and ends at zero and the gate is isolated, $P_1 n$ is conserved, so its net prescribed-switch work is exactly zero:

$$W_R = \hbar \int \dot{\chi}\, \langle P_1 n\rangle dt = 0.$$

The oscillator's stored energy, gate duration and controller are not zero. The integrated expected interaction energy is $\pi\hbar\langle P_1 n\rangle$; $\chi \le \chi_{\max}$ gives duration at least $\pi/\chi_{\max}$. If the pulse area has error δ, its half-trace error, with any reference and an input occupation bound $\langle n\rangle \le \bar{n}$, is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$. Indeed the relative unitary is $e^{-i\delta P_1 n}$, and the purified vector error squared is bounded by $2|\delta|\langle P_1 n\rangle$, using $1-\cos x \le |x|$. This is an energy-constrained error, not an unrestricted diamond bound. It gives an explicit retained-oscillator gate account; it does not price trapping a broadband travelling mode or operating the classical switch.

A. Thermal inquiry with a binary record and a finite bank

The echo gives an actual thermal inquiry protocol on the retained oscillator. Let its input be the centered Gibbs state τ_T of $H_b = \hbar\omega n$, with $n_T = (e^{\hbar\omega/(k_B T)} - 1)^{-1}$ and $\nu_T = 2n_T + 1$. This input is independent of the held body, its old reference and earlier records. Both parity gates in $RU_m R$ are used: the first is generally consequential on a thermal input. The result is $\rho_{L\mathrm{old}} \otimes D_m \tau_T D_m^\dagger$, where $D_m = e^{-im\sqrt{k}Q}$. Thus the body and its entire old reference are preserved, while the oscillator has

$$\langle P\rangle = -2m\sqrt{k}, \qquad \mathrm{Var}(P) = \nu_T, \qquad E_b - E_{b,\mathrm{in}} = \hbar\omega k m^2.$$

The displaced Gibbs state has the same entropy as τ_T.

A quadrature mean is not yet a saved record. Prepare a fresh degenerate qubit C in $\eta_v = (I + vZ_C)/2$, $0 < v \le 1$, independently of the oscillator and all earlier outputs. Couple it through

$$V_\kappa = e^{-i\kappa P \otimes Y_C}, \qquad \kappa > 0,$$

then measure X_C and store the outcome $Y = \pm 1$. The preparation, orientation, measurement and storage are declared operations.

Proposition 19 (Thermal binary inquiry)**.** *For this protocol the instrument on the held body and its old reference is*

$$\mathcal{J}_y(\rho_{L\mathrm{old}}) = \frac{1 + y\mu_T(m)}{2}\rho_{L\mathrm{old}}, \qquad \mu_T(m) = -ve^{-2\kappa^2\nu_T}\sin(4\kappa m\sqrt{k}). \tag{38}$$

The read interaction adds oscillator energy $\hbar\omega\kappa^2$ before conditioning on the record. If $m = \pm m_0$ with known $m_0 > 0$, $0 < \kappa \le \pi/(8m_0\sqrt{k})$ gives sign gain $\gamma_T = ve^{-2\kappa^2\nu_T}\sin(4\kappa m_0\sqrt{k}) > 0$. For J fresh trials, the sign estimate from $-\sum_j Y_j$ has error at most $e^{-J\gamma_T^2/2}$, including ties as errors.

Proof. The echo acts only on the independent oscillator, so all subsequent oscillator–port operations factor from $\rho_{L\mathrm{old}}$. Conjugation gives $V_\kappa^\dagger X_C V_\kappa = \cos(2\kappa P)X_C + \sin(2\kappa P)Z_C$. The Gibbs characteristic function is $\langle e^{itP}\rangle = e^{-t^2\nu_T/2}$, and displacement changes it by $e^{-2itm\sqrt{k}}$. Taking the qubit expectation proves Eq. (38). Also $V_\kappa^\dagger Q V_\kappa = Q + 2\kappa Y_C$, while P is fixed. Since the independent port has $\langle Y_C\rangle = 0$ and $Y_C^2 = I$, $n = (Q^2 + P^2 - 2)/4$ increases in mean by κ^2. Each sign-aligned binary variable has mean γ_T; Hoeffding's inequality gives the stated bound. Independence here follows from fresh inputs and the renewal below, rather than from reusing a disturbed mode. □

This record reads a sign that parity cannot resolve. Its orientation reference is an additional permission. The sine is periodic, so one setting does not identify arbitrary continuous m; an unknown initial quadrature mean also confounds it. If the mode initially shares correlations with the body, the echo still preserves the unconditional body state, but measuring the mode may steer it. That case does not give Eq. (38).

The pulse can have bounded amplitude and finite duration. In the laboratory frame choose a smooth real $g(t)$ of compact time support, with $\int g(t)dt = \kappa$, and set

$$H_{\mathrm{read}}(t) = \hbar\omega n + \hbar g(t)[P\cos(\omega t) + Q\sin(\omega t)] \otimes Y_C. \tag{39}$$

Its interaction-picture generator is exactly $\hbar g(t) P \otimes Y_C$. The final laboratory evolution includes a known free rotation; subsequent pulses and read axes use the same clock. Completing the square gives $H_{\mathrm{read}}(t) \ge -\hbar g(t)^2/\omega$. Adding the scalar $\hbar g(t)^2/\omega$ makes it nonnegative and changes only an overall phase; the scalar vanishes at both endpoints. For $|g| \le g_{\max}$ the duration is at least $\kappa/g_{\max}$. The net prescribed-switch work is the endpoint free-Hamiltonian energy increase $\hbar\omega\kappa^2$ in this isolated, degenerate-port model.

The echo's displacement admits the same finite pulse construction, with laboratory force $\hbar m\sqrt{k} f(t)[Q\cos(\omega t) - P\sin(\omega t)]Z_L$ and $\int f(t)dt = 1$. Completing the square bounds its Hamiltonian below by $-\hbar k m^2 f(t)^2/\omega$; a finite command range keeps this bound uniform. The source pulse supplies $\hbar\omega k m^2$ on the Gibbs input. The two parity gates have the finite dispersive realization above. These rotating profiles supply phase and clock access explicitly; their implementation by an autonomous controller is a further physical question.

Proposition 20 (Finite thermal renewal with retained outputs)**.** *Take a bank of J independent same-frequency oscillators in τ_T, independent mixed read ports, and storage for J binary records. After each echo and read, couple b to the next fresh bank oscillator through $\hbar h(t)(b^\dagger c + c^\dagger b)$ with area $\pi/2$. The active oscillator returns exactly to τ_T, independent of all earlier retained outputs. Its entire post-read state and correlations are transferred to the spent bank oscillator, up to known phase rotations. After summing over the new outcome, the entire old body/reference/history state is preserved, including for commands selected from earlier records. If $|h| \le h_{\max} < \omega$, the total contact Hamiltonian is bounded below and its duration is at least $\pi/(2h_{\max})$. A pulse off at both endpoints has zero net prescribed-switch work. For fixed m and κ, the spent oscillator's mean excess energy above its incoming Gibbs value is*

$$\Delta E_{\text{spent}} = \hbar\omega(km^2 + \kappa^2). \tag{40}$$

This is a finite J-trial realization of Eq. (38)*.*

Proof. The contact sends $b \mapsto -ic$, $c \mapsto -ib$. For any joint post-read state ρ_{bA}, where A contains the held body, old reference, port and records, the output is $\tau_{T,b} \otimes \rho'_{cA}$. Gibbs states are invariant under the known phase rotation. This identity transfers correlations instead of tracing them out. The two instantaneous normal frequencies are $\omega \pm h$. Total occupation and $b^\dagger c + c^\dagger b$ are conserved during the contact, which proves the stability and zero switch-work statements as in Proposition 21. The preceding two displacement accounts prove Eq. (40) after averaging over the retained record. For an old classical history r, the new record appends the kernel $p(y \mid m_r)$ to its existing conditional body/reference state. Summing over y leaves each old block, and hence their full joint state, unchanged. Induction proves renewal and this nonselective preservation for adaptive commands. Conditioning only on a new outcome can reweight old history blocks and change the reader's body/reference state; it need not preserve that conditional state. For a fixed unknown sign and fixed settings the new inputs give independent trials. □

For example, $k = .2$, $m_0 = .7$, $v = .6$ and $\kappa = \min\{\pi/(8m_0\sqrt{k}), 1/(2\sqrt{\nu_T})\}$ give sufficient trial counts 178, 247 and 593 for error .05 at $n_T = .2$, .5 and 2. The respective excess oscillator energies per trial, in units of $\hbar\omega$, are .276571, .223 and .148. Higher temperature weakens the record even though this choice reduces the read-pulse energy. A bank consumes one fresh Gibbs oscillator and one mixed read port per trial, retains the spent oscillator and a binary record, and pays both parity gates, the source/read profiles and the resonant contact duration. The initial active Gibbs oscillator is needed in addition to the J fresh bank inputs. Preparation, transport, readout, storage, orientation and the clock/support must be charged in addition.

Neither term in Eq. (40) is an entropy-export certificate. The source displacement preserves entropy; the read operation changes correlations with its port and record. Replenishing the bank or erasing spent outputs requires their joint state, including side information. This construction supplies repeated thermal inquiry for each finite bank. It does not supply the chiral capacity of Section VII, an infinite thermal line or the total covariant stress of a controller.

For a centered independent input the original outgoing source is $m^2 F_f$, $F_f = (f')^2/4$. The coherent echo leaves this coupling-added stress in the field while preserving content. Its implementation requires an available mode and controlled parity; it supplies neither a local capture apparatus for a whole continuous field nor a sustained thermal exhaust. The finite oscillator identity and the moving-field apparatus are different levels of the construction.

IX. RELATIONS TO OTHER WORK

A. Q-RAIF Papers A and C

Paper A [30] looks for the relative sign in translation-invariant local unitary propagation. When all bands share one quadratic shell, the propagation symbol obeys Euclidean Clifford relations. Its form $\eta = \omega^2 - Q(q)$ has one positive frequency direction and d negative spatial directions; the mass term together with the spatial generators yields $Cl(1, d)$. Reversing the overall metric convention reverses these signs, not their relative opposition. By Sylvester's law neither positive G nor K of Section V can be brought to that indefinite form.

Section V relates the spatial shell to measured work under explicit alignment, while keeping their domains and the indefinite spacetime form distinct. Paper C [31] asks what a coupling supplies in place of an assumed representation map. For the coupling XQ_g of this paper the algebra represented exactly is the commutant of X, for every drive state, and the defect on the remaining controls reproduces the energy–accuracy bound of Theorem 8.

The mass–energy paper [32] now uses the same quadrature interaction to realize a quantum-walk coin. With a fresh Gaussian drive each step, its mean angle fixes the rest resonance, while drive fluctuations set its decay rate. At input occupation N the smallest Gaussian rate is $\theta^2/[2\tau N(N+1)]$, attained by a displaced squeezed state. This connects the price of control to the persistence of a spectral readout. The rate is an open-channel linewidth, not a new spectral mass, and shift, preparation and port reset costs are separate.

B. Quantum control theory

Lie-algebraic control characterizes reachability through the algebra generated by drift and control Hamiltonians [14]. Stabilization against perturbations also needs magnitudes and costs. In the realized actuator these come from the physical coupling: the rotation angle from the drive amplitude, the error from the drive's quadrature fluctuations, and the cost from the field stress. A Clifford quadratic form is not needed to supply them.

C. Decoherence-free subspaces

Decoherence-free subspaces [19, 20] are protected passively by symmetry. The feedback task gives the active counterpart. The held-code instrument, coherent echo and finite collection protocols also join passive encoding to priced inquiry and correction: their exact identities protect a supplied code, while collection loss and gate error determine what survives. Brain Anchor [4] uses records of this interface for finite-window inquiry and maintenance decisions. Forming the code and its physical permission from a prior history is a further mechanism, not a consequence of encoding it.

D. The Imprint Chain and HAFF

The Imprint Chain [36] supplies the instrument, the calibration and the task, with an exact supplied flip. Replacing that flip by the field coupling raises the calibrated loss from 0.199631 to 0.202127 at $N = 16$ and makes the drive energy part of the feasibility condition. Each coupling also adds the stress F_g to the outgoing field. The HAFF gravity paper [29] uses this outgoing field as a matter source in a Jackiw–Teitelboim model; there, calibration changes the mean source through the coupling rate p of Eq. (28). Selecting the coupled projection changes its represented algebra and source even at fixed activation schedule. A boundary readout can then update the next projection, with known probe stresses and bounded errors included in the inverse. Section VII F supplies one energy constraint on a specified realization of that return.

E. Quantized control fields

The effect of field quantization on gate accuracy has a substantial literature. For Jaynes–Cummings-type couplings, conservation of excitation number leads to error bounds controlled by photon-number fluctuations, of order $1/\bar{n}$ for coherent fields [24]; coherent-field control limits for such models were analysed in Ref. [25], and simultaneous driving of several atoms by one quantized pulse in Ref. [26]. The coupling studied here is a quadrature coupling of a degenerate bit and conserves no excitation number, which is why squeezing changes the scaling from ϵ^{-1} to $\epsilon^{-1/2}$. Theorem 8 gives the quadrature-specific bound and explicit near-attainment under these assumptions. A covariant stress tensor for a finite detector, including its localization, is constructed in Ref. [23]; it shows why the apparatus stress cannot simply be omitted.

X. DISCUSSION

For gradient-type feedback the algebra must compose associatively and the descent metric must be positive. When control is realized by a field, the structure that limits persistence is the canonical commutation relation of the drive quadratures: $[Q, P] = 2i$ fixes the trade-off between flip accuracy and drive occupation in Theorem 8, and it is the reason squeezed drives do better than classical ones. For the fixed Pauli flip the commutation relation of the driving mode sets the accuracy cost. For a family of control operators, Proposition 6 additionally identifies when their quadratic structure fixes a state-independent centered-input work form. Persistence has a closed-loop proof in this model, with an explicit Lyapunov function, and the drive energy and calibration records enter one feasibility condition.

Several inputs remain supplied. The free chiral field, the degenerate bit, the clock and switch, the drive ports and the instrument family are given. The feedback task also supplies the success probability for compensated randomization and scheduled beams. Its analysis concerns Z-diagonal bit states and flip channels; complex $\chi_\sigma(2\lambda)$ instead rotates Z into Y. The held-content construction concerns arbitrary logical inputs. Its finite-band realization prepares and collects a specified packet, with a finite waveform, clock tolerances and independent vacuum exhausts. It returns a

coherent meter; the carrier, detector, port selection and vacuum supply remain separate inputs. The stress of the switch and its support is not computed; outgoing-field stress is used outside the apparatus.

The thermal bank identifies what renewal has to preserve. A fresh input restores the oscillator only because the previous state, including its record correlations, moves to a spent output. Bank replenishment and memory erasure must therefore be priced on that joint output, as in the information thermodynamic account [9, 10]. The chiral capacity bound constrains a specified entropy flow; the finite oscillator construction gives an implementable inquiry and transfer, rather than that flow's attainability.

Repeated propagation in one field remains a stronger problem. Proposition 12 proves that shared drives can preserve each marginal channel while correlating failures. A persistence calculation on that joint history may therefore have a different energy scaling from the fresh-drive task. The other unresolved connection is formation: which interactions and retained records produce the code, access permissions and controller used here? T-DOME, Imprint and Brain Anchor address that mechanism; the present results constrain the resources of the operations it must supply.

XI. CONCLUSION

The connection between control and propagation becomes precise only after a physical condition is specified. In the centered-field calibration studied here, a linear Hermitian control family has state-independent switching work exactly when it has a scalar quadratic square. A calibrated use of those same generators for propagation then gives a positive pullback relation between work and the spatial shell. The time direction and mass anticommutator enter separately. Persistence alone supplies none of these identifications.

The same field coupling gives a finite actuator whose stress, accuracy and drive energy can be calculated together. Squeezed inputs approach the $\epsilon^{-1/2}$ occupation bound, while classical drives require order ϵ^{-1}. Calibration and physical actuation enter a single feedback feasibility condition. Content-preserving inquiry, finite port renewal and the separate delay-readout bound extend the accounting to both directions of an interface. The remaining formation problem is which physical histories produce the aligned couplings, permissions and controller, rather than merely supplying them. The conditional bridge locates that question at the coupling itself; it does not replace it with the existence of a well-priced device.

Appendix A: Physical collection of the inquiry mode

These constructions support the mode access used in Section VIII. They distinguish a resonant stored mode, a finite packet and a finite-band travelling input; none substitutes a free capture or renewal operation for a physical interaction.

1. A resonant collection port

The collection fraction can itself be realized by a finite-duration interaction. This construction starts with a retained source oscillator b and a receiver oscillator c of the same frequency ω, with c initially in vacuum. It does not assume that a broadband travelling mode has already been trapped by this interaction. In the interaction picture put

$$H_{\mathrm{cap}}(t) = \hbar g(t) C, \quad C = b^\dagger c + c^\dagger b, \quad \theta = \int g(t)\,dt, \quad U_{\mathrm{cap}} = e^{-i\theta C}.$$

Proposition 21 (Finite resonant collection)**.** *For $0 \le \theta \le \pi/2$, the receiver collects $\xi = \sin^2\theta$ of the source mode. Corrected receiver-parity readout realizes Eq. (35) with this ξ. If $|g| \le g_{\max} < \omega$, the total Hamiltonian $\hbar\omega(n_b + n_c) + H_{\mathrm{cap}}$ is bounded below and the pulse needs duration at least $\theta/g_{\max}$. A pulse starting and ending at zero does no net prescribed-switch work in this isolated two-mode model. For the independently prepared vacuum encounter, the mean occupations after collection are ξkm^2 and $(1-\xi)km^2$. Their sum is km^2.*

A pulse-area error δ changes ξ by at most $|\delta|$. For any input with $\langle n_b + n_c\rangle \le \bar{n}$, including a retained reference, its half-trace error is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$.

Proof. The Heisenberg transforms are $b \mapsto b\cos\theta - ic\sin\theta$ and $c \mapsto c\cos\theta - ib\sin\theta$. The conditional coherent states produced by U_m therefore split into receiver and remainder amplitudes with squared fractions $\xi, 1-\xi$. The receiver's known phase does not change parity. Applying Proposition 17 proves the instrument and the occupation account, for any logical input.

The normal modes $(b \pm c)/\sqrt{2}$ have instantaneous frequencies $\omega \pm g$, which proves the lower bound. The amplitude bound proves the duration statement. Both C and total occupation commute with the Hamiltonian, so $\int \langle \partial_t H_{\mathrm{cap}} \rangle dt = \hbar \langle C \rangle [g(t_{\mathrm{end}}) - g(t_{\mathrm{start}})] = 0$. Finally $|\sin^2(\theta + \delta) - \sin^2 \theta| \le |\delta|$ and $|C| \le n_b + n_c$. The purified-vector estimate $2\langle 1 - \cos(\delta C) \rangle \le 2|\delta|\bar{n}$ proves the energy-constrained error bound. □

This account includes transferred energy and a gate clock. Receiver parity control, reading and retaining a syndrome, preparing the next empty receiver, and supplying the switching apparatus remain separate resources. Emptying the receiver exports energy; it is not included in the zero net switch work above. Restoring a degraded collection setting also needs a stated physical permission and price. Brain Anchor [4] uses such a permission in a finite model where the same content survives inquiry, operation and maintenance. Resonant port capture, local capture of a continuous field and sustained thermal export are three different implementation questions.

Whole-mode control can be reduced to successive port contacts in a finite packet model. Take n source oscillators of frequency ω and one vacuum receiver c. For real $f_j \ge 0$, $\sum_j f_j^2 = 1$, form the packet mode $b_f = \sum_j f_j b_j$. Sequential source interactions $e^{-im\sqrt{k} f_j Z_L (b_j + b_j^\dagger)}$ produce the same packet displacement as $e^{-im\sqrt{k} Z_L (b_f + b_f^\dagger)}$. After contacting port j, couple only that port to c through the resonant Hamiltonian above, with area $0 \le \theta_j \le \theta_{\max} \le \pi/2$. Released ports are inaccessible. The receiver's packet amplitude, up to its known common phase, obeys

$$y_0 = 0, \qquad y_j = y_{j-1} \cos \theta_j + f_j \sin \theta_j.$$

Proposition 22 (Sequential local-port collection)**.** *The receiver collects fraction $\xi = y_n^2$ of the packet, and its corrected parity instrument is Eq.* (35)*. Among these permitted real resonant contacts, final collection is maximized by*

$$\theta_j = \min\{\theta_{\max}, \arctan(f_j / y_{j-1})\},$$

with the first angle $\pi/2$ before applying the cap when $y_{j-1} = 0 < f_j$. When $y_{j-1} = f_j = 0$, choose $\theta_j = 0$. For a uniform packet and $\theta_{\max} = .5$, the first five contacts are capped and the remaining ones are unconstrained. For $n \ge 5$ the exact optimum is

$$\xi = 1 - \frac{d}{n}, \qquad d = 5 - \cot^2(.25)(1 - \cos^5(.5))^2 = 1.47392688\ldots.$$

Thus 148 uniform ports suffice for collection at least .99, and fewer do not under the same cap. With no cap below $\pi/2$, the angles $\theta_1 = \pi/2$, $\theta_j = \arctan(1/\sqrt{j-1})$ collect a uniform finite packet completely.

Proof. Passive mode mixing sends the packet's conditional coherent displacement to receiver amplitude $-iy_n$ and a remainder of squared norm $1 - y_n^2$. Orthogonal input modes are vacuum. Their overlaps prove the instrument. Each step maximizes $y \cos\theta + f_j \sin\theta$ at the stated angle. The maximum is nondecreasing in $y \ge 0$, because all permitted cosines are nonnegative. Induction therefore makes this stepwise optimum a final optimum. For a uniform packet, the first capped steps give $y_j = n^{-1/2} \cot(.25)(1 - \cos^j(.5))$. After the fifth, the interior optimum has $y_j^2 = y_{j-1}^2 + 1/n$. This proves the formula and threshold. The uncapped recursion gives $y_j = \sqrt{j/n}$. □

Each contact has a Hamiltonian bounded below when $g_{\max} < \omega$, needs duration at least $\theta_j / g_{\max}$ and conserves total occupation. The source prepares mean total occupation km^2, the receiver holds ξkm^2 and the released ports hold $(1-\xi)km^2$. Parity control is needed only on the retained receiver. For contact-area errors δ_j, the half-trace error is at most $\min\{1, \sum_j \sqrt{2|\delta_j|\bar{n}}\}$ when total occupation is bounded in mean by $\bar{n}$; telescope the preceding energy-constrained bound. Source switching, receiver reuse and detector errors require their own account. These are local contacts in a finite collision architecture, not a proof that a specified relativistic wave packet admits that architecture or that its clocks and references can be sustained indefinitely.

2. Positive-frequency continuous collection

A specified continuous field can also supply the collection port. Let $\omega = \Omega + \delta$, $|\delta| \le B < \Omega$, and couple it to a retained oscillator c. With $[a_\delta, a_{\delta'}^\dagger] = \delta(\delta - \delta')$, take a real even form factor g, $J = g^2$ and $G = \|g\|_2$. Put

$$H(h) = \hbar\Omega n_c + \hbar \int_{-B}^{B} (\Omega + \delta) a_\delta^\dagger a_\delta d\delta + \hbar h(t) \int_{-B}^{B} g(\delta)(c^\dagger a_\delta + a_\delta^\dagger c) d\delta, \qquad |h| \le 1. \tag{A1}$$

This is an excitation-preserving effective Hamiltonian with a specified band-smeared coupling. Its frequency form factor gives spatial tails; it is not a strictly point-local relativistic detector. No negative physical frequencies or Markov

extension of the band are used. Matched pulse absorption by a tunable resonator is an established mechanism [2]; the question here is how its finite-band realization enters the held-content instrument and source account.

In the frame rotating at Ω, the one-particle operator is

$$A = \begin{pmatrix} 0 & \langle g| \\ |g\rangle & \delta \end{pmatrix}, \quad S = e^{-iAT}, \quad r(T) = \langle c|S|c\rangle.$$

The capture pulse has $h = 1$ on $[0, T]$ and is otherwise off. Deviations from this prescribed pulse are bounded below.

Proposition 23 (Matched finite-band capture)**.** *If $\Omega > B + G$, every Hamiltonian in Eq.* (A1) *is bounded below by $\hbar(\Omega - B - G)N$. Write $\xi_0 = 1 - |r(T)|^2 > 0$. Among normalized incoming field modes, the mode*

$$f = -P_{\text{field}} S^\dagger |c\rangle / \sqrt{\xi_0}$$

maximizes collection and is collected with fraction ξ_0. It can be prepared from independent vacuum by a finite prescribed body-controlled linear drive with envelope $u(s) = r(T - s)$, $0 \le s \le T$. For even J, r and this envelope are real. The prepared field amplitudes are

$$m\beta w_u(\delta), \qquad w_u(\delta) = -ig(\delta) \int_0^T u(s) e^{i\delta s} ds, \quad w_u = -P_{\text{field}} S^\dagger |c\rangle, \quad \beta = \sqrt{k/\xi_0}. \tag{A2}$$

For an arbitrary prepared envelope and $w_u \neq 0$, replace k by $k_{\text{eff}} = \beta^2 \|w_u\|^2$ and use $\xi = |\langle c|S(0, w_u)\rangle|^2 / \|w_u\|^2$. Corrected receiver-parity readout then realizes Eq. (35) *with these actual parameters.*

Proof. $\|A\| \le B + G$ and second quantization prove the lower bound. The norm of the field part of the receiver row is $\sqrt{\xi_0}$; Cauchy–Schwarz proves the optimum. Its orthogonal complement has zero collection, so the statement is not uniform over unknown packets.

Eliminating the field component of $S^\dagger|c\rangle$ gives Eq. (A2). Reflection $\delta \mapsto -\delta$, field sign reversal and complex conjugation define an antiunitary taking A to $-A$ and leaving its propagator invariant. Hence r is real. One explicit preparation places the band-smeared source at $x = -T$ and the receiver at $x = 0$, with propagation speed one. Drive the source during $t \in [-T, 0]$ with envelope $u(t+T)$ and a prescribed carrier phase Ω. Explicitly its interaction-picture Hamiltonian is

$$H_s^I(t) = \hbar m \beta Z_L u(t + T) \int_{-B}^{B} g(\delta)[a_\delta^\dagger e^{i\delta(t+T)} + a_\delta e^{-i\delta(t+T)}] d\delta.$$

Its displacement is $m\beta w_u$; the receiver is coupled only after this drive. The source Hamiltonian is linear in the band field and proportional to mZ_L. Its time-ordering commutator is a scalar times Z_L^2, so it changes only an overall phase. Completing the square bounds its instantaneous energy below since $\Omega - B > 0$ and $\int J/\omega$ is finite.

Passive field evolution splits the conditional coherent amplitudes into a receiver component of squared fraction ξ and a remainder of fraction $1 - \xi$. Their vacuum overlaps give the claimed instrument for arbitrary logical input and retained reference. Initial independence and parameters fixed within an encounter are required. □

The propagation prescription is within the band-smeared model: a strictly band-limited field operator has no compact spatial support. The construction supplies a drive and a receiver, not their formation from unknown interactions. For real envelopes and even g, the free input and output energies are

$$E_{\text{source}} = \hbar\Omega k_{\text{eff}} m^2, \qquad E_c = \hbar\Omega\xi k_{\text{eff}} m^2, \qquad E_{\text{free,out}} = \hbar\Omega(1 - \xi) k_{\text{eff}} m^2. \tag{A3}$$

The same reflection symmetry makes the mean detuning energy zero before and after collection. Consequently the net prescribed capture-switch work is zero in this family. The complete source drive supplies E_{source}. For other packets the capture work need not vanish: at mean occupation $\bar{n}$, its magnitude is at most $2\hbar B\bar{n}$. Detector, carrier, clock and support costs are separate from these accounts.

Proposition 24 (A finite waveform certificate)**.** *Let $J(\delta) \le J_0$. Approximate $r(T - s)$ by linear interpolation of node values on L uniform intervals, each rounded on a prescribed grid of spacing q. Include a further uniform envelope error η_u. With $\nu = \sqrt{\xi_0}$,*

$$\sigma = \sqrt{2\pi J_0 T} \left[\frac{G^2 (T/L)^2}{8} + \frac{q}{2} + \eta_u \right]$$

bounds the prepared field-vector error relative to Eq. (A2)*. An interval enclosure error for node values is added inside the brackets. Let duration and arrival errors be bounded by* d_T, d_t*, and the capture pulse have* $\int |h_{\mathrm{actual}} - h|dt \le d_h$*. Put* $\varepsilon = (B+G)d_T + Bd_t + Gd_h$*. If* $\sigma < \nu$*, then*

$$1 - \xi \le \left[\frac{|r|\nu + \sigma + \varepsilon(\nu+\sigma)}{\nu - \sigma}\right]^2, \qquad \left(1 - \frac{\sigma}{\nu}\right)^2 \le \frac{k_{\mathrm{eff}}}{k} \le \left(1 + \frac{\sigma}{\nu}\right)^2. \tag{A4}$$

Arrival and duration bounds here concern collection and the logical parity instrument, which are insensitive to the common carrier phase.

Proof. $|r''(s)| = |\langle Ac|e^{-iAs}|Ac\rangle| \le G^2$ gives the interpolation bound. Plancherel over the entire frequency line, followed by restriction to the band, gives $\|w_u - w_{\mathrm{ideal}}\| \le \sigma$. Duhamel bounds the carrier-referenced propagator error by ε. The ideal outgoing field remainder has norm $|r|\nu$. Adding preparation and propagator errors to this norm and dividing by the actual input norm, at least $\nu - \sigma$, proves the first bound. The second follows by the same input-norm bounds. □

For a concrete continuum certificate use

$$J(\delta) = \frac{1}{2\pi}\left[1 - (\delta/B)^2\right]^2, \quad \Omega = 20, \quad B = 5, \quad T = 16.$$

Frequencies are measured in units of $2\pi J_0$ and times in its reciprocal. Then $G^2 = 8/(3\pi)$. The exact even bath moments are $\mu_{2j} = 16J_0B^{2j+1}/[(2j+1)(2j+3)(2j+5)]$. If $M_n = \langle c|A^n|c\rangle$, then

$$M_0 = 1, \qquad M_{2n} = \sum_{j=0}^{n-1} \mu_{2j} M_{2(n-1-j)}, \qquad r(t) = \sum_{n=0}^{\infty} \frac{(-1)^n M_{2n} t^{2n}}{(2n)!}.$$

Decomposing each departure and return to c proves the recurrence. The degree-P Taylor remainder is at most $[(B+G)T]^{P+1}/(P+1)!$ by self-adjoint functional calculus. Directed intervals at 120 decimal digits and $P = 400$ give $r(16) = 0.000061743299929498335 73\ldots$ with remainder below 1.53×10^{-79} and $\xi_0 > 0.99999999618$.

Choose $L = 512$, $q = 1/65536$, $\eta_u = d_T = d_t = d_h = 10^{-5}$. The 513 rounded node values fit the grid $[-1/64, 1]$, whose 66561 labels need 17 bits each. The fixed table uses 8721 logical bits, apart from its decoder, interpolator, carrier and clock. A separately encoded normalization amplitude rounded on a 2^{-24} grid contributes less than 10^{-7} relative error at nominal $k = .2$; it fits within η_u and uses 24 more label bits. Eq. (A4) gives $1 - \xi < 4.42935 \times 10^{-7}$ and $k_{\mathrm{eff}}/k \in [.999030268, 1.000970203]$. For $\alpha = 1.1$, $h = .5$, $N = 4096$ probes at each of $\pm h$ and nominal $k = .2$, the calibration's collection-dependent dephasing error is below .000946. This contrasts with the loss of content at $\xi = .99$ above; it does not include parity-gate or detector errors.

A second numerical representation replaces frequency multiplication by cell midpoints while retaining g itself. The invariant vector in each cell is $g\mathbf{1}_{\mathrm{cell}}/\|g\mathbf{1}_{\mathrm{cell}}\|$, with coupling $\sqrt{\int_{\mathrm{cell}} J}$. Its finite matrix approximates the continuum propagator with operator error at most TB/M for M cells. These cells are a numerical resolution, not fresh physical ports. Dense and sparse matrix calculations agree with the moment enclosure and the prepared waveform. The recurrence and rounding rule specify the waveform without a fitted infinite-precision function.

After an encounter, swapping c with an independent same-frequency vacuum oscillator returns c exactly to vacuum, even when its old state is correlated with a reference. The swap exports E_c and has the finite duration and occupation account of Proposition 21. It supplies a reusable receiver for a finite bank of encounters, not a free thermal sink. Repeating the query law needs independent field preparations as well as independent vacuum exhausts. Independent field sectors selected one at a time give a finite-bank realization whose Hamiltonian remains bounded below. 8192 queries consume 8192 such preparations and exhausts, while sharing the waveform program. In this calibration example the nominal source energy is $E_{\mathrm{probes}}/[\hbar(2\pi J_0)] = 4096(.2)(20)[(1 - .55)^2 + (1 + .55)^2] = 42680.32$. It is predominantly collected and later exported, not eliminated by the small content-damage bound. Serial preparation and capture take $8192(2T) = 262144$ time units before parity control and receiver reset. Time windows in one band-limited waveguide are not automatically independent; that alternative requires a return/reset bound. Thermal exhaust resets c to a thermal state and changes the readout law. For example, for the exactly matched mode, a thermal receiver of mean occupation n_T gives even-parity probability

$$\frac{1}{2}\left[1 + \frac{e^{-2\xi k m^2/[1+2(1-\xi)n_T]}}{1 + 2(1-\xi)n_T}\right]$$

after mixing with the vacuum source packet, rather than the vacuum formula. For an unmatched packet, the thermal occupation term instead uses $|\langle c|S|c\rangle|^2 n_T$, which need not equal $(1-\xi)n_T$.

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Representing Control in the World: Retained History, Physical Sources and Renewal in Q-RAIF

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October 2026

Abstract

What does a physical coupling actually represent? For a system coupled to a discarded port, the reduced map is completely positive but need not be multiplicative. We determine its exact algebra for finite-spectrum dephasing: a non-atomic port leaves precisely the coupling operator's commutant, which can be noncommutative. Retaining an oscillator permits exact closed-loop gates; retaining translational degrees of freedom gives a faithful representation in the closed propagation carrier. A single quadratic shell supplies the spatial Clifford relations, while evolution time and the mass anticommutator enter separately in the Lorentzian symbol.

A centered-input charging condition connects these structures without identifying their quadratic forms. Under an actual calibrated alignment of control and propagation generators, the work form is a positive pullback of the spatial shell. The same encounter can satisfy that all-state charge condition while its reduced exact algebra remains only a commutant. Neither persistence nor a positive cost alone selects a spacetime representation.

The physical realization retains the records and resources that the algebraic statement omits. A positive Gaussian material Hamiltonian supports acquisition, record-controlled force, sensing and another action, with bounds on physical energy moments and smooth interacting sources. A paid Gibbs bank renews a specified receiver, and one finite cursor approximates a sequence while retaining the complete old history. Full-field ring renewal, neutral compression and thermal sorting, local massive-body gates and a spatial work clock provide further finite implementations in their stated layouts. A common action includes clock and support reaction. Matching these quantum layouts to one covariant supported apparatus, sustained thermal renewal and jointly sharp exhaust remains a separate construction.

Keywords: Q-RAIF, physical representation, multiplicative domain, propagation symbol, quantum records, material response, thermal renewal

Technical appendices and their original numbering are retained in Q-RAIF C technical supplement (`QRAIF_C_supplement.pdf`). The certificate guide at the end of this paper identifies their claims.

1 Introduction

A control algebra specifies how operations compose. A physical interface must also say which operations its coupling transports exactly, what happens to the degrees of freedom carrying its records, and what resources allow another operation. An assumed homomorphism answers the first algebraic question by definition; it does not construct the interface.

Q-RAIF A identifies a Lorentzian read-out under propagation-symbol conditions, Q-RAIF B derives actuation and work from a field coupling, and the mass–energy paper relates spectral mass to finite read-outs [25, 26, 27]. Here we put the representation question between these structures and their physical implementation. The distinction is whether the relevant degrees of freedom are discarded or retained, not whether the environment has a large Hilbert space.

Exact transport. Sections 2–4 begin with the algebra supplied by a coupling. Its reduced Heisenberg map has an exact algebra, the multiplicative domain [29, 30, 31]. Theorem 4 computes it for a dephasing interaction. With a non-atomic discarded port it is the commutant of the coupling observable, including its full degeneracy blocks. The field-coupled flip then relates its remaining defect to drive energy and escaped information. These are architecture-specific statements, not a universal conservation-law prohibition on control [34, 35, 36].

Propagation and charging. When translations remain in the closed carrier, the internal algebra embeds faithfully. Section 5 distinguishes that exact embedding from the quadratic-space structure of a first-order propagation symbol. The latter requires a single shell, and its Lorentzian extension uses the supplied time and a mass anticommutator. Section 6 brings in Q-RAIF B's conditional work bridge: one centered independent field input charges every unknown carrier state equally precisely when each control generator combination has a scalar square. An actual calibration between these generators and the propagation symbol then relates the positive work form to the spatial shell. It does not turn that form into an indefinite spacetime metric. We show why the same calibrated encounter still need not represent the full control algebra after the port is discarded.

Retention and physical return. Section 7 gives the two decisive controls on the discarded-port argument: a genuinely fresh lost port leaves an irreversible defect, whereas a retained oscillator closes exact Weyl loops. The continuum loop and its finite adaptive error budget are developed in Appendix B. The retained case requires coherent routing, quadratures and a clock; an abstract embedding supplies none of those permissions. These are two realizations, not an exhaustive classification of physical interfaces.

The material construction in Sections 8–10 then follows one complete Gaussian Hamiltonian through acquisition, force, sensing, receiver replacement and another action. Records remain operators in their old joint history. Physical energy moments and smooth-source forms control unbounded field readings, rather than inferring stress convergence from trace distance alone. Receiver replacement uses a prepared Gibbs bank and transfers the spent correlations into its ledger.

The longer certificates are grouped by the physical interface they establish. Appendix A constructs the charged regulator, a finite mode with its remainder and the same-Hamiltonian source bounds. Appendix C treats the complete ring body, neutral gates, packet loading and thermal sorting. Appendix D treats the spatial work field and the separate massive-body gate. The remaining appendices retain the battery, outlets, thermal preparations, contact recoil, actual diagnostic return and current-stress compiler. Their distinct carriers are not identified by a common vocabulary. Section 11 gives a common action with clock/support reaction and states the matching still required.

Three resource limits matter. Bounded-energy receiver renewal takes longer and uses more clock states; fixed-duration finite rounds use growing clock energy; a spatial work clock uses horizon-dependent inertia and a supplied material frame. The ring converter's complete-output thermal deficit can approach zero in a prescribed finite layout. These results do not yet form one indefinitely renewed, fixed-capacity, covariant apparatus.

Unless stated otherwise, $\hbar = c = k_B = 1$. Energies expressed as frequencies acquire a factor $\hbar$ when units are restored, times and pulse areas use the corresponding frequency units, and entropy is in nats. The combinations $b + b^\dagger$, $i(b^\dagger - b)$ and occupation numbers are dimensionless; material smearings use the normalizations specified with their Hamiltonians below. All joint-state assertions include the stated old references; a preserved nonselective marginal need not remain unchanged after conditioning on a new record.

2 The algebraic statement

Let (V, q) be a finite-dimensional real vector space with a non-degenerate quadratic form and $Cl(V, q)$ the real Clifford algebra generated by V with $v^2 = q(v)\mathbf{1}$. When generators are to act as operators, those with $q(e) > 0$ are taken Hermitian and those with $q(e) < 0$ anti-Hermitian; this fixes an involution with $e^* = \operatorname{sign}(q(e))\, e$ on an orthogonal basis.

Definition 1 (Representation input)**.** A *representation* of a control algebra $\mathcal{O}_{\text{int}} \cong Cl(V, q)$ in an environmental C^*-algebra $\mathcal{A}_{\text{ext}}$ is a unital $*$-homomorphism $\phi : \mathcal{O}_{\text{int}} \to \mathcal{A}_{\text{ext}}$.

Lemma 2 (Semisimplicity)**.** *$Cl(V, q)$ is semisimple. If $\dim V$ is even it is simple, and every nonzero algebra homomorphism out of it is injective.*

Proof. By the classification of real Clifford algebras [40, 41], $Cl(V, q)$ is a matrix algebra over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$, or a direct sum of two such algebras, according to signature and dimension modulo eight. For even $\dim V$ it is a single matrix algebra. The kernel of a homomorphism is a two-sided ideal, and a simple algebra has none besides 0 and itself. □

Proposition 3 (Image closure)**.** *Let ϕ be as in Definition 1. Then $\operatorname{Im}\phi$ is a $*$-subalgebra of $\mathcal{A}_{\text{ext}}$ in which the Clifford relations hold. If $Cl(V, q)$ is simple, in particular when $\dim V$ is even, ϕ is injective. If it is the sum of two simple algebras, its image is isomorphic to the full algebra or to one of the two summands.*

Proof. Images of $*$-homomorphisms are $*$-subalgebras and relations descend. Simplicity gives injectivity. The only proper ideals of a sum of two simple algebras are zero and either summand; a unital representation has a proper kernel. □

Non-degeneracy is needed: for $q(x,y)=x^2$ on $\mathbb{R}^2$, the map $e_1 \mapsto \mathrm{diag}(1,-1)$, $e_2 \mapsto 0$ is a nonzero unital homomorphism into $M_2(\mathbb{R})$ that is not injective. An algebra homomorphism also does not compare signatures. That requires a quadratic-space morphism $j : (V,q) \to (W,Q)$ with $Q(jv) = q(v)$, and, where relevant, preservation of the involution and grading by the induced map of Clifford algebras. Proposition 3 provides neither the map ϕ nor such a j. Sections 3–5 ask where they can come from.

3 Couplings give channels

Let a finite-dimensional system S interact with an environment $\mathcal{E}$ through a unitary U on $\mathcal{H}_S \otimes \mathcal{H}_\mathcal{E}$, with the environment in a normal state σ. The induced Heisenberg map on system observables is

$$\Phi^*(A) = \mathrm{tr}_\mathcal{E}\big[(\mathbf{1}\otimes\sigma)\,U^*(A\otimes\mathbf{1})U\big]. \tag{1}$$

It is unital and completely positive, and by Stinespring's theorem every such map arises in this way [29]. It is in general not multiplicative. Its *multiplicative domain*

$$\mathcal{M}_\Phi = \{A :\ \Phi^*(A^*A) = \Phi^*(A)^*\Phi^*(A),\ \Phi^*(AA^*) = \Phi^*(A)\Phi^*(A)^*\} \tag{2}$$

is a C^*-subalgebra on which Φ^* is a $*$-homomorphism, and $\Phi^*(AB) = \Phi^*(A)\Phi^*(B)$ whenever $A \in \mathcal{M}_\Phi$ [30, 31]. It is the largest part of the system's algebra that the coupling represents exactly. For other elements we measure the failure by the *defect*

$$\delta(A) = \Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A) \ \geq 0. \tag{3}$$

Theorem 4 (Exactly represented algebra of a dephasing coupling)**.** *Let $X = \sum_x x P_x$ have finite spectrum, let E be self-adjoint on $\mathcal{H}_\mathcal{E}$, and $U = \exp(-iX\otimes E)$. Write $\chi(t) = \mathrm{tr}(\sigma e^{-itE})$ and $A_{xx'} = P_x A P_{x'}$. Then*

$$\Phi^*(A) = \sum_{x,x'} \chi(x'-x)\, A_{xx'}, \tag{4}$$

the relation $x \sim x'$ defined by $|\chi(x'-x)| = 1$ is an equivalence relation on the spectrum of X, and

$$\mathcal{M}_\Phi = \{A :\ A_{xx'} = 0 \text{ whenever } |\chi(x'-x)| < 1\}. \tag{5}$$

On $\mathcal{M}_\Phi$, Φ^* is an inner automorphism $A \mapsto W^*AW$ with $W = \sum_x u_x P_x$ for phases u_x. If $|\chi(t)| < 1$ for every nonzero difference t of eigenvalues of X, then $\mathcal{M}_\Phi = \{X\}'$ and Φ^* is the identity on it.

Proof. $U = \sum_x P_x \otimes e^{-ixE}$, so $U^*(A\otimes\mathbf{1})U = \sum_{x,x'} A_{xx'} \otimes e^{ixE}e^{-ix'E}$, and taking the expectation in σ gives Eq. (4).

If $|\,\mathrm{tr}(\sigma e^{-itE})| = 1$, write $\sigma = \sum_i p_i |\psi_i\rangle\langle\psi_i|$; each $|\langle\psi_i|e^{-itE}|\psi_i\rangle| \leq 1$, so equality forces every ψ_i to be an eigenvector of e^{-itE} with one common eigenvalue $\chi(t)$. Hence e^{-itE} acts as the scalar $\chi(t)$ on the support of σ. If this holds for t_1 and t_2, it holds for $t_1 + t_2$ with $\chi(t_1+t_2) = \chi(t_1)\chi(t_2)$. Together with $\chi(0) = 1$ and $\chi(-t) = \overline{\chi(t)}$ this shows that $\sim$ is an equivalence relation, and that within a class with base point x_0 the phases $u_x = \chi(x - x_0)$ satisfy $\chi(x'-x) = \overline{u_x}u_{x'}$.

The diagonal blocks of the defect are

$$P_y\big[\Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A)\big]P_y = \sum_x \big(1 - |\chi(y-x)|^2\big)A^*_{xy}A_{xy}, \tag{6}$$

a sum of positive operators with nonnegative coefficients. It vanishes for all y only if $A_{xy} = 0$ whenever $|\chi(y-x)| < 1$; the condition with AA^* is the same with the roles of rows and columns exchanged. Conversely, if A has this block pattern, Eq. (4) gives $\Phi^*(A) = \sum \overline{u_x}u_{x'}A_{xx'} = W^*AW$, and these A form a $*$-algebra on which Φ^* is multiplicative. If no nonzero difference has $|\chi| = 1$, the classes are single eigenvalues, the algebra is $\bigoplus_x P_xB(\mathcal{H}_S)P_x = \{X\}'$, and W is scalar on each block. □

Corollary 5 (Continuous spectrum forbids exact coupled control)**.** *If the spectral distribution of E in σ has no atoms, then $|\chi(t)| < 1$ for every $t \neq 0$, so $\mathcal{M}_\Phi = \{X\}'$. In particular this holds for every normal state when E has purely continuous spectrum. Every exactly represented element A commutes with X, and hence with the interaction on its natural domain. No nonzero rescaling of the coupling changes this conclusion.*

Proof. The equality $|\int e^{-ite}\,d\mu(e)| = 1$ requires the phase e^{-ite} to be constant μ-almost everywhere. For $t \neq 0$ its level sets are translates of the countable lattice $(2\pi/t)\mathbb{Z}$, and a non-atomic probability measure gives each such set measure zero. Absolute continuity is sufficient but unnecessary. □

A field quadrature, or the smeared current Q_g of Paper B, has purely absolutely continuous spectrum, so Corollary 5 applies to every normal drive state, whatever its energy. For other couplings, $|\chi(t)| = 1$ requires support in an eigenspace of e^{-itE}, not necessarily an eigenstate of E. Periodic spectra can give exact revivals: with $X = \pi Z$, $E = \mathrm{diag}(0,1)$ and $\sigma = |+\rangle\langle+|$, the interaction unitary is $I \otimes \mathrm{diag}(1,-1)$ and the system channel is identity, although σ is not an E eigenstate. A Fock state in a number-type coupling gives another finite-energy possibility. The quadrature obstruction concerns the specified single coupling; extending it to composite local controls requires a separate argument.

4 The field-coupled flip

Paper B couples a degenerate bit to one mode of a free chiral field through $U = \exp(-i\lambda XQ)$, $\lambda > 0$, with $Q = b + b^\dagger$, $P = i(b^\dagger - b)$ and $[Q,P] = 2i$ [26]. Here $E = \lambda Q$ and the eigenvalues of X are ± 1. Write $\chi = \mathrm{tr}(\sigma e^{-2i\lambda Q})$ and $\sigma_\pm = e^{\mp i\lambda Q}\sigma e^{\pm i\lambda Q}$ for the field states conditioned on $X = \pm 1$. A flip of Z-diagonal states fails with probability $\epsilon = (1 + \mathrm{Re}\,\chi)/2$.

Theorem 6 (Representation defect of the field-coupled flip)**.** *Let the drive mode be in a normal state σ with mean occupation $N = \mathrm{tr}(\sigma b^\dagger b) < \infty$.*

1. *In the X eigenbasis, Φ^* multiplies the two off-diagonal entries by χ and $\overline{\chi}$, and $\mathcal{M}_\Phi = \{X\}'$, the algebra generated by X.*

2. *For every A with vanishing diagonal in the X basis, in particular Z and Y, $\delta(A) = (1 - |\chi|^2)\,\mathrm{diag}(|A_{-+}|^2, |A_{+-}|^2)$; thus $\delta(Z) = \delta(Y) = (1 - |\chi|^2)\mathbf{1}$.*

3. $1-|\chi|^2 \;\geq\; \dfrac{\arccos^2(\operatorname{Re}\chi)\,|\chi|^2}{(2N+1)^2}$.

4. $1-|\chi|^2 \geq D(\sigma_+,\sigma_-)^2$*, with equality when* σ *is pure, where* D *is the trace distance.*

Proof. Item 1 is Theorem 4 with eigenvalue differences ± 2 and Corollary 5. Item 2 follows from Eq. (4) by multiplying out 2×2 matrices.

Item 3 uses Paper B's two quadrature bounds, valid for a normal (density-operator) drive state of finite occupation and $[Q,P]=2i$ [26]: with $G(c)=\arccos^2 c$,

$$\langle Q^2\rangle \geq \frac{G(\operatorname{Re}\chi)}{4\lambda^2}, \qquad (\Delta P)^2 \geq \frac{4\lambda^2|\chi|^2}{1-|\chi|^2}.$$

The first is spectral Jensen; the second is the unitary uncertainty bound. Since $4N+2=\langle Q^2+P^2\rangle$,

$$\frac{4\lambda^2|\chi|^2}{1-|\chi|^2} \leq \langle P^2\rangle = 4N+2-\langle Q^2\rangle \leq 4N+2-\frac{G(\operatorname{Re}\chi)}{4\lambda^2}.$$

With $u=4\lambda^2$, $a=G(\operatorname{Re}\chi)$ and $A=4N+2$, this reads $1-|\chi|^2 \geq u^2|\chi|^2/(Au-a)$ for some $u>a/A$. The right side is minimized over u at $u=2a/A$, where it equals $4a|\chi|^2/A^2$.

Item 4. If $\sigma=|\psi\rangle\langle\psi|$, the conditioned states are pure with overlap $\langle e^{i\lambda Q}\psi, e^{-i\lambda Q}\psi\rangle=\chi$, and $D^2=1-|\chi|^2$. For mixed σ, applying $e^{\mp i\lambda Q}\otimes\mathbf{1}$ to a purification of σ gives purifications of $\sigma_\pm$ with overlap χ, so Uhlmann's theorem gives fidelity $F(\sigma_+,\sigma_-)\geq|\chi|$, and $D\leq(1-F^2)^{1/2}$ [32, 33]. □

Corollary 7 (The energy–accuracy bound as a representation statement)**.** *For a flip with failure probability* $\epsilon\in(0,\frac{1}{2})$ *and* $\theta_\epsilon=\arccos(2\epsilon-1)$*,*

$$N \;\geq\; \frac{\theta_\epsilon(1-2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}}-\frac{1}{2}. \tag{7}$$

Proof. $\operatorname{Re}\chi=2\epsilon-1$ gives $|\chi|^2\geq(1-2\epsilon)^2$, hence $1-|\chi|^2\leq 4\epsilon(1-\epsilon)$. Combining with item 3 of Theorem 6 gives $4\epsilon(1-\epsilon)\geq\theta_\epsilon^2(1-2\epsilon)^2/(2N+1)^2$. □

This is exactly the occupation bound of Paper B. Read in the present terms, an accurate flip requires the coupling to represent Z almost multiplicatively, and the defect $1-|\chi|^2$ cannot vanish at finite occupation. Item 4 identifies the defect with information: for pure drives it is the squared distinguishability of the two field states that record the value of X. The field is then a fragment in the sense of the HAFF paper on measurement, with conditional overlap $|\chi|$; that paper determines which pointer effects such a fragment can implement [28], and the same overlap fixes the representation defect here. The exactly represented algebra $\{X\}'$ is the pointer algebra of the decoherence literature [37]: the control interface transports exactly the part of the system that the environment records.

For the phase-fixed half-turn squeezed family, Paper B gives occupations 2.958, 7.330 and 24.329 at $\epsilon=.05,.01,.001$, against bound values 2.28, 6.74 and 23.80. Its separately optimized finite quadrature-grid values are variational minima, not certified global optima; the continuous supporting ground-state argument has its own assumptions. Mixtures of coherent states require occupation of order $\pi^2/(16\epsilon)$ [26].

4.1 Same clock as mode matching

The requirement that control and environment share a clock, so that their operational windows overlap, has a precise form in the field-coupled flip. The switch g defines a normalized mode function f and the mode operator $b_f = \int \overline{f(\omega)}\, a_\omega \, d\omega$ with $Q_g = \lambda(b_f + b_f^\dagger)$ [26]. A drive is prepared in some normalized mode f', by a source with its own clock.

Proposition 8 (Mode overlap)**.** *Let the drive occupy mode f' with mean occupation N', all modes orthogonal to f' being in the vacuum, and let $c = \int \overline{f(\omega)} f'(\omega)\, d\omega$. Then the bit channel equals the channel of Theorem 6 for the reduced state of mode f, which is the output of a pure-loss channel of transmissivity $|c|^2$ and has mean occupation $|c|^2 N'$. A flip with failure ϵ therefore needs*

$$N' \;\geq\; \frac{1}{|c|^2}\left[\frac{\theta_\epsilon(1-2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}\right]. \tag{8}$$

If $f'(\omega) = f(\omega)e^{i\omega\Delta}$, a copy of f delayed by Δ, then $c = \int |f(\omega)|^2 e^{i\omega\Delta} d\omega$. For a positive-frequency field this integral is over $\omega > 0$. Extending a narrowband Gaussian spectral density of width Ω to the full real line gives $|c| \simeq e^{-\Omega^2\Delta^2/2}$ and an occupation penalty $e^{\Omega^2\Delta^2}$ in the range where this approximation controls the overlap. The exact positive-frequency integral, not this Gaussian approximation, determines the cost at long delays.

Proof. Write $f = \bar{c} f' + r$ with $\int \overline{f'}\, r\, d\omega = 0$ and $\|r\|^2 = 1 - |c|^2$. Since $g \mapsto b_g$ is antilinear, $b_f = c\, b_{f'} + b_r$, where $[b_r, b_r^\dagger] = 1 - |c|^2$ and b_r commutes with $b_{f'}$ and $b_{f'}^\dagger$. The mode $r/\|r\|$ is orthogonal to f' and in the vacuum, independent of mode f', so the state of mode f is that of a beam splitter of transmissivity $|c|^2$ with vacuum in the other port, and its occupation is $|c|^2 N'$. Q_g involves only mode f, so the bit channel depends only on this reduced state, and Corollary 7 applies to it. Translation gives the exact Fourier overlap. The Gaussian specialization follows only after the stated extension of the frequency support. □

The prepared drive has energy $N'\bar{\omega}'$, while only the fraction $|c|^2$ of its occupation reaches the selected mode. This is a charged preparation resource, not automatically dissipated heat. The source/switch clock offset therefore enters Paper B's feedback acceptance as a mode-matching loss.

5 A representation generated by propagation

The situation is different when the "environment" is part of the same closed dynamics. In the local propagation model of Paper A, a particle with internal space $\mathbb{C}^s$ moves on a lattice Λ by a unitary step $W = \sum_{h \in N} T_h \otimes A_h$, where T_h translates by h [25]. The internal algebra is represented by

$$\phi : M_s(\mathbb{C}) \to B\big(\ell^2(\Lambda) \otimes \mathbb{C}^s\big), \qquad \phi(A) = \mathbf{1} \otimes A, \tag{9}$$

a faithful unital $*$-homomorphism, and $[\phi(A), W] = \sum_h T_h \otimes [A, A_h]$ is nonzero unless A commutes with every A_h. The requirement of Definition 1 together with nontrivial coupling is met exactly, because nothing is traced out: the translational degrees of freedom and the internal ones evolve under one unitary. The contrast with Section 4 is the content of

Corollary 5: discarding a non-atomic port restricts exact multiplicativity to the coupling commutant.

The propagation also supplies the quadratic-space morphism that Section 2 found missing. Near a band point the step has a first-order symbol $H(q) = \sum_i q_i\alpha_i + \mu\beta$ with Hermitian coefficients [25].

Proposition 9 (The propagation symbol as a quadratic-space morphism)**.** *Let $j(q) = \sum_i q_i\alpha_i$ with $\alpha_1, \dots, \alpha_d$ linearly independent. Then $j(q)^2 = Q(q)\mathbf{1}$ for a quadratic form Q if and only if, for $\mu = 0$, every eigenvalue of $H(q)$ lies on one shell $\omega^2 = Q(q)$. In that case Q is positive definite, j is an isometry from $(\mathbb{R}^d, Q)$ into* $\mathrm{Herm}(s)$ *with the normalized Hilbert–Schmidt form, and j extends to a unital $*$-homomorphism $Cl(\mathbb{R}^d, Q) \to M_s(\mathbb{C})$. If moreover $\mu \neq 0$, $\beta^2 = \mathbf{1}$ and $\{\alpha_i, \beta\} = 0$, then*

$$J(\omega, q) = \omega\beta - \sum_i q_i\beta\alpha_i \tag{10}$$

satisfies $J(\omega, q)^2 = (\omega^2 - Q(q))\mathbf{1}$ and extends to a unital homomorphism $Cl(\mathbb{R}^{1+d}, \eta) \to M_s(\mathbb{C})$ with $\eta(\omega, q) = \omega^2 - Q(q)$, which preserves the involution in which β is Hermitian and the $\beta\alpha_i$ are anti-Hermitian.

Proof. The first statement is the single-shell theorem of Paper A [25] with $\mu = 0$; isometry follows from $\frac{1}{s}\operatorname{tr} j(q)^2 = Q(q)$, and the extension from the universal property of Clifford algebras. For J, expand J^2 using $\beta^2 = \mathbf{1}$ and $\beta\alpha_i\beta = -\alpha_i$: the cross terms $\omega q_i\{\beta, \beta\alpha_i\}$ vanish and $\sum q_i q_j \frac{1}{2}\{\beta\alpha_i, \beta\alpha_j\} = -Q(q)$. Hermiticity of β and anti-Hermiticity of $\beta\alpha_i = -\alpha_i\beta$ follow from $(\beta\alpha_i)^* = \alpha_i\beta$. □

J is the map $j : (V, q) \to (W, Q)$ of Section 2 with $(V, q) = (\mathbb{R}^{1+d}, \eta)$, a Lorentzian quadratic space, and W the real span of β and the $\beta\alpha_i$ in $M_s(\mathbb{C})$, with the form defined by $Y^2 = Q_W(Y)\mathbf{1}$. The spatial scalar-square condition is equivalent to the specified single-shell property, so no internal band has a different squared dispersion. The Lorentzian extension additionally uses the time parameter and the mass anticommutator. A massless one-dimensional carrier can have only one chiral branch; a full two-sided cone requires both frequency signs, as Q-RAIF A explains [25].

The canonical inclusion ϕ remains exact on the lattice. It is the linear quadratic-space description of its propagation symbol that generally holds only at leading order. For $s = 2$ and $W(k) \in SU(2)$, the effective generator $H_{\mathrm{eff}}(k) = (i/\tau)\operatorname{Log} W(k)$ is traceless, so $H_{\mathrm{eff}}(k)^2 = \omega(k)^2\mathbf{1}$ holds at every k; what fails is linearity and isotropy of $k \mapsto H_{\mathrm{eff}}(k)$. In the Weyl automaton of Paper A, $H_{\mathrm{eff}}(k) = c\,\sigma \cdot k$ up to corrections of relative order $|k|\ell$, which include a parity-odd anisotropic term. In the split-step walk the quadratic identity $\omega^2 = c_{\mathrm{eff}}^2 q^2 + \mu^2$ fails at order $(qa)^4$, with coefficient $-\theta^2/45 + O(\theta^4)$ in lattice units [27]. These orders are the precise sense in which a lattice propagation gives an approximate representation.

6 Calibrated work and exact access

The propagation symbol and an actuator can use the same internal generators without being the same physical system. Q-RAIF B supplies a specific way to test their alignment [26].

Let $A(a) = \sum_i a_i A_i$ be a fixed real linear family of dimensionless Hermitian operators. During an encounter the command a is held fixed, the carrier free Hamiltonian is zero or commutes with $A(a)$, and a nonzero real smooth compact switch g couples it to one chiral current as $g(t)A(a)j(t)$. The field input is independent, centered and of finite energy. With $\kappa_g = \frac{1}{4}\int g'(u)^2 du > 0$, the same encounter has

$$W_\rho(a) = \kappa_g \operatorname{tr}[\rho A(a)^2]. \tag{11}$$

This is field energy delivered by switching, not a total-device heat charge. A nonzero commuting carrier Hamiltonian has no energy change during this encounter; cancelling a non-commuting drift is an additional operation.

The charge is independent of every carrier state ρ for every command precisely when $A(a)^2 = Q_{\mathrm{ctrl}}(a)I$. Polarization gives $\{A_i, A_j\} = 2(Q_{\mathrm{ctrl}})_{ij} I$. For an injective family Q_{ctrl} is positive definite. The implication is from an all-state charging requirement, not from persistence, reachability or positivity of an average cost. The field shift is $j(u) \mapsto j(u) + A(a)g'(u)/2$. The second Magnus term can be a system operation $e^{i\vartheta A(a)^2}$ rather than a global phase; it commutes with the field and does not change this work calculation [26].

To relate it to propagation, write the spatial symbol as $j_{\mathrm{prop}}(p) = \sum_i p_i \alpha_i$, with energy units and $j_{\mathrm{prop}}(p)^2 = Q_{\mathrm{prop}}(p)I$. Let C be the measured map from commands to momenta and $E_* > 0$ an energy scale. The actual coupling must satisfy

$$A(a) = j_{\mathrm{prop}}(Ca)/E_*.$$

The equality specifies a physical alignment. It cannot be inferred by renaming two positive forms.

Corollary 10 (Aligned charging and the reduced exact algebra)**.** *Under these encounter and alignment assumptions,*

$$W(a) = a^T K_{\mathrm{op}} a, \qquad K_{\mathrm{op}} = \frac{\kappa_g}{E_*^2} C^T Q_{\mathrm{prop}} C. \tag{12}$$

For a normal field input and a discarded field port, the encounter's multiplicative domain is nevertheless

$$\mathcal{M}_a = \{j_{\mathrm{prop}}(Ca)\}'. \tag{13}$$

If C is invertible, the common exact algebra for all commands is $\bigcap_a \mathcal{M}_a = \{\alpha_1, \ldots, \alpha_d\}'$. For an irreducible complex matrix representation this common algebra is scalar.

Proof. The scalar-square identity inserted in Eq. (11) gives Eq. (12), the aligned-work result of Q-RAIF B. The nonzero switched current is a field quadrature with non-atomic spectral distribution in every normal input. Corollary 5 applies with $X = A(a)$ and $E = Q_g$. The commuting Magnus factor is scalar for this aligned family, so it does not change the domain. Commuting free carrier evolution preserves the same block algebra. Commuting with every linear combination is equivalent to commuting with each generator when C is invertible. The last assertion is the matrix commutant criterion for irreducibility. □

The corollary combines the charging result of Q-RAIF B with the multiplicative-domain calculation; it is not a further selection principle for Clifford algebras. For one command the domain retains full degenerate eigenspace blocks and can be noncommutative. Over all commands it can become much smaller. Exact closed-carrier representation, state-independent work and exact reduced control are therefore different properties.

Two failures keep the calibration operational. For $A(a) = a_1 Z + a_2 I$, the maximally mixed state costs $\kappa_g(a_1^2 + a_2^2)$, but the two Z eigenstates cost $\kappa_g(a_1 \pm a_2)^2$; one average-state test does not establish the all-state condition. For $j_{\mathrm{prop}}(p) = p_1 X + p_2 Z$ and $C = \mathrm{diag}(1, 2)$ in units $E_* = 1$, the unchanged Euclidean propagation shell gives $K_{\mathrm{op}}/\kappa_g = \mathrm{diag}(1, 4)$. Without C, work does not identify the propagation form.

The time direction and mass anticommutator remain the additional conditions of Proposition 9. In $\omega^2 - Q_{\mathrm{prop}}(p)$, ω is conjugate to the supplied evolution time. An independent, noncentered drive or a correlated input has a stress cross term $g'(u)\langle A(a)j(u)\rangle$; a coherent gate cannot inherit the centered calibration price for all states. Nor does the calibration construct a shared propagation medium, pause its noncommuting drift for free, or maintain its controls indefinitely.

7 Discarded ports and coherent return

A discarded record port and a retained environment permit different controls. Write $\chi(t) = \mathrm{Tr}(\sigma_E e^{-itE})$ for the port's spectral characteristic function. Section 3 derives the single-coupling channel and its multiplicative domain; the following bound survives arbitrary later controls that cannot reach the lost port.

Proposition 11 (A fresh port leaves an irreversible defect)**.** *Suppose a finite-dimensional system first undergoes ideal unitary controls and then one coupling $e^{-iX\otimes E}$ to an initially independent port, with X having at least two distinct eigenvalues and the environment's spectral distribution non-atomic. Let F collect every auxiliary port accessible later; just before this coupling the joint state is $\rho_S \otimes \sigma_E \otimes \tau_F$ for every system input. If E is thereafter inaccessible, no subsequent system channel can make the complete channel unitary on the full system algebra. More precisely, let Λ include all the preceding and subsequent controls, and let $\mathcal{V}$ be any unitary channel. For every pair of distinct eigenvalues x, x',*

$$\frac{1}{2}\|\Lambda - \mathcal{V}\|_\diamond \;\geq\; \frac{1 - |\chi(x' - x)|}{2}. \tag{14}$$

Proof. Choose eigenvectors $|x\rangle, |x'\rangle$ and the orthogonal states $|\pm\rangle = (|x\rangle \pm |x'\rangle)/\sqrt{2}$. Their trace distance is one before the coupling and $|\chi(x' - x)| < 1$ afterwards. All subsequent operations induce a system channel and cannot increase this distance. Pull the two inputs back through the preceding unitaries and call them $\rho_\pm$. If $e_\pm = D(\Lambda\rho_\pm, \mathcal{V}\rho_\pm)$, the triangle inequality gives $1 \leq e_+ + |\chi(x' - x)| + e_-$. Thus $\max e_\pm \geq (1 - |\chi|)/2$, which also bounds half the diamond norm from below. □

The independence concerns the entire later-accessible auxiliary system jointly, including any purification of the discarded port. A later register correlated with that port can reveal which phase was applied and allow its reversal; the system's reduced channel alone then

does not describe the available recovery operation. For a qubit and no later noise, the bound is attained: remove the phase of χ, and the remaining dephasing is the mixture $(1+|\chi|)\mathcal{I}/2 + (1-|\chi|)\mathcal{Z}/2$. Its half diamond distance from $\mathcal{I}$ is $(1-|\chi|)/2$.

Proposition 12 (An exact interface from a retained oscillator)**.** *Let $Q = b+b^\dagger$, $P = i(b^\dagger - b)$ satisfy the Weyl relations, so $[Q,P] = 2i$, and let X be Hermitian with finite spectrum. Then*

$$e^{-i\lambda XQ}e^{-i\mu XP}e^{i\lambda XQ}e^{i\mu XP} = e^{-2i\lambda\mu X^2} \otimes I. \tag{15}$$

Every initial oscillator state is returned exactly. The induced system map has multiplicative domain $M_s(\mathbb{C})$, including operators that do not commute with the coupling observable X.

Proof. In an X eigenblock, the two pulse generators have scalar commutator

$$[-i\lambda xQ, -i\mu xP] = -2i\lambda\mu x^2.$$

The Weyl group commutator is therefore that scalar phase. Summing the spectral blocks gives (15). □

For a qubit choose $X = (I+\sigma_x)/2$. With $\lambda\mu = \pi/2$, the loop is $e^{-i\pi X} = -\sigma_x$: an exact corrective flip, even with a vacuum or thermal oscillator input. The intermediate system–oscillator correlations are returned, not discarded. Conditional displacement loops of this kind are established geometric gates [24]; the point here is their implication for the interface. Continuous spectrum alone cannot prohibit exact control of a retained port.

Corollary 13 (One retained oscillator gives universal qubit control)**.** *Suppose the four pulses of Proposition 12 can be applied through either $P_z = (I+\sigma_z)/2$ or $P_x = (I+\sigma_x)/2$, with arbitrary real pulse areas. Every qubit unitary, up to global phase, is a sequence of at most three closed loops, and the oscillator returns after each loop.*

Proof. A loop with $2\lambda\mu = \alpha$ implements $e^{-i\alpha P_z} = e^{-i\alpha/2}e^{-i\alpha\sigma_z/2}$, and similarly for P_x. The z–x–z Euler decomposition of a qubit unitary therefore gives three loops. For example the three angles $\pi/2, \pi/2, \pi/2$ give a Hadamard up to global phase. Each identity holds on the full system–oscillator tensor product, so composition returns even an oscillator initially correlated with another system. □

For a larger system the same construction supplies the one-parameter groups e^{-itP_j}. Full unitary control follows when the real Lie algebra generated by iP_j is $\mathfrak{u}(s)$; generation of the associative matrix algebra alone is not the required controllability condition. The pulse permissions, including the choice between noncommuting projectors, remain physical inputs.

The loop is not costless. For a centred input and a mode of frequency ω, the largest conditional occupation increase along the four corners is $(\lambda^2+\mu^2)x^2$. Minimizing it for the qubit flip gives $\lambda = \mu = \sqrt{\pi/2}$ and peak mean stored energy $\pi\omega\langle X\rangle \le \pi\omega$. The mode's final energy is unchanged; an ideal reversible switch can recover the intermediate work. This is a stored-energy statement, not a total-device dissipation estimate. It assumes access to both quadratures, their controlled pulse order, and coherent retention until closure. A travelling field that has left the apparatus need not supply these permissions. An implementation with smooth local spacetime couplings must establish them separately.

Thus a record carried away by an inaccessible port constrains future control, while a coherently returned record can disappear without thermal erasure. Retaining an interface can replace a preparation-accuracy requirement by a closure requirement. The oscillator, its routing, clock and switching controls are part of that retained interface; they are not supplied by an abstract algebra homomorphism.

8 What must be retained by a material interface

The representation results answer an algebraic question about a specified interaction. A material interface must additionally let an acquired record change a later operation on the same body. The source then belongs to that joint dynamics; it is not assigned from the program's name.

For the next construction, fix a finite Dirichlet interval and retain all modes of the positive Gaussian operator K built in Appendix A.1. Its energy form contains the complete carrier square $\|\phi' + a\Theta\|^2$, the material gradient and a strictly positive material mass matrix. The interval gives a spectral gap. The field Hamiltonian is $H_F = d\Gamma(\sqrt{K})$ with one fixed vacuum subtraction. Projecting onto the bound material mode does not remove the orthogonal modes: their reaction is explicitly bounded in Proposition 23. The charged regulator in that appendix is a separate finite-resolution construction, not a proof that this static Gaussian realization equals the nonlinear charged continuum.

The following section keeps this complete K, the finite controller, its records and all old references. Completing the controlled field squares gives a positive Hamiltonian and a physical rms bound. The operator retarded equation makes acquisition affect the force, the actual field affects another sensor, and its new record can control the next force. The smooth interacting-source estimate retains both cross terms and connected fluctuations.

Receiver renewal has a narrower but attainable target. A prepared matched-gap Gibbs cell can replace one force-idle receiver, moving its old correlations into the spent bank. The material and the bank's purifier remain present. Section 10 gives this certificate and compiles a specified finite sequence on one cursor. Renewing the entire interacting equilibrium while preserving arbitrary held quantum content is a different task, with a product-extension obstruction.

The longer implementations establish additional permissions: full-ring reset and spectral sorting (Appendix C), a spatial work field and a separate massive-body gate (Appendix D), a guarded battery (Appendix E), local export (Appendix F), thermal preparation (Appendix G), and contact reaction/recoil (Appendix H). Their lattice, ring, copropagating and bidirectional carriers have different supports and Hamiltonians. Their estimates must be matched before they describe one apparatus.

9 Acquired quantum records on the complete material

The force calculation in Appendix A.2 conserves a classical record during its drive. An acquired record need not be classical in the full old history. Keep the complete finite-interval K and $H_F = d\Gamma(\omega)$, $\omega = \sqrt{K}$, of Section A.3. Let the finite controller include probes,

record cells and program cursor. Its nonnegative Hamiltonian H_D need not commute with the force amplitudes. The following construction therefore includes acquisition and a later record-controlled response within one autonomous dynamics.

9.1 A positive common Hamiltonian and a uniform physical moment

Let B_a be finitely many commuting bounded self-adjoint controller operators and h_a real smooth compact material smearings. Define

$$F = \sum_a B_a h_a, \qquad C_{\min} = \frac{1}{2}\sum_{ab} B_a B_b \langle h_a, K^{-1} h_b\rangle,$$
$$H_{\rm tot} = H_D + H_F - \sum_a B_a Q(h_a) + C_{\min}. \tag{16}$$

The full canonical material fields obey $[Q(h), P(k)] = i\langle h, k\rangle$, with $Q(h) = [a(\omega^{-1/2}h) + a^*(\omega^{-1/2}h)]/\sqrt{2}$. This is the canonical normalization of the material Hamiltonian, distinct from the earlier dimensionless $[b + b^\dagger, i(b^\dagger - b)] = 2i$ convention. The amplitudes commute at equal Schrödinger time, not necessarily at different Heisenberg times. In particular a cursor-controlled record force and a probe force can commute while acquisition edges in H_D change those same records.

Proposition 14 (Positive quantum interface and physical rms bound)**.** *The Hamiltonian* (16) *is self-adjoint and nonnegative. Its domain is equivalent to* $\operatorname{Dom} H_F$*. For a normalized initial vector* ψ *in that domain, including all old references, put*

$$L_F = \sqrt{2}\sum_a \|B_a\|\|\omega^{-1}h_a\|, \qquad c_F = \frac{1}{\sqrt{2}}\sum_a \|B_a\|\|\omega^{-1/2}h_a\|,$$
$$A_F = \|H_{\rm tot}\psi\| + \|H_D\| + \|C_{\min}\| + c_F, \qquad M_F = \left[\frac{L_F + \sqrt{L_F^2 + 4A_F}}{2}\right]^2.$$

Then $\|H_F e^{-itH_{\rm tot}}\psi\| \le M_F$ *for every* t*. If the initial cursor lies in a force-idle region and* $\|H_D\|$*, coupling norms and initial physical rms stay fixed, this certificate does not grow with the finite program length.*

Proof. Resolve the common spectrum of the B_a into projections P_b and write $f_b = \sum_a b_a h_a$, $z_b = -\omega^{-3/2} f_b/\sqrt{2}$. The finite interval has a gap; hence the finite controlled Weyl unitary $D = \sum_b P_b \otimes W(z_b)$ exists and $D^* H_F D = H_F - \sum B_a Q(h_a) + C_{\min}$. It follows that $H_{\rm tot} = D^* H_F D + H_D \ge 0$. The linear field bound

$$\|Q(h)\zeta\| \le \sqrt{2}\|\omega^{-1}h\|\|H_F^{1/2}\zeta\| + \|\omega^{-1/2}h\|\|\zeta\|/\sqrt{2}$$

also makes the force infinitesimally H_F-bounded, giving the graph domain statement by the closed operator realization. For $\psi_t = e^{-itH_{\rm tot}}\psi$ and $x = \|H_F\psi_t\|$, conservation of $\|H_{\rm tot}\psi_t\|$ gives $x \le A_F + L_F\sqrt{x}$, since $\langle H_F\rangle_t \le x$. Solving this quadratic inequality proves the bound. Spectral-core approximation extends the calculation to the stated domain. In a force-idle initial region, $\|H_{\rm tot}\psi\| \le \|H_F\psi\| + \|H_D\|$. Every step tensors with the actual reference identity. □

The displacement D proves positivity; it is not a free preparation of a dressed vacuum. This finite-interval argument is not an infinite-line static-dressing theorem. A positive nearest-neighbor Feynman cursor with bounded edge gates can have $\|H_D\| \le 4J + \|H_{\mathrm{data}}\|$ independently of its site count [4]. This removes a program-length factor from the physical moment bound, not from cursor space, execution time or preparation cost. For the uniform-J cursor, Foundation III's lemma "Wholly idle input and a finite completion window" in the appendix "A Finite Arrival Bound at Fixed Hopping Energy" bounds arrival for a packet supported wholly in the input idle region of the pure gate-dressed hopping comparison [50]. Cell gaps and forces require the subsequent Duhamel control; this is neither the spin-clock nor the spatial work-clock estimate below. Linear finite-system/field Hamiltonians and their domain analysis are established tools [5]; the issue here is their common retained-record interface.

The field obeys the operator retarded equation

$$Q(t) = \cos(\omega t)Q(0) + \omega^{-1}\sin(\omega t)P(0) + \int_0^t \omega^{-1}\sin[\omega(t-s)]F(s)\,ds. \tag{17}$$

It follows by solving $\dot{Q} = P$, $\dot{P} = -KQ + F(t)$ as weak smeared identities on the energy core. The $B_a(s)$ in $F(s)$ are the actual joint Heisenberg operators. Their expectations are not an externally supplied force history. With material-only force, the exterior carrier formula of Section A.2 holds with the complete operator $\Delta\Theta$:

$$\Delta\phi_R(u) = -\tfrac{1}{2}\int a(y)\Delta\Theta(y, u+y)\,dy, \qquad \Delta\phi_L(v) = \tfrac{1}{2}\int a(y)\Delta\Theta(y, v-y)\,dy.$$

An additional carrier-smeared probe contributes its own retarded source. Quadratic stress uses correlations of these operators, not only the square of their mean.

9.2 Actual acquisition, sensing and the next action

For a probe P and a blank record R, the unitary $C_Y = P_-^Y \otimes I_R + P_+^Y \otimes X_R$ acquires the probe's Y reading. Retain P, R and their old references. During a subsequent block that conserves R, the force of Section A.2 acts as $V = \sum_{r=0,1} |r\rangle\langle r| \otimes e^{ir^2\Phi_h} W(rz_h)$. For actual initial material/reference blocks $\sigma_{rr'}$, the output is

$$\sum_{rr'} |r\rangle\langle r'| \otimes e^{i(r^2 - r'^2)\Phi_h} W(rz_h)\sigma_{rr'}W(r'z_h)^*. \tag{18}$$

This is not a sampled classical force. For a pure complete input with both record branches present, dephasing R changes the full retained state by half trace distance $\sqrt{p(1-p)}$, where $p = \Pr(R=1)$. Indeed the deleted off-diagonal block connects two orthogonal vectors of norms $\sqrt{p}, \sqrt{1-p}$ and has two eigenvalues $\pm\sqrt{p(1-p)}$.

Now let a fresh sensor S start in $|+x\rangle$, couple by $e^{igZ_S Q(b)}$ to the material after its actual propagation, and acquire its Y value into a cell E. The latter starts with thermal occupation ν, independently prepared with its purifier retained. For a centered Gaussian

material initially independent of the first probe, let μ_r be the actual conditional mean of $Q(b)$ and V_b its conditional variance after the force. Then

$$\Pr(E = 1 \mid R = r) = \frac{1}{2} - \frac{1 - 2\nu}{2} e^{-2g^2 V_b} \sin(2g\mu_r). \tag{19}$$

The sensor's Y mean is $-\operatorname{Im}\langle e^{2igQ(b)}\rangle$; the Gaussian characteristic function and the mixed cell's visibility $1-2\nu$ give the formula. The acquisition acts on the same disturbed material, and the actually acquired n_E can control the next force. Its energy and source are part of that dynamics. For a correlated or non-Gaussian input use its actual characteristic function, not this Gaussian specialization.

Before sensing, a conserved source record and centered product input give $\langle \Delta j\rangle = p j_h$ but $\Delta\langle : j^2 : \rangle = p j_h^2$, where j_h is the complete driven carrier current. The connected contribution is $p(1-p)j_h^2$, not an extra energy charge. Similarly the full added material/carrier energy is pE_h, of which $p^2 E_h$ is the energy of the mean displacement. Record dephasing leaves this immediate field marginal unchanged while changing the old joint history. Equal mean source therefore does not establish equal later control.

The formulas describe the sequential block. Under the autonomous Hamiltonian (16), acquisition, force and sensing instead occupy cursor regions that a dispersive cursor can revisit. Its deadline statistics require the actual joint evolution, without postselection or assignment of the sequential probabilities.

A one-mode check retains the old probe reference and the second cell's purifier: $p = .2766071$, $\Pr(E = 1 \mid R = 1) = .4267590$, dephasing distance .4473205, and first-acquisition probe/reference change below 4.1×10^{-16}. Cutoffs $6, 10, 14$ reduce Gaussian-law error from 1.04×10^{-7} to 1.11×10^{-16}. Field energy .0418368 splits into mean .0115724 and connected .0302645; sensing adds $.04805 = g^2/2$ at $g = .31$. This checks the sequence, not the full spatial K theorem.

A separate 31-site positive cursor diagnostic executes actual first acquisition, record force, sensing, second acquisition, receiver/bank SWAP and a later force controlled by the spent cell's actual record. With both cell purifiers retained, its unpostselected deadline $t = 40$ has field energy .0300724, physical rms at most .186211 against the uniform bound 3.01330, and energy residual below 1.9×10^{-13}. But only .857524 of its cursor probability is after the SWAP region; its receiver occupation is .157712, exceeding the prepared bank value .13 by .027712. Unfinished and returning cursor branches retain pre-SWAP data and can revisit acquisition and force regions. The program therefore fails an exact deadline-renewal claim. The force-idle window condition and its completion error must be certified separately, as done below; the sequential probability formula is not assigned to this autonomous output.

9.3 The local field source during interaction

The free-output qualification in Corollary 26 is unnecessary for a specified smooth test under the same H_{tot}. Let $B(x)$ be a real symmetric smooth matrix, supported strictly inside the Dirichlet interval, with $\|B\|_\infty = G$; let $w \in C_c^\infty$ be real and supported in $[0, T]$. Use the same free-K vacuum subtraction and energy factors $\mathcal{Y} = (P, L_s Q)$, $L_s^* L_s = K$. Define

$$O_{\mathrm{int}} = \int w(t) \int [\mathcal{Y}(t,x)^T B(x) \mathcal{Y}(t,x)/2 - \text{free vacuum}_K]\, dx\, dt.$$

This is the material/carrier field source, not the controller's stress. By (17),

$$\mathcal{Y}(t) = \mathcal{Y}_0(t) + \Delta\mathcal{Y}(t), \qquad \Delta\mathcal{Y}(t) = \sum_a \int_0^t M_{t-s} h_a B_a(s)\, ds,$$

where $M_\tau h = (\cos(\omega\tau)h, L_s\omega^{-1}\sin(\omega\tau)h)$ and $\|M_\tau h\| = \|h\|$. As a vector of joint-state operators, $\|\Delta\mathcal{Y}(t)\psi\| \le W(t)\|\psi\|$, $W(t) = t\sum_a \|B_a\|\|h_a\|$.

Proposition 15 (An interacting-window field-source bound)**.** *Assume smooth coefficients of K and the smooth compact h_a above. For each a, define finite constants*

$$D_{P,a} = \sup_{0\le\tau\le T} \|\omega^2 [B M_\tau h_a]_P\|,$$
$$D_{S,a} = \sup_{0\le\tau\le T} \|\omega^2 L_s^* [B M_\tau h_a]_S\|.$$

With $\omega_i \ge c_- i$ as in Section A.3, set

$$C_0(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\|[D_{P,a} c_-^{-3/2}\zeta(3/2) + D_{S,a} c_-^{-5/2}\zeta(5/2)],$$
$$C_1(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\|[D_{P,a} c_-^{-2}\zeta(2) + D_{S,a} c_-^{-3}\zeta(3)].$$

For any $\Lambda > 0$, O_{int} is a Hermitian form with

$$\|(H_F + \Lambda)^{-1/2} O_{\mathrm{int}} (H_F + \Lambda)^{-1/2}\| \le C_{\mathrm{int}}, \tag{20}$$

where

$$C_{\mathrm{int}} = C_O + \int |w(t)| \left[\frac{C_1(t)}{\sqrt{\Lambda}} + \frac{C_0(t) + GW(t)^2/2}{\Lambda}\right] dt.$$

Here C_O is the free source constant with time test w. All references and the finite controller are retained. For two initial joint vectors of rms at most M_F and vector distance e, their actual interacting-window source difference is at most

$$2C_{\mathrm{int}}\sqrt{\Lambda e^2 + 2M_F e}\sqrt{\Lambda + M_F}. \tag{21}$$

Proof. The free part is Proposition 25. The response-square term is bounded by $GW(t)^2/2$ without assuming commuting response components. For the symmetrized cross term, let $v_i(t,x)$ be the free annihilation coefficient of mode i and $C_i(t) = \int v_i(t,x)^T B(x)\Delta\mathcal{Y}(t,x)\, dx$. Projection of the two response components onto e_i and insertion of ω^2 give

$$\|C_i(t)\| \le \frac{t}{\sqrt{2}} \sum_a \|B_a\|[D_{P,a}\omega_i^{-3/2} + D_{S,a}\omega_i^{-5/2}].$$

Interior multiplication by smooth B removes boundary compatibility conditions. Iterating the elliptic operator $K = -I\partial_x^2 +$ smooth lower-order terms controls the necessary Sobolev orders of propagated compact h_a, proving finiteness of the displayed graph norms uniformly over the compact time interval. Thus $\sum_i \|C_i(t)\| \le C_0(t)$ and $\sum_i \|C_i(t)\|/\sqrt{\omega_i} \le C_1(t)$.

For $E = \langle H_F\rangle_\psi$, $\|a_i\psi\| \le \sqrt{E/\omega_i}$ and $\|a_i^*\psi\| \le \sqrt{E/\omega_i} + 1$. Bounding $\langle a_i C_i\rangle$ by $\|a_i^*\psi\|\|C_i\psi\|$, and $\langle a_i^* C_i^*\rangle$ by $\|a_i\psi\|\|C_i^*\psi\|$, bounds the cross expectation by $2C_1(t)\sqrt{E}+C_0(t)$. No commutation between C_i and a_i has been used. The C_0 term is the creation contribution; treating the record as a c-number would incorrectly remove it. Apply $2\sqrt{E} \le (E+\Lambda)/\sqrt{\Lambda}$ and add the free and response bounds. Absolute convergence of the cross terms, the bounded response vector and the established free form give spectral-core convergence. Polarization proves (20). The weighted-form difference estimate of Corollary 26, with $\|H_F(\psi-\varphi)\| \le 2M_F$, proves (21). □

The constants do not depend on cursor site count if its coupling norms stay fixed. They do depend on the box, smooth test, profiles and time window. Rough spatial multipliers, boundary distributions and unsmeared stress are not covered. This result compares retained inputs under one common $H_{\rm tot}$; it does not compare arbitrary different apparatus Hamiltonians or supply the controller's missing stress.

9.4 A local contact energy and the interacting thermal reference

The inverse-K counterterm in (16) is useful for proof but is not a local physical density. For smooth compact material profiles $w_a(x)$ choose $h_a = (0, M(x)w_a(x))$ and replace it by

$$C_{\rm loc} = \frac{1}{2}\sum_{ab} B_a B_b \int w_a^T M w_b \, dx.$$

The actual mass-potential density becomes

$$V_{\rm loc}(x) = \tfrac{1}{2}[u - \sum_a B_a w_a]^T M [u - \sum_a B_a w_a]. \tag{22}$$

All gradient and current squares of the complete material remain. For each joint spectral branch $w = \sum_a b_a w_a$, the inverse-form variational formula gives

$$\langle (0,Mw), K^{-1}(0,Mw)\rangle = \sup_X \{2\langle w, Mu\rangle - q[X]\} \le \langle w, Mw\rangle,$$

because $q[X] \ge \langle u, Mu\rangle$. Consequently $C_{\rm loc} - C_{\rm min} \ge 0$ is a bounded controller operator. The actual local-contact Hamiltonian is $D^* H_F D + H'_D$ with $H'_D = H_D + C_{\rm loc} - C_{\rm min} \ge 0$. The positivity, rms and source proofs therefore apply to this physical contact, using its actual $C_{\rm loc}$ and the adjusted controller norm. No local stress is assigned to a nonlocal inverse operator.

Proposition 16 (Interacting equilibrium and retained record capacity)**.** *For either contact, put $d = \dim \mathcal{H}_D$, $d_0 = \|H'_D\|$, and $Z_F(\beta) = \operatorname{Tr} e^{-\beta H_F}$. Then the actual interacting Gibbs state exists and*

$$\begin{aligned} d e^{-\beta d_0} Z_F(\beta) \le Z_{\rm tot}(\beta) \le d Z_F(\beta), \\ \langle H_{\rm tot}^2\rangle_\beta \le \frac{16 e^{\beta d_0}}{e^2\beta^2} \frac{Z_F(\beta/2)}{Z_F(\beta)}. \end{aligned} \tag{23}$$

If independent uniform input bits $X_1, \dots, X_N$ are retained in the accessible controller and material with field mean energy at most M, and each is later readable with error $\epsilon_j \le 1/2$, then

$$\sum_j [\log 2 - h_2(\epsilon_j)] \le \log d + \inf_{\beta>0} [\beta M + \log Z_F(\beta)]. \tag{24}$$

Any decoder-accessible outside reference is an additional resource.

Proof. The eigenvalues of $D^* H_F D + H'_D$ lie between those of $D^* H_F D$ and those eigenvalues plus d_0, by min–max. Summing scalar exponentials proves the partition bounds; operator monotonicity of the exponential is neither available nor needed. For $x \ge 0$, $x^2 e^{-\beta x/2} \le 16/(e^2\beta^2)$. The partition ratio then proves the moment bound; its controller multiplicity cancels. Subadditivity and the field Gibbs variational inequality give $S(\rho) \le \log d + \beta M + \log Z_F(\beta)$. Binary Fano, conditional entropy subadditivity for the independent classical inputs, and data processing give $\sum_j [\log 2 - h_2(\epsilon_j)] \le I(X^N : \text{apparatus}) \le S(\rho)$. Take the infimum. The frequency bound also yields $\log Z_F(\beta) \le q/(1-q)^2$, $q = e^{-\beta c_-}$, so the capacity is finite for fixed resources. □

The same-H Gibbs purification is a graph-domain input, not the product Gaussian in (19). Stationary equilibrium supplies neither directed cursor nor blank records; $\beta^{-1} D(\rho_{\text{prep}} \| \rho_\beta)$ is a finite running state's free-energy excess, not its preparation protocol. At fixed energy, cursor dimension increases record capacity only logarithmically. Any outside history bank or entropy channel remains a resource.

10 Finite renewal and continuing return

10.1 Replacing a port is not renewing the interacting material

There are three distinct thermal operations below: replacement of a specified finite port; local thermal restriction on a free-carrier window; and preparation of the whole interacting Gibbs state. Only the first two have the stated finite operating protocols.

Proposition 17 (Universal held-content preservation forces a product extension)**.** *Let a channel $\mathcal{E}$ append a finite-dimensional port B to a held d-dimensional system A, whose marginal is the identity channel on every input, including all reference extensions. Then $\mathcal{E}(\rho) = \rho \otimes \tau$ for one fixed port state τ. Thus such a universal nondisturbing preparation cannot reproduce a correlated interacting Gibbs state. For $H = J Z_A Z_B$, $x = \beta J$, its Gibbs state is $\gamma = (I - \tanh(x) Z_A Z_B)/4$. At held input $I/2$ every such extension has half trace distance from γ at least $|\tanh x|/2$, attained by $\tau = I/2$.*

If instead only the classical Z_A label is conserved while conditional port Gibbs states $\tau_\pm = (I \mp \tanh(x) Z_B)/2$ are installed, the largest retained coherence factor is $\operatorname{sech} x$*. After the best held-system phase correction, an input $|+\rangle$ therefore suffers half trace distance at least $(1 - \operatorname{sech} x)/2$.*

Proof. Dilate the channel and purify its fixed ancilla. A pure held input must have a pure unchanged held marginal; hence the isometry sends $|\psi\rangle$ to $|\psi\rangle|e_\psi\rangle$. Applying this to basis

states and their superpositions forces one common environment vector. Tracing the environment gives the product extension. Equivalently the identity channel has rank-one Choi state. For the Gibbs example the observable $Z_A Z_B$ has norm one, expectation zero in $(I/2)\otimes\tau$ and $-\tanh x$ in γ, giving the distance bound; direct diagonal subtraction proves attainment. In a Z-nondemolition conditional preparation the two environment/port vectors purify τ_+, τ_-. Their overlap is at most their root fidelity, $\sum_z \sqrt{p_+(z)p_-(z)} = \operatorname{sech} x$, with equal Schmidt bases attaining it. That overlap multiplies the held off-diagonal matrix element. The qubit trace-distance formula proves the last assertion. □

This is a universal identity-channel condition, not a prohibition on preparing one specified state or maintaining a selected classical content. It supplies no universal constant for arbitrary approximate identity channels. An entire-body SWAP can prepare a fresh correlated copy, but it moves the old body's quantum content and correlations to the spent copy; it is not that identity channel.

For an attainable finite update, specify instead one receiver cell E with $H_E = \delta n_E$ and a fresh bank cell B with the same gap, Gibbs state η_β, and retained purifier. All held material, earlier records and old references form $\mathcal{O}$; the previous state of E may be correlated with all of them. The true S_{EB} commutes with $H_E + H_B$. The positive pulse $g(t)(I - S_{EB})$, $g \geq 0$, $\int g = \pi/2$, implements the true SWAP at finite duration, at least $\pi/(2g_{\max})$. At its endpoint E is η_β independently of the old $\mathcal{O}E$ state. That complete old state now belongs to $\mathcal{O}B$, with its correlations retained. The new cell remains correlated with its own Gibbs purifier. The $\mathcal{O}$ marginal is unchanged nonselectively; old records stored in E have moved to the spent slot rather than being erased. For an isolated prescribed pulse its switch work is zero because the bare cell energy is conserved and the pulse vanishes at the endpoints. This is not zero total controller or preparation cost.

Such a SWAP is also a bounded edge gate in H_D of the positive common material Hamiltonian. The cell must be force-idle while it is replaced. If n_E is still coupled to the material, $[S_{EB}, n_E Q(h)] \neq 0$; the matched gap alone gives no exact reset for that interacting contact. A finite-window estimate can retain this imperfection explicitly.

Proposition 18 (Finite-window port renewal with a live material)**.** *Compare the actual autonomous interface with an ideal renewal evolution having the same H_D, matched cell gaps and SWAP edge gate, but no material force in its renewal window $[0, T_R]$. Suppose their difference on the reference trajectory is $-\sum_a \Delta B_a Q(h_a) + \Delta C$, where $\Delta B_a = P_{\text{leak}} \Delta B_a$, $\Delta C = P_{\text{leak}} \Delta C$ and these controller operators commute with H_F. Here P_{leak} is a controller orthogonal projection; the differences are self-adjoint, so their displayed one-sided support is also two-sided support. Write $b_a = \|\Delta B_a\|$, $c = \|\Delta C\|$, $p(s) = \|P_{\text{leak}} \psi_s^i\|^2$, and assume $\|H_F \psi_s^i\| \leq M$. Define*

$$L_\Delta = \sqrt{2} \sum_a b_a \|\omega^{-1} h_a\|, \qquad c_\Delta = c + \sum_a b_a \|\omega^{-1/2} h_a\| / \sqrt{2}.$$

Including the bank and every old reference, the vector error is at most

$$\epsilon_R \leq \int_0^{T_R} [L_\Delta \sqrt{M}\, p(s)^{1/4} + c_\Delta \sqrt{p(s)}]\, ds. \tag{25}$$

Any specified incomplete ideal-SWAP or clock error is added separately. The same bound controls half trace distance of the renewed cell and held-reference marginal from their ideal outputs.

Proof. Duhamel bounds the vector difference by the integrated norm of the Hamiltonian difference on ψ_s^i. Apply the linear-field estimate to $P_{\text{leak}}\psi_s^i$. Its norm is $\sqrt{p}$ and

$$\|H_F^{1/2} P_{\text{leak}}\psi_s^i\|^2 \le \|P_{\text{leak}}\psi_s^i\|\|H_F P_{\text{leak}}\psi_s^i\| \le M\sqrt{p}.$$

The field and counterterm contributions give (25). Pure-vector distance bounds half trace distance, which contracts under the stated restrictions. □

Corollary 19 (A constructive finite receiver-renewal family)**.** *Fix $A > 0$, $0 < p < 1/2$, a matched receiver/bank pair and a material smearing h. For even m, use the positive spin-history clock with $J = A/m$, a true receiver/bank SWAP on the first edge and identities thereafter. Put $\chi_m = \mathbf{1}_{\{j \ge m/2\}}$ and*

$$H_m = H_{\text{hist}} + H_E + H_B + H_F - \chi_m n_E Q(h) + \tfrac{1}{2}\chi_m n_E \langle h, K^{-1}h\rangle.$$

Its initial cursor is $j = 0$; the material/old receiver state and all old references may be correlated, with initial physical rms $M < \infty$. The independent bank is Gibbs with its own purifier retained. This is a positive autonomous Hamiltonian. At $t_R = (m/A)\arcsin\sqrt{p}$ its joint half trace distance from a completed receiver renewal, with an explicitly specified comparison clock, is at most

$$2(1-p)^{m/2} + \frac{m \arcsin\sqrt{p}}{A}[L_\Delta \sqrt{M}\, e^{-mD/4} + c_\Delta e^{-mD/2}], \quad D = D_{\text{Ber}}(1/2\|p) > 0. \tag{26}$$

It tends to zero while the clock mean energy A, its norm bound $2A$ and the common-H physical rms certificate remain bounded independently of m. Clock space and duration grow; the bank is not cycled for free.

Proof. The field/contact terms are the positive controlled-displacement Hamiltonian of Proposition 14; the history clock and matched cell energies are nonnegative. For the force-free reference, each prefix is either the true SWAP or identity and commutes with $H_E + H_B + H_F$. Prefix conjugation therefore gives the same binomial cursor law for every correlated input. For $s \le t_R$ its binomial parameter is at most p; Chernoff gives $\Pr(j \ge m/2) \le e^{-mD}$. The ideal physical rms is the unchanged M. Proposition 18 proves the second term. The no-SWAP probability is $(1-p)^m$. As comparison use the same untrimmed force-free cursor with the completed SWAP and freely evolved material/reference state. Only its $j = 0$ branch differs from the ideal joint state; the retained-purification vector error is at most $2\sqrt{(1-p)^m}$. Half trace distance is no larger. No clock outcome is postselected. Triangle inequality proves (26). Finally $\|H_{\text{hist}}\| \le 2A$ and its initial mean is A. The cell norms and force/counterterm norms are fixed, so the initial total rms and the physical bound of Proposition 14 are uniform. □

This condition has a finite nonempty realization. Take the positive spin-history clock with $m = 64$, $J = .02$, the matched SWAP on its first edge, and a material force $-.2\,\mathbf{1}_{\{j\ge 32\}} n_E Q(h)$

with $\omega = 1$ and unit one-mode smearing. At $T_R = \arcsin\sqrt{.1}/J = 16.08753$ the force-free reference cursor has Bin$(64, .1)$ law; throughout this window its source tail is at most 7.04652×10^{-16}. Its uncompleted first-edge probability is $.9^{64} = .00117902$. For vacuum material input $M = 0$, (25) is below 6.9×10^{-8}. Completing the ideal SWAP on the same untrimmed cursor costs at most $2\sqrt{.9^{64}} = .068674$ in vector norm. The clock's mean energy is 1.28. Direct sparse evolution of the full force-on Hamiltonian at Fock cutoffs $6, 10, 14$, retaining a correlated held-body/receiver input, its old reference and the fresh bank purifier, gives vector difference 6.54×10^{-10} from the same force-free reference. The receiver's half trace distance from its bank state is .00043624; the held-body/reference marginal changes by less than 2.5×10^{-13} and energy conservation residual is below 4.1×10^{-13}. These are finite-regulator diagnostics of the nonempty renewal window, not a continuum proof. This is a paid receiver-renewal block with given correlated input, not an assertion that a previous program produces its cursor for free.

Actual leakage and completion require paid force-idle padding. The receiver certificate preserves held material and old content, but neither prepares its interacting Gibbs state nor implements an entropy outlet. A J-cell bank retains J spent slots, outputs and purifiers; each isolated replacement transfers excess energy $\delta(\langle n_E\rangle - \nu)$ of either sign. Preparation, clock, storage and removal remain charged. Q-RAIF B's oscillator-bank formulas [26] concern its different material. Repeated noisy readings of one command also do not justify $J \log 2$ in the independent-bit capacity bound (24).

10.2 Several rounds on one running material

The next reading can be installed on the same cursor that performs replacement. Fix a finite sequence of permitted data gates U_b, including receiver–bank swaps, and a nonnegative data energy H_0 with $[U_b, H_0] = 0$. Degenerate logic and matched-cell exchanges meet this condition. Keep the body, all banks, their purifiers, previous records and reversible workspace in the input. Between gates use the linear material forces and completing terms of Eq. (16).

Choose rational $0 < a_1 < \cdots < a_q < 1$ and integers m for which $k_b = ma_b$ are integral. Put U_b on edge k_b of the spin clock; other edges are identities. Each cursor interval has its own fixed force amplitudes, commuting within that interval. Orthogonal cursor supports make the full amplitudes commute. With J fixed, the resulting $H_m \geq 0$ is time independent and has common domain $\operatorname{Dom} H_F$. Its comparison schedule is

$$p(t) = \sin^2(Jt), \qquad t_b = J^{-1} \arcsin\sqrt{a_b}, \qquad T = \pi/(2J).$$

It applies the specified gates at t_b while the same complete material evolves throughout. Only the comparison uses these switching times; the actual apparatus has its gates and forces fixed in space along its cursor.

Proposition 20 (A finite sequence with its history retained)**.** *Let the smooth force profiles satisfy $h_\ell, \omega^{1/2} h_\ell \in L^2$, and fix an input purification with $\|H_F\psi\| \leq M_0$, including arbitrary old references. The actual H_m approximates the scheduled sequence jointly with its complete, untrimmed cursor, at every fixed checkpoint away from the t_b. At T all data gates have been included. The joint vector error tends to zero as $m \to \infty$, uniformly on this input class.*

Both actual and comparison material rms energies are bounded by a constant independent of m.

More explicitly, set

$$L_P = \sqrt{2}\sum_\ell \|B_\ell\|\|h_\ell\|, \qquad c_P = \frac{1}{\sqrt{2}}\sum_\ell \|B_\ell\|\|\omega^{1/2}h_\ell\|.$$

For any energy $E_* > 0$ a common rms bound is

$$M_T = \left[\sqrt{M_0 + E_*} + \frac{T}{2}\left(L_P + \frac{c_P}{\sqrt{E_*}}\right)\right]^2 - E_*. \tag{27}$$

In the prefix gauge, let H_b be the material/data Hamiltonian in interval b, and choose K_b so that $\|(H_b - H_{b-1})\zeta\| \le K_b$ for normalized vectors with $\|H_F\zeta\| \le M_T$. These constants are finite by the linear field bound. Then the gauge-state error satisfies

$$e_m(t) \le \sum_{b=1}^{q} K_b \int_0^t e^{-m[p(s)-a_b]^2}\, ds. \tag{28}$$

At a physical checkpoint add $2\sum_b e^{-m[p(t)-a_b]^2}$; this additional term is zero at T. For $0 < \alpha_b < \min(a_b, 1-a_b)$ and $v_b = 2J\sqrt{(a_b-\alpha_b)(1-a_b-\alpha_b)}$,

$$e_m(T) \le \sum_b K_b \left[\frac{\sqrt{\pi}}{v_b\sqrt{m}} + Te^{-m\alpha_b^2}\right]. \tag{29}$$

Proof. The prefix conjugation used for the history clock transforms its hopping term to $mJ + 2JJ_x$ and each force to the corresponding prefix-conjugated data amplitude. The transformed material Hamiltonian is piecewise constant in j. Compare it with the product of the untrimmed spin-clock state and the data/material sequence H_b. The clock distribution is $\mathrm{Bin}(m, p(t))$. Writing both piecewise Hamiltonians as sums of their jumps gives

$$H_m^{\mathrm{gauge}} - H_{\mathrm{cmp}}(t) = \sum_b (\mathbf{1}_{\{j\ge k_b\}} - \mathbf{1}_{\{t\ge t_b\}})(H_b - H_{b-1}).$$

The norm of each clock factor on the comparison state is the square root of its wrong-side probability. Hoeffding's bound is $e^{-m[p(t)-a_b]^2}$. Duhamel's formula therefore proves (28). On $|p - a_b| \le \alpha_b$ one has $p' \ge v_b$; integrate the Gaussian there and bound its complement by $Te^{-m\alpha_b^2}$ to obtain (29). Undoing the gauge at a checkpoint costs at most twice the square root of the probability of an incorrect prefix. At T the reference cursor is exactly $|m\rangle$, giving no prefix error.

For the energy estimate, all controller terms commute with H_F, even though their norms grow with m. On the energy core, $[H_F, Q(h)] = -iP(h)$ and

$$\left\|\sum_\ell B_\ell P(h_\ell)\zeta\right\| \le L_P\|H_F^{1/2}\zeta\| + c_P\|\zeta\|.$$

For $x(t) = \|H_F\psi_t\|$ this gives $D^+x \le L_P\sqrt{x} + c_P$. Hence $D^+\sqrt{x + E_*} \le (L_P + c_P/\sqrt{E_*})/2$, which proves (27). The data gates commute with H_F and cause no jump in this bound. Finite-mode energy-core regularization gives the same estimate before removing the cutoff; the common-domain propagator limit and closed-operator lower semicontinuity extend it to $\operatorname{Dom} H_F$. The linear field bound then supplies the K_b used above. Every estimate acts on the full purification and tensors with the old reference identity. □

For prescribed finite N, compile acquisition, decision, record force and bank swap on this path. Proposition 58 supplies one decision circuit from actual meter records and retained posterior weights; its repair flag can drive the force. Registers and thermal pairs are prepared initially, and spent correlations remain. At checkpoints, (21) and (27) certify a later reading and source test under the same subsequent H_m.

As a finite diagnostic, two probe readings of one oscillator, an intervening receiver–bank swap and a later force controlled by both records give endpoint joint vector errors .0701594, .0355814, .0192136, .0107206, .00393182 at $m = 16, 32, 64, 128, 512$, with $J = 1$. An old probe reference and the spent bank are retained. At $p = 7/16$, after the swap and before the second probe interval, receiver excitation is .0321012 for $m = 128$ and .00089552 for $m = 512$; endpoint accuracy alone would miss this timing error. The $m = 128$ final coordinate is .0393364 against the scheduled .0393304; removing the record-controlled forces gives -2.89×10^{-6} against zero. Fock cutoffs 8 and 10 change the endpoint vector error by 1.1×10^{-11}. These are regulator checks of the construction.

The clock uses $m + 1$ states, mean energy mJ, variance mJ^2, norm $2mJ$ and maximal edge coefficient at most $J(m+1)/2$. Fixed duration thus costs growing energy; fixed-energy renewal instead costs growing duration. Neither is indefinite fixed-capacity operation, and this material-source bound excludes controller covariant stress.

Appendix K gives a further use of this program: local linear-current pulses implement stress flows in the neutral $c = 1$ carrier, uniformly on bounded mean-energy inputs. The construction supplies a physical permission used in finite modular-geometry tasks, with explicit control and coherence conditions.

The comparison clock need not have the actual clock's energy. For one oscillator take $H_F = \omega a^\dagger a$, $Q = (a + a^\dagger)/\sqrt{2}$, $V = -bQ + C$ and $C = b^2/(2\omega)$. Use identity clock edges and turn on V for cursor sites $j \ge m/2$, with even m. The positive Hamiltonian is $H_m = H_{\mathrm{clock,m}} + H_F + \mathbf{1}_{j\ge m/2}V$. Starting in $|0\rangle_{\mathrm{clock}}|0\rangle_F$, its conserved total energy is mJ. The scheduled endpoint is $|m\rangle \otimes \zeta$ up to phase, where $\zeta = e^{-i(T/2)(H_F+V)}|0\rangle$ and $\langle H_F + V\rangle_\zeta = C$. Consequently

$$\langle H_m\rangle_{\mathrm{scheduled}} - \langle H_m\rangle_{\mathrm{actual}} = C \quad \text{for every even } m. \tag{30}$$

The joint-state error still tends to zero. The finite work is paid by the actual clock and missed by its unperturbed product comparison.

This defect can be repaired for the total-energy observable. Rotate the terminal spin toward negative S_x by an angle θ. Its clock energy is $mJ - mJ\sin\theta$, and its vector distance from $|m\rangle$ is at most $|\theta|\sqrt{m}/2$. If $\epsilon_m(\theta)$ is its probability below $m/2$, the product state's total-energy excess is $-mJ\sin\theta + C - \epsilon_m(\theta)\langle V\rangle_\zeta$. It equals C at zero and is negative at $\theta = \arcsin[2C/(mJ)]$ for sufficiently large m: the last probability is at most $e^{-3m/8}$ once $\theta \le \pi/6$. Continuity therefore supplies an exactly energy-matched comparison state with an additional $O(m^{-1/2})$ vector error. This is a physical comparison vector, not an extra gate

supplied by the original dynamics. Matching one energy expectation does not match an extended stress distribution. HAFF-D uses the mass-work identity of its common action to keep this debit in a finite source comparison [23].

10.3 Local thermal windows and a continuing diagnostic return

Port replacement is distinct from using a later region of a correlated field. Appendix I.1 executes a local thermal preparation with a paid finite autonomous clock. Appendix I.2 proves its record-selected same-field conditional error from vacuum mutual information and controls the source with second moments. The complete outside purification stays present; a spatially separate window is not declared globally fresh. These free-CAR certificates require a new thermal matching proof before use on the nonlinear interacting material.

Appendix J.1 gives a separate finite-horizon return in which the actual meter word, not an inaccessible fault label, selects repair of the held body and a thermal exchange. Appendix J.2 supplies an actual terminal record and replacement of its specified probe. The full input bank, posterior arithmetic, reversible garbage, clock energy and retained output history belong to that horizon. The actual mixed-port quantum-material update above does not inherit their different deadline or mean-support guarantees.

11 A common covariant ledger: controller, clock and support

The local potential (22) has potential stress $-g_{\mu\nu}V_{\mathrm{loc}}$ if its variables are Lorentz scalars. That term alone omits the mechanism which moves its profile and changes its amplitude. A common local action makes this omission explicit and offers a candidate completion at the classical action and regulated canonical levels.

Take dynamical scalars $X = (\phi_1, \phi_2, Z, \theta, y, \chi, r_1, \dots, r_d)$ on a $(1+1)$-dimensional background of signature $(-,+)$. Here Z is a support texture, χ a clock coordinate, and r_j memory/control coordinates. Specify the positive target coframe

$$\begin{aligned} e_{\phi_a} &= d\phi_a + \lambda_a \theta f(Z) dZ, & e_Z &= m_Z dZ, \\ e_\theta &= \sqrt{I}\, d\theta, & e_y &= \sqrt{J}\, dy, \\ e_\chi &= \sqrt{I_\chi}\, d\chi, & e_{r_j} &= \sqrt{I_j}\, dr_j. \end{aligned}$$

with all diagonal coefficients strictly positive, and put $G = \sum e \otimes e$. Let $b_a(\chi, r)$ be smooth bounded functions, $w_a(Z)$ smooth profiles compact in the texture value, and $M(Z)$ a smooth strictly positive material matrix. With $u = (\sqrt{I}\theta, \sqrt{J}y)$ and $v = u - \sum_a b_a(\chi, r) w_a(Z)$, choose

$$S = -\int \sqrt{-g}\, [\tfrac{1}{2} G_{AB}(X) g^{\mu\nu} \partial_\mu X^A \partial_\nu X^B + V(X)]\, d^2x, \tag{31}$$

$$V = V_Z(Z) + V_{\mathrm{mem}}(r) + V_{\mathrm{clock}}(\chi) + \tfrac{1}{2} v^T M(Z) v \geq 0. \tag{32}$$

The last square replaces the material mass potential rather than adding a second copy of it. Z, χ, r are actual fields in this candidate, not prescribed positions, waveforms or cursor

amplitudes. A prepared finite-energy texture can localize the contact; localization is not produced by writing $w_a(Z)$.

Proposition 21 (Common-action stress and clock reaction)**.** *For smooth finite-energy classical data of* (31)*, metric variation gives*

$$T_{\mu\nu} = G_{AB}\partial_\mu X^A \partial_\nu X^B - g_{\mu\nu}[\tfrac{1}{2} G_{AB}\partial X^A \cdot \partial X^B + V]. \tag{33}$$

The Euler–Lagrange equations are

$$\nabla_\mu (G_{AB}\partial^\mu X^B) - \tfrac{1}{2}(\partial_A G_{BC})\partial X^B \cdot \partial X^C - \partial_A V = 0. \tag{34}$$

They imply $\nabla^\mu T_{\mu\nu} = 0$ for the complete matter action. In specified Minkowski coordinates its canonical energy is

$$H = \int [\tfrac{1}{2}\Pi^T G^{-1}\Pi + \tfrac{1}{2} X'^T G X' + V]\, dx \geq 0.$$

At finite Fourier resolution, the following specified canonical ordering has a nonnegative Friedrichs realization on the supplied charge sectors and finite real coordinate space.

Proof. Vary the inverse metric with all scalar variables held fixed and use $T_{\mu\nu} = -2(\delta S/\delta g^{\mu\nu})/\sqrt{-g}$. Integration by parts in a field variation gives (34); contraction with $\partial_\nu X^A$ gives the stress divergence. In Minkowski coordinates $\Pi_A = G_{AB}\dot{X}^B$, so the Legendre transform has the displayed positive squares. The coframe is nonsingular and triangular. For a spatial Fourier regulator, let q denote its finite real coordinates and compact scalar zero modes. The reduced Lagrangian has positive kinetic matrix $G_K(q)$; put $A_K = G_K^{-1}$ and let $W_K(q) \geq 0$ be its integrated gradient and potential energy. Use Lebesgue measure on the real coordinates and the scalar zero-mode line bundles of Proposition 22, with their charge/winding direct sum. On smooth compactly supported sections choose the divergence-form kinetic ordering

$$\mathfrak{q}_K[\psi] = \tfrac{1}{2}\int (D_i\psi)^*(A_K)^{ij}(D_j\psi)\, dq + \int W_K|\psi|^2\, dq.$$

The D_i are coordinate derivatives with the stated zero-mode boundary conditions. Smooth locally elliptic coefficients make this densely defined nonnegative form closable; its Friedrichs realization is the specified operator. If the interaction acts only on retained low modes, the complete free carrier tail is added positively, as in the charged regulator. This choice of canonical measure and ordering is not a claim of regulator-independent covariant quantization. □

In particular,

$$I_\chi \Box\chi = \partial_\chi V = V'_{\text{clock}} - \sum_a (\partial_\chi b_a) w_a^T M v.$$

The amplitude change reacts on the clock. The memory equations have the corresponding $\partial_{r_j} b_a$ force, and the support equation includes derivatives of w_a, M and the target coframe. Dropping these terms returns an externally driven model. For a free massless clock initially

having $\partial_t \chi = \Omega$ over a proper-length interval ℓ, its initial kinetic energy is at least $I_\chi \Omega^2 \ell/2$. Smooth spatial tapers and contact reaction add costs and change that free trajectory. A locally timelike clock gradient, texture, memory preparation and boundary conditions are specified initial data; they are not guaranteed throughout an arbitrary run.

The action includes clock/support reaction; canonical positivity is only in the chosen coordinates and finite regulator. Curved-space quantum positivity, renormalized continuum stress and causal propagation after cutoff removal require further proofs. Matching (16) requires quantum acquisition and transported gates, the prepared texture/charge sectors, and a common finite-cursor/source estimate. The Gaussian K needs static supplied Z and controlled freezing or small perturbations of r, χ; dynamical cross terms and backreaction otherwise change H. That approximation needs an error bound, as does matching HAFF D's dilaton coordinates and full supported quantum state.

12 Discussion

The exact representation problem and the material return problem now have different tests. A discarded-port channel is checked by its multiplicative domain; a closed propagation symbol by its shell and coefficient relations. An aligned actuator can share the latter generators and charge their spatial form while still losing exact reduced control. The common-action and finite-return constructions ask whether the records, permissions and sources needed for these operations can be implemented together.

Acquisition is a physical change of the old history, not a supplied source label. The actual record drives the same complete material and produces another reading. Its connected fluctuations contribute to the source; dephasing can preserve an immediate field marginal while changing later joint control. Physical moments and smooth-source forms certify that return without replacing an unbounded stress by a trace-distance estimate.

Renewal depends on what is held. Universal identity-channel preservation cannot append a correlated interacting equilibrium. A specified receiver can nevertheless be replaced by a paid Gibbs bank, with spent correlations retained; the full ring can be reset and its mixed target reinstalled through four thermal outlets. Neutral compression, packet loading and spectral sorting supply a complete-output converter, not merely thermal mean energy. Isospectral axes need not retain a heat-source contrast after optimal sorting, and accurate direct reset alone leaves its purifier redundancy. Persistent sharp-exhaust gain remains a joint implementation question.

The clock distinctions in the introduction fix the resource claims: bounded-energy slow receiver renewal, fixed-duration energy-growing finite rounds, and a spatial work field with horizon-dependent inertia. The full joint-state estimate for the latter retains every old reference and charges finite clock work in the source comparison. Static frame/endpoints and dynamical support are still different problems. The neutral-current stress compiler gives another finite physical control permission, with coherence and precision budgets; it is not noise-protected control over every charged sector.

Foundation III supplies a complementary finite semiclassical mean-action scalar/JT return at common initial constraints: a learned frame protects old content, its record changes the material source, and a later X-basis reading of the old probe carries the geometric re-

Interface question	Established construction	Required input or remaining bridge
Exact operations	Multiplicative domain; propagation symbol; retained loops	Spectral law, access and quadratic-space conditions
Calibrated charging	State-independent work with actual generator alignment	Centered independent input; commuting drift; a measured command–momentum map
Actual quantum return	Acquisition, full material response and second record under one positive H	Prepared ports, allowed gates, physical transport and finite cursor
Local source	Same-H smooth interacting field bound; local positive contact potential	Smooth interior tests and moments; controller/support stress is separate
Finite maintenance	Receiver-bank SWAP; correlated-window and diagnostic protocols	Paid bank, storage and clock; not full interacting-material Gibbs renewal
Repeated body renewal	Full ring contraction, mixed-target preparation, neutral code, packet loading and finite-carrier sorting	Four thermal outlets, paid buffer, prescribed gates and energy-constrained capacity
Massive spatial body	Guarded local full-angle gate and weighted physical stress tests	Paid mass profile and state; actual outlet renewal and moving support remain
Field–clock evolution	Retained-state finite schedule with matrix Ward and delayed boundary	Static walls/frame; smooth input norms and horizon-dependent inertia
Geometric flow tasks	Local linear-current synthesis of neutral $c = 1$ stress gates	Circle recurrence, coherent control and growing precision budgets
Covariant apparatus	Common local action with dynamic clock and support reaction	Quantum gate matching, charged continuum and renormalized total stress

Table 1: The main interfaces and their distinct burdens. Auxiliary apparatus proofs are collected in the appendices.

sponse into the spent cell [50]. Its first-order signal arises after acquisition; the cursor and old purifications remain. Matching that run to the complete quantum material here requires a physical state/source bridge. HAFF D's rest-frame mean-stress proposition identifies one excess-radiation sector under fixed covariance and linear contact assumptions, not the entire acquired-record apparatus. Neither construction treats a spent port as a fresh bath.

A world model must retain the carrier and conserved total source; an interface must specify actual records and executable access; a continuing observer must maintain the content and resources those records serve. The common-action candidate connects these responsibilities at the classical and regulated canonical levels. Its quantum acquisition, charged continuum, moving supports and total renormalized stress still need matching. Neither exact multiplicativity nor finite repair derives experience or public geometric agreement. Conservation-law and multimode bounds [36, 38, 39] do not settle all these architectures at once. Split/modular inclusions and virtual subsystems [43, 44, 42] organize retained access; formation and maintenance of its permissions remain questions for T-DOME and Imprint.

13 Conclusion

A coupling represents an algebra exactly only under specific conditions. After a non-atomic dephasing port is discarded, its multiplicative domain is the coupling commutant. Retention can change that answer: closed oscillator loops return their port exactly, while closed propagation retains a faithful copy of the whole internal algebra. The single-shell condition supplies the spatial quadratic structure, and the supplied time and mass anticommutator supply its Lorentzian extension.

An actual alignment of propagation and actuator generators gives a conditional work–shell relation. It does not identify a positive cost with an indefinite metric, and it does not remove the reduced operation's multiplicative defect. This separates a testable physical compatibility condition from an assumed representation map.

The finite material construction makes the records, source and next action part of one retained quantum history. Receiver renewal and smooth-source accuracy have explicit clock, preparation and energy conditions. The further ring, massive-body and spatial-clock certificates extend what can be implemented, but their distinct layouts still need a common supported quantum realization. Foundation III's finite semiclassical return and HAFF-D's sustained renewal and joint sharp-exhaust questions retain their separate burdens. Exact representation is an interface condition; it does not by itself generate the world, a permanent apparatus or a subject.

Certificate guide: Q-RAIF C technical supplement

The representation bridge, physical mechanism and core results remain in this paper. The complete technical appendices are in `QRAIF_C_supplement.pdf`, *Retained Material, Work Clocks and Renewal Certificates: Technical Supplement to Q-RAIF C*. This is supplementary documentation, not a separate paper. Appendix letters and all existing result and equation numbers are retained. References below link to the supplement, which links back to the main argument.

Main claim or question	Technical certificate	Scope of support
Complete charged material	Appendix A; Propositions 22, 25	Regulated carrier and smooth source tests with weighted tails.
Finite thermal-field return	Appendix B; Propositions 27, 28	The same field and old correlations are retained.
Ring-body renewal and conversion	Appendix C; Propositions 29, 30, 32	Paid outlets and buffers; finite-carrier assumptions.
Spatial work clock and massive gate	Appendix D; Propositions 35, 37	Finite joint-state control under the supplied walls and frame.
Conserved control and local outlet	Appendices E, F; Propositions 38, 41	Battery, timing, output window and outlet costs are retained.
Preparation, contact and recoil	Appendices G, H; Propositions 45, 54	The selected source is not its mean covariance; reaction is physical.
Actual acquired-record return	Appendices I, J; Propositions 57, 58	Actual programs, delayed readings and charged finite windows.
Neutral stress-flow control	Appendix K; Proposition 60	Local current synthesis with stated recurrence and precision resources.

These certificates concern distinct finite layouts. They do not establish one supported quantum apparatus with sustained renewal and jointly sharp informative exhaust; the remaining matching conditions are stated in the main discussion.

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